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NWS MG Series SECS/GEM Documentation V1.1.18

NexGen Wafersystems GmbH

1 Documentation History

Version	Date	Modification/Description	Author
1.0.0	10.02.2017	Initial version	Dominik Mueller
1.0.1	11.06.2018	Additional events and Variables	Dominik Mueller
1.0.2	2.10.2018	• 6.6 - Process Program Management Formatted Process Program streams are not supported	Dominik Mueller
1.0.3	25.10.2018	Updated Equipment Constants, Alarms and Data/Status-variables Added definitions for PM/CHC/Control/Process/Port states	Dominik Mueller
1.0.4	04.03.2019	Updated definition for ControlState and ProcessState	Dominik Mueller
1.0.5	13.05.2019	• 3.3 – Process State Model Updated CEIDs for the given examples • 8.2 – Data Variables (DVVALs) Added a CIED column for the DataVariables to clarify when they are valid.	Dominik Mueller
1.0.6	15.07.2019	• 6.3 – Exception Handling Added S5,F7 and S5,F8 Description • 8.2 – Status Variables Updated AlarmsSet Status	Dominik Mueller
1.0.7	19.02.2020	Updated following sections: • 6.8 – Stream 14: Object Services • 6.9 – Stream 16: Processing Management • 7.9 – Processing Management • 8.1 – Collection Events • 8.2 – Data Variables	Iwin Lee
1.1.0	26.03.2020	Updated following sections: • 2.2 – SEMI Standards • 3.4 – Load Port Transfer State Model • 3.5 – Load Port Access Mode State Model • 3.6 – Carrier State Model • 3.7 – Process Job State Model • 3.8 – Control Job State Model • 4 – Message Summary • 6 – Message Details • 8.1 – Collection Events • 8.2 – Data Variables	Iwin Lee
1.1.1	27.03.2020	Added Alarms/Events/Variables for HPC/BEM/ATMSi • 8.1 – Collection Events • 8.2 – Data Variables	Dominik Mueller

		• 8.2 – Status Variables • 8.5 – Alarms	
1.1.2	09.04.2020	Updated following: • 2.2 – GEM300 Compliance • 3.6 – Load Port Reservation State Model • 3.7 – Load Port/Carrier Association State Model • 4.3 – Stream 3 • 6.3 – Stream 3 • 6.9 – Stream 10 • 6.10 – Stream 16 • 8.1 – Collection Events • 8.2 – Data Variables	Iwin Lee
1.1.3	31.08.2020	Additional Data and Status Variables	Iwin Lee
1.1.4	31.08.2020	Updated: - Status Variables added for BEM flow in Status Variables (SVs) table. - Data Variables added for BEM flow in Data Variables (DVVALs) table.	Slaven Jelinic
1.1.5	24.09.2020	Added DVVAL 2103-2108, 2113-2118, 2210 Added SV 4999-5039. Added CEID 862.	Iwin Lee
1.1.6	25.01.2021	Added DVVAL 2301-2315, 2401-2415 for ATMSi	Iwin Lee
1.1.7	25.03.2021	Added DVVAL 2212,2213. Updated Data Variables (DVVALs) table.	Iwin Lee
1.1.8	11.05.2021	Added SV 5100-5141.	Iwin Lee
1.1.9	08.06.2022	Added DV 2317, 2417.	Iwin Lee
1.1.10	15.08.2022	Added SV 5300, 5301.	Iwin Lee
1.1.11	17.02.2023	Added CEID 531-532, DV 3001-3109 for IRM.	Iwin Lee
1.1.12	04.09.2023	Added LowFlow variables. • 8.1 – Collection Events • 8.2 – Status Variables	Josef Fritzer
1.1.12	19.01.2024	Added CEID 600, DV 2900 for PreAligner	Iwin Lee
1.1.13	29.05.2024	Added SV 4450 for diwO3Conc	Iwin Lee
1.1.14	13.06.2024	Removed SV 3905 and added DV 1914-1916, 2014-2016, SV 3521, SV 3721 for DiwO3	Iwin Lee
1.1.15	07.09.2024	Added CEID 880-883 for Canister Fill Pause	Iwin Lee
1.1.16	13.11.2024	Add 3540-3545, 3740-3745 PM CO2 and Exhaust SVIDs Add 4451-4456 DiwO3 Concentration SVIDs	Iwin Lee
1.1.17	11.03.2025	Updated 5.2 with latest HCACK description	Iwin Lee
1.1.18	01.04.2025	Added 9.2 GEM300 Lot Start sequence example	Iwin Lee

Contents

2 Introduction

This SECS-GEM documentation applies to the following equipment plattforms:

- MG21
- MG22
- MG22-300

While it is intended that the information contained herein will accurately correspond to the current version of the equipment software, the SECS-GEM package may be subject to modifications without prior notice. Data types, numerical constants such as collection event IDs (CEIDs), Variable IDs (VIDs), and the numbers assigned to equipment processing states may change.

2.1 GEM Compliance

Fundamental GEM Requirements	Implemented	GEM Compliant
State Models	☑ Yes	☑ Yes
Equipment Processing States	☑ Yes	
Host-Initiated S1,F13/F14 Scenario	☑ Yes	
Event Notification	☑ Yes	
Online Identification	☑ Yes	
Error Messages	☑ Yes	
Documentation	☑ Yes	
Control (Operator-Initiated)	☑ Yes	

Additional Capabilities	Implemented	GEM Compliant
Establish Communication	☑ Yes	☑ Yes
Dynamic Event Report Configuration	☑ Yes	☑ Yes
Variable Data Collection	☑ Yes	☑ Yes
Trace Data Collection	☑ Yes	☑ Yes
Status Data Collection	☑ Yes	☑ Yes
Alarm Management	☑ Yes	☑ Yes
Remote Control	☑ Yes	☑ Yes
Equipment Constants	☑ Yes	☑ Yes
Formated Process Programm Management	☑ No	☑ No
Unformatted Process Programm Management	☑ Yes	☑ Yes
Material Movement	☑ Yes	☑ Yes
Equipment Terminal Services	☑ No	☑ No
Clock	☑ Yes	☑ Yes
Limits Monitoring	☑ Yes	☑ Yes

Spooling	☑ No	☑ No
Control (Host-Initiated)	☑ Yes	☑ Yes

2.2 GEM300 Compliance

GEM300 is only supported on equipment platforms equipped with FOUP.

GEM300 Requirements	Implemented	GEM300 Compliant
Carrier Management	☑ Yes	☑ Yes
Process Job Management	☑ Yes	
Control Job Management	☑ Yes	
Substrate Tracking	☑ Yes	

2.3 SEMI Standards

The following SEMI standards have been referenced for the implementation and terminology of this interface documentation:

Standard	Description	Revision
E4	SEMI Equipment Communications Standard 1Message Transfer (SECS-I)	0307
E5	SEMI Equipment Communications Standard 2 Message Content (recipe-II)	0707
E37	High Speed SECS Message Services (HSMS) Generic Services.	0303
E30	Generic Model for Communications and Control of Manufacturing Equipment (GEM)	0307
E39	Object Services Standard	1218
E40	Process Job Management	1218
E87	Carrier Management	0619
E90	Substrate Tracking	0312
E94	Control Job Management	0314

2.4 State Chart Notation

State charts used to describe depict states, events, and conditions of the equipment.

3 Equipment Overview

3.1 Communications State Model

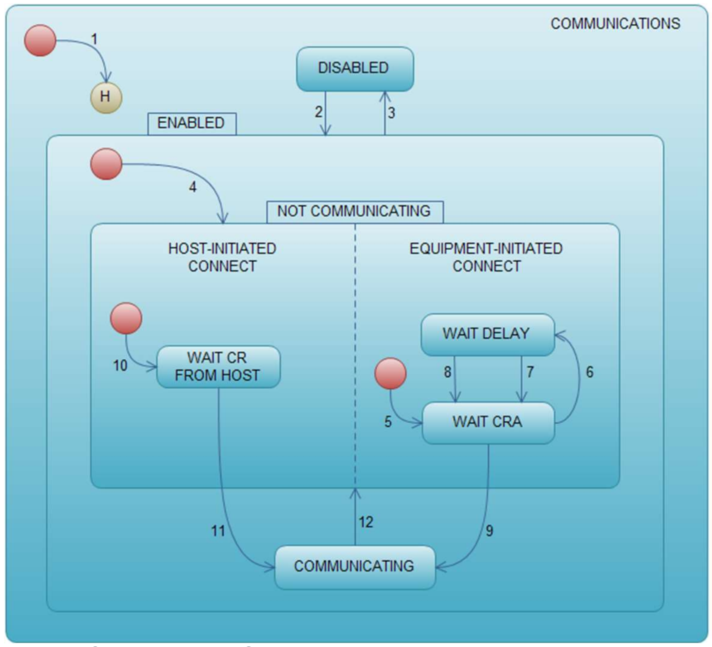


Figure 1 Communications State Diagram

Communications State Definitions

3.1.1.1 ENABLED

In this state the equipment is either communicating with the host or actively attempting to establish communications. Correspondingly, the ENABLED state has two sub states, COMMUNICATING and NOT COMMUNICATING. At entry to the ENABLED state, the sub state NOT COMMUNICATING is active until communications are established. This state is a potential system default.

3.1.1.2 DISABLED

The DISABLED state is characterized by no equipment/host communications. Open transactions or conversations are terminated, and output message queues are flushed. This state is a possible system default.

3.1.1.3 ENABLED/NOT COMMUNICATING

Only messages of type S1,F13, S1,F14, and S9,Fx can be sent in this sub state. All other messages are ignored. Similarly, only S1,F13 and S1,F14 messages are processed by the equipment during this sub state. The equipment waits for an S1,F13 from host to go in the COMMUNICATING state.

3.1.1.4 ENABLED/COMMUNICATING

Host/equipment communications have been established and must be maintained. In the event of communications failure, the equipment returns to the NOT COMMUNICATING sub state.

3.1.1.5 NOT COMMUNICATING/HOST-INITIATED CONNECT

This state describes the 11ignalled of the system in response to a host-initiated S1,F13 while NOT COMMUNICATING is active.

3.1.1.6 NOT COMMUNICATING/HOST-WAIT CR FROM HOST

The equipment waits for an S1,F13 from the host. If the S1,F13 is received, the equipment attempts to send and S1,F14 with COMMANK=0

3.1.1.7 NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT

This state describes the 11ignalled of the system to a equipment initiated connection scenario. When this state becomes active a transition to WAIT CRA occurs, the CommDelay timer is set to "expired" and an attempt to send S1,F13 is made.

3.1.1.8 NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT/WAIT CRA

A S1,F13 message has been sent to the host. The equipment waits for the host to acknowledge the request.

3.1.1.9 NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT/WAIT DELAY

A connection transaction failure has occurred. The CommDelay timer has been initialized. The equipment waits for the timer to expire.

Communications State Transition Table

#	Current State	Trigger	New State	Action	Comment
1	Entry to COMMUNICATIONS	System initialization	Default state	None	
2	DISABLED	User switches from DISABLED to ENABLED	ENABLED	None	SECS-II communications is enabled
3	ENABLED	User switches from ENABLED to DISABLED	DISABLED	None	SECS-II communications is disabled
4	Entry to ENABLED	Any entry to the ENABLED state	NOT COMMUNICATING	None	May enter form the system initializations or User switches to ENABLED
5	Entry to EQUIPMENT-INITIATED CONNECT	Any entry to NOT COMMUNICATING	WAIT CRA	Set CommDelay timer to "expired". Send S1,F13	Begin attempt to establish communications.
6	WAIT CRA	Connection transaction failure	WAIT DELAY	Initialize CommDelay timer.	Wait for timer to expire
7	WAIT DELAY	CommDelay	WAIT CRA	Send	Wait for

		timer expired.		S1,F13	S1,F14. May receive S1,F13 from host.
8	WAIT DELAY	Received a message other than S1,F13	WAIT CRA	Discard message. No reply. Set CommDelay timer “expired”. Send S1,F13.	Indicates opportunity to establish communications.
9	WAIT CRA		COMMUNICATING		
10	Entry to HOST-INITIATED CONNECT	Any entry to NOT COMMUNICATING	WAIT CR FROM HOST	None	Wait S1,F13 from host
11	WAIT CR FROM HOST	Receive S1,F13 from host	COMMUNICATING	Send S1,F14 with COMMACK =0	Communications are established
12	COMMUNICATING	Communication failure.	NOT COMMUNICATING	Discard all messages queued to send	

3.2 Control State Model

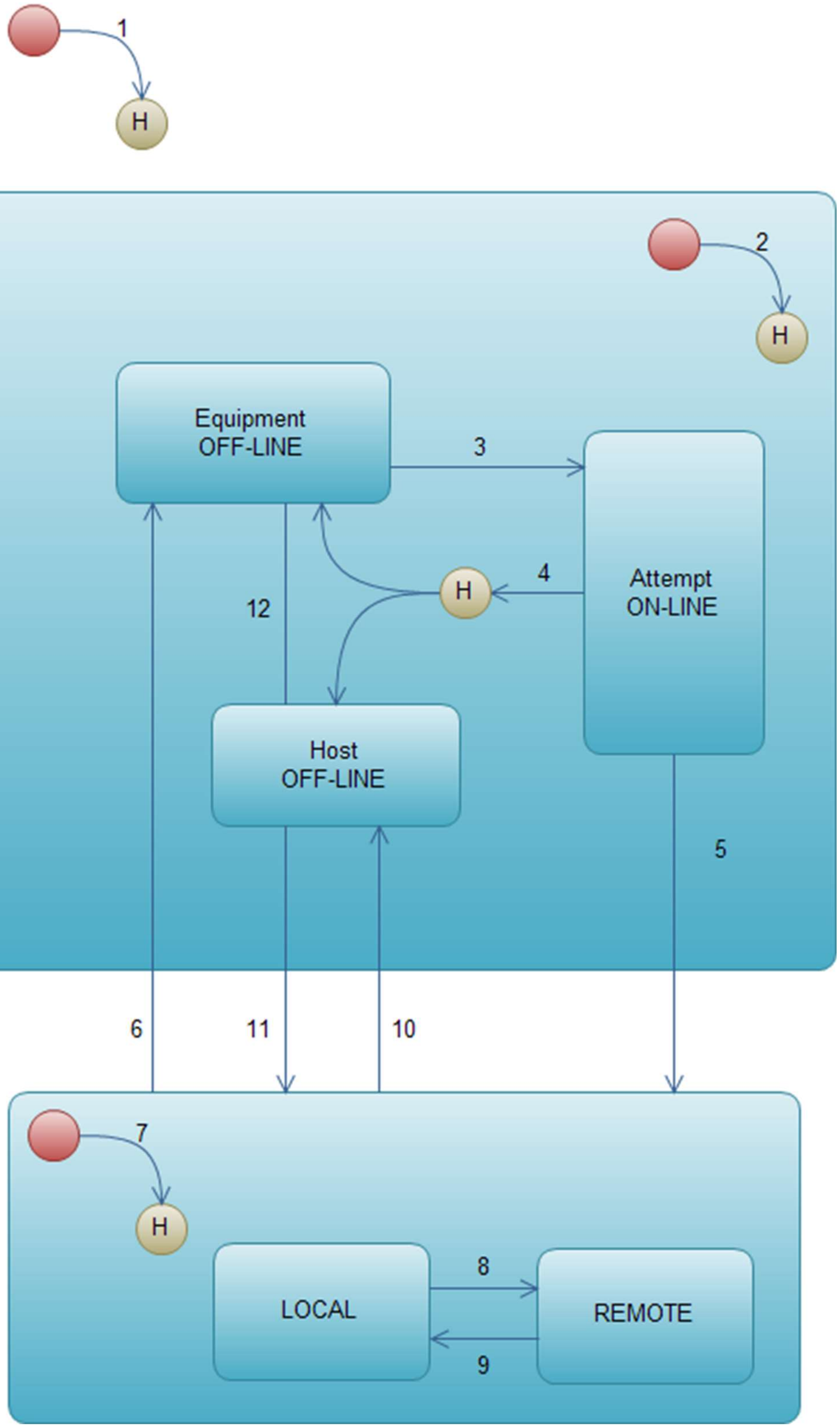


Figure 2 Control State Model

Control State Definitions

OFF-LINE

When the OFF-LINE state is active, operation of the equipment is performed by the operator at the operator console. While the equipment is OFF-LINE, message transfer is possible. However the use of messaging for any automation purpose is severely restricted. While the OFF-LINE state is active, the equipment will only respond to those messages used for the establishment of communications or a host request to activate the ON-LINE state.

While OFF-LINE, the equipment will respond with an Sx,F0 to any primary message from the host other than S1,F13 or S1,F17. It will process and respond to S1,F13 and S1,F17. S1,F17 is used by the host to request the equipment to transition to the ON-LINE state. The equipment will accept this request and send a positive response only when the HOST OFF-LINE state is active (see transition 11 definition below). While the OFF-LINE state is active, the equipment shall attempt to send no primary message other than S1,F13,10S9,Fx,11and S1,F1 (see ATTEMPT ON-LINE sub state). If the equipment receives a reply message from the host other than S1,F14 or S1,F2, this message is discarded.

OFF-LINE has three sub states:

- EQUIPMENT OFF-LINE
- ATTEMPT ON-LINE
- HOST OFF-LINE.

OFF-LINE/EQUIPMENT OFF-LINE

While this state is active, the system maintains the OFF-LINE state. It awaits operator instructions to attempt to go
ON-LINE.

OFF-LINE/ATTEMPT ON-LINE

While the ATTEMPT ON-LINE state is active, the equipment has responded to an operator instruction to attempt to go to the ON-LINE state. Upon activating this state, the equipment attempts to send an S1,F1 to the host. Note that when this state is active, the system does not respond to operator actuation of either the ON-LINE or the OFF-LINE switch.

OFF-LINE/HOST OFF-LINE

While the HOST OFF-LINE state is active, the operator's intent is that the equipment be ON-LINE. However, the host has not agreed. Entry to this state may be due to a failed attempt to go ON-LINE or to the host's request that the equipment go OFF-LINE from ON-LINE (see the transition table for more detail). While this state is active, the equipment shall positively respond to any host's request to go ON-LINE (S1,F17). Such a request shall be denied when the HOST OFF-LINE state is not active.

ON-LINE

While the ON-LINE state is active, SECS-II messages may be exchanged and acted upon. Capabilities that may be available to the host should be similar to those available from the operator console wherever practical. The use of Sx,F0 messages is not required while the ON-LINE state is active. Their use is discouraged in this case.

The only allowed use is to close open transactions in conjunction with message faults.

ON-LINE/LOCAL

Operation of the equipment is implemented by direct action of an operator. All operation commands shall be available for input at the local operator console of the equipment.

ON-LINE/REMOTE

For equipment which supports the GEM capability of remote control (see § 4.4), while the REMOTE state is active, the host shall have access, through the communications interface, to the necessary commands to operate the equipment through the full process cycle in an automated manner. The

equipment does not restrict any host capabilities when REMOTE is active. The degree of control executed by the host may vary from factory to factory.
In some cases, the operator maybe required to interact during remotely controlled processes. This interaction may involve set-up operations, operator assist situations, and others. This state is intended to be flexible enough to accommodate these different situations.

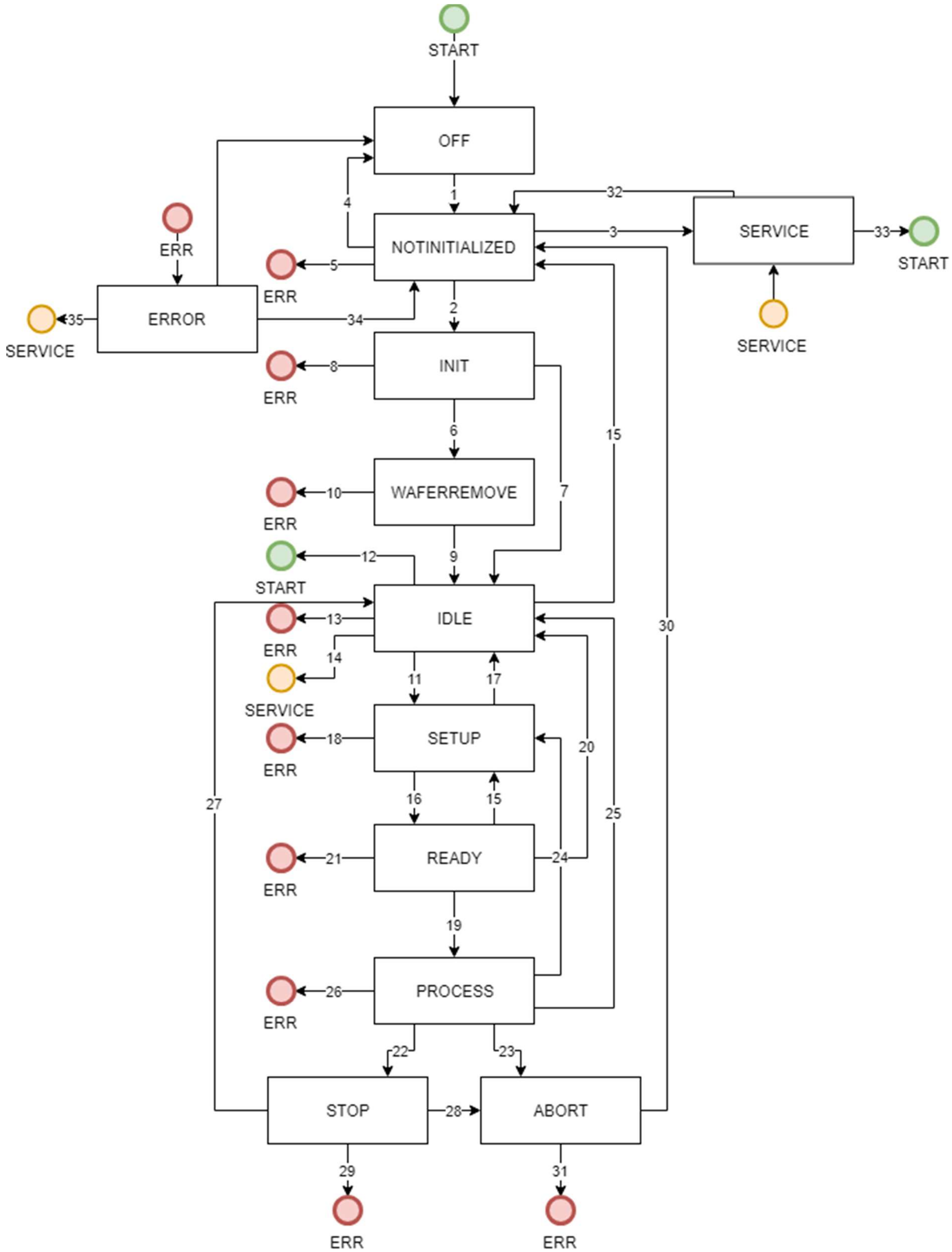
Control State Transition Table

#	Current State	Trigger	New State	Action	Comment
1	(Undefined)	Entry into CONTROL state (system initialization).	CONTROL (Sub state conditional on configuration).	None	Equipment may be configured to default to ON-LINE or OFF-LINE
2	(Undefined)	Entry into OFF-LINE state.	OFF-LINE (Sub state conditional on configuration).	None	Equipment may be configured to default to any sub state of OFF-LINE.
3	EQUIPMENT OFF-LINE	Operator actuates ON-LINE switch.	ATTEMPT ON-LINE	None	Note that an S1,F1 is sent whenever ATTEMPT ON-LINE is activated.
4	ATTEMPT ON-LINE	S1,F0	New state conditional on configuration.	None	This may be due to a communication failure, reply timeout, or receipt of S1,F0. Configuration may be set to EQUIPMENT OFF-LINE or HOST OFFLINE.
5	ATTEMPT ON-LINE	Equipment receives expected S1,F2 message from the host.	ON-LINE	None	Host is notified of transition to ON-LINE at transition 7.
6	ON-LINE	Operator actuates OFF-LINE switch.	EQUIPMENT OFF-LINE	None	“Equipment OFF-LINE” event occurs. Event reply will be discarded while OFF-LINE is active.
7	(Undefined)	Entry to ON-LINE state.	ON-LINE (Sub state conditional on REMOTE/LOCAL switch setting.)	None	“Control State LOCAL” or “Control State REMOTE” event occurs. Event reported based on actual ON-LINE sub state activated.
8	LOCAL	Operator sets front panel switch to REMOTE.	REMOTE	None	“Control State REMOTE” event occurs.

9	REMOTE	Operator sets front panel switch to LOCAL.	LOCAL	None	“Control State LOCAL” event occurs.
10	ON-LINE	Equipment accepts “Set OFF-LINE” message from host (S1,F15).	HOST OFF-LINE	None	“Equipment OFF-LINE” event occurs.
11	HOST OFF-LINE	Equipment accepts host request to go ON-LINE (S1,F17).	ON-LINE	None	Host is notified to transition to ON-LINE at transition 7.
12	HOST OFF-LINE	Operator actuates OFF-LINE switch.	EQUIPMENT OFF-LINE	None	“Equipment OFF-LINE” event occurs.

3.3 Process State Model

The actual process state of the equipment is indicated in status variable SVID 15 (processState), via this variable the process state is indicated in 1-byte integer (unsigned) U1 format according to the following diagram.



Process state definitions

0 - OFF
In this state the equipment is switched off and not ready for operation. Control system and PC are running.

1 - ERROR
In this state the equipment has an uncleared alarm pending. This alarm can be a process alarm or hardware alarm depending on the equipment and monitoring configuration.

2 - SERVICE
Equipment is in service mode only when service personal activates service switch into service position. Within this state teaching and maintenance work on the equipment is possible.

3 - INIT
The equipment is initializing all hardware sub-modules.

4 - NOT INITIALIZED
The equipment is waiting for hardware module initialization.

5 - IDLE
In this state the equipment completely initialized and no critical alarm is active.

7 - PROCESS SETUP
In this state the equipment is setting up all necessary sub- and external modules for processing (e.g. medium temperature), based on the loaded/selected recipe.

8 - PROCESS READY
In this state all external conditions necessary for process execution are satisfied, such as ensuring material is present at the equipment, input/output ports are in the proper state, parameters such as temperature and pressure values are within limits, etc.

In this state all conditions to start processing are met. The equipment is ready for processing and will start to process a job automatically.

9 - PROCESSING
Executing is the state in which the equipment is executing a process recipe automatically and continues to do so without external intervention.

10 - STOP
Equipment has received a stop request from operator.
If there was a wafer in the process chamber when the stop process was triggered the wafer in the process chamber was processed and handled back to its destination slot. If there were remaining wafers in the handling system (robot or ecos) they have been handled back to their load slot and have not been processed.

12 - ABORT
Equipment has received an abort request from operator, host or due to limit deviations.
In this state an alarm with abort condition occurred or it was triggered by operator when processing a lot/wafer. Depending on the trigger condition, there are two possibilities of behaviour. When operator or equipment controller triggers the abort, in both chamber clean

program is immediate started and all wafers from handling and in the end the wafer from chamber handled back to there cassette slots. When a process module triggers the abort, in this process module the clean program is immediately started, and wafer is cleaned with clean program, the wafer in the other chamber will normally processed and finished. All wafers from handling and chamber are handled back to their cassette slots.

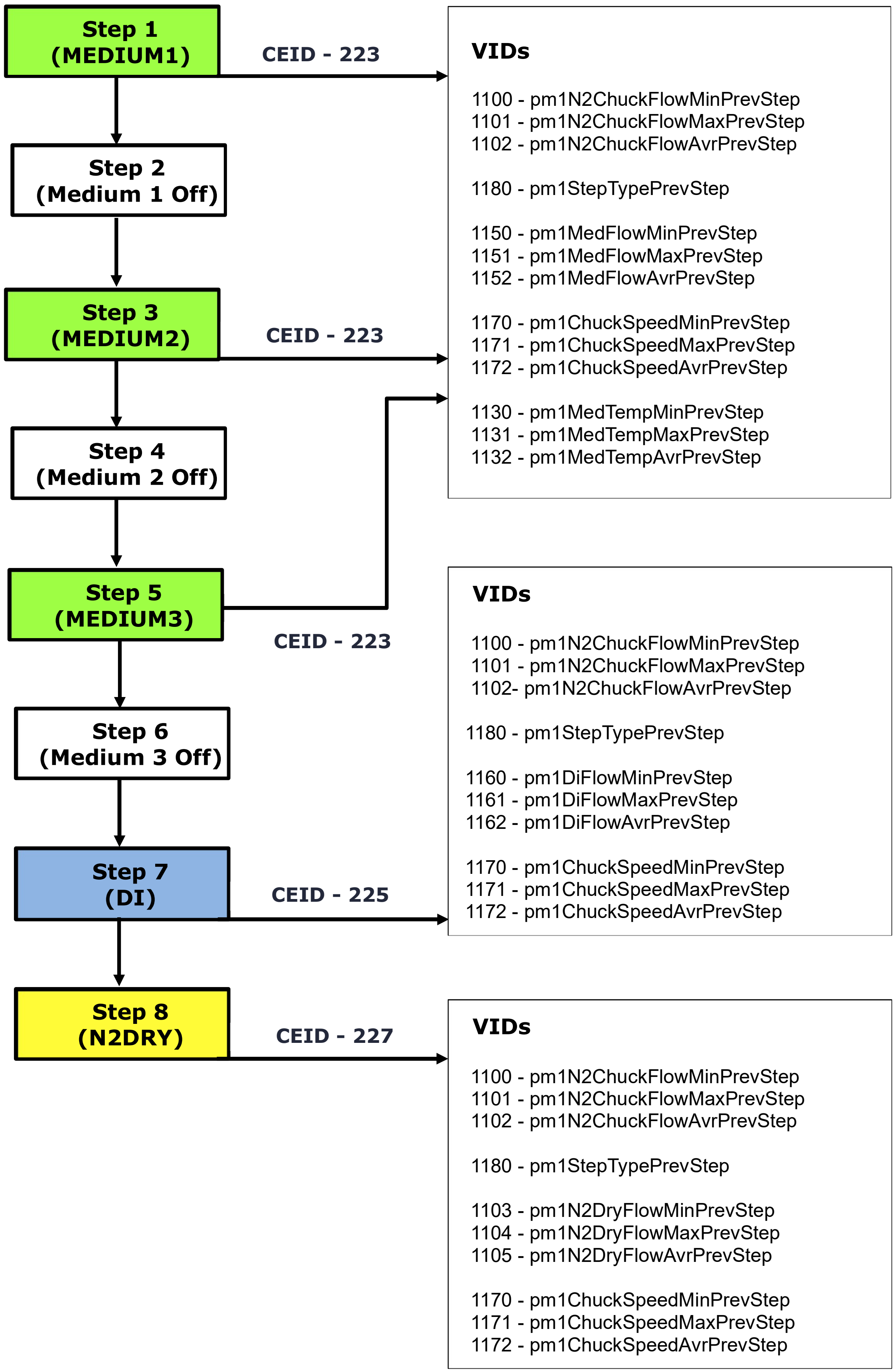
20 - WAFER REMOVE
Equipment is ready to remove wafer that are already in the system after initialization.

Process step related events (CEIDs) and variables (VIDs)

The following illustration shows events and variables that are available to collect process (recipe) step related information for PM1. The given example shows a recipe with eight steps including medium 1, medium 2, medium 3, associated spin off steps, di water rinse and the dry step.

Used CEIDs:
223 - pm1MediumStepFinished
225 - pm1DiStepFinished
227 - pm1N2DryStepFinished

Used VIDs:
1100 - pm1N2ChuckFlowMinPrevStep
1101 - pm1N2ChuckFlowMaxPrevStep
1102 - pm1N2ChuckFlowAvrPrevStep
1103 - pm1N2DryFlowMinPrevStep
1104 - pm1N2DryFlowMaxPrevStep
1105 - pm1N2DryFlowAvrPrevStep
1130 - pm1MedTempMinPrevStep
1131 - pm1MedTempMaxPrevStep
1132 - pm1MedTempAvrPrevStep
1150 - pm1MedFlowMinPrevStep
1151 - pm1MedFlowMaxPrevStep
1152 - pm1MedFlowAvrPrevStep
1160 - pm1DiFlowMinPrevStep
1161 - pm1DiFlowMaxPrevStep
1162 - pm1DiFlowAvrPrevStep
1170 - pm1ChuckSpeedMinPrevStep
1171 - pm1ChuckSpeedMaxPrevStep
1172 - pm1ChuckSpeedAvrPrevStep
1180 - pm1StepTypePrevStep


Step 1
(MEDIUM1)
CEID - 223
VIDs
1100 - pm1N2ChuckFlowMinPrevStep
1101 - pm1N2ChuckFlowMaxPrevStep
1102 - pm1N2ChuckFlowAvrPrevStep
1180 - pm1StepTypePrevStep
1150 - pm1MedFlowMinPrevStep
1151 - pm1MedFlowMaxPrevStep
1152 - pm1MedFlowAvrPrevStep
1170 - pm1ChuckSpeedMinPrevStep
1171 - pm1ChuckSpeedMaxPrevStep
1172 - pm1ChuckSpeedAvrPrevStep
1130 - pm1MedTempMinPrevStep
1131 - pm1MedTempMaxPrevStep
1132 - pm1MedTempAvrPrevStep
Step 2
(Medium 1 Off)
Step 3
(MEDIUM2)
CEID - 223
Step 4
(Medium 2 Off)
Step 5
(MEDIUM3)
CEID - 223
VIDs
1100 - pm1N2ChuckFlowMinPrevStep
1101 - pm1N2ChuckFlowMaxPrevStep
1102- pm1N2ChuckFlowAvrPrevStep
1180 - pm1StepTypePrevStep
1160 - pm1DiFlowMinPrevStep
1161 - pm1DiFlowMaxPrevStep
1162 - pm1DiFlowAvrPrevStep
1170 - pm1ChuckSpeedMinPrevStep
1171 - pm1ChuckSpeedMaxPrevStep
1172 - pm1ChuckSpeedAvrPrevStep
Step 6
(Medium 3 Off)
Step 7
(DI)
CEID - 225
Step 8
(N2DRY)
CEID - 227
VIDs
1100 - pm1N2ChuckFlowMinPrevStep
1101 - pm1N2ChuckFlowMaxPrevStep
1102 - pm1N2ChuckFlowAvrPrevStep
1180 - pm1StepTypePrevStep
1103 - pm1N2DryFlowMinPrevStep
1104 - pm1N2DryFlowMaxPrevStep
1105 - pm1N2DryFlowAvrPrevStep
1170 - pm1ChuckSpeedMinPrevStep
1171 - pm1ChuckSpeedMaxPrevStep
1172 - pm1ChuckSpeedAvrPrevStep

3.4 Load Port Transfer State Model

A load port is used to load and unload carriers to and from the equipment. A load port is generally designed to handle one specific carrier type, such as substrate cassettes, leadframe magazines, SMIF pods, or FOUPs. The purpose of the Load Port Transfer State Model is to define the host view of a carrier transfer, which includes the host interactions with the equipment necessary to transfer carriers to and from equipment load ports. Each load port on the equipment shall maintain an independent instance of this state model.

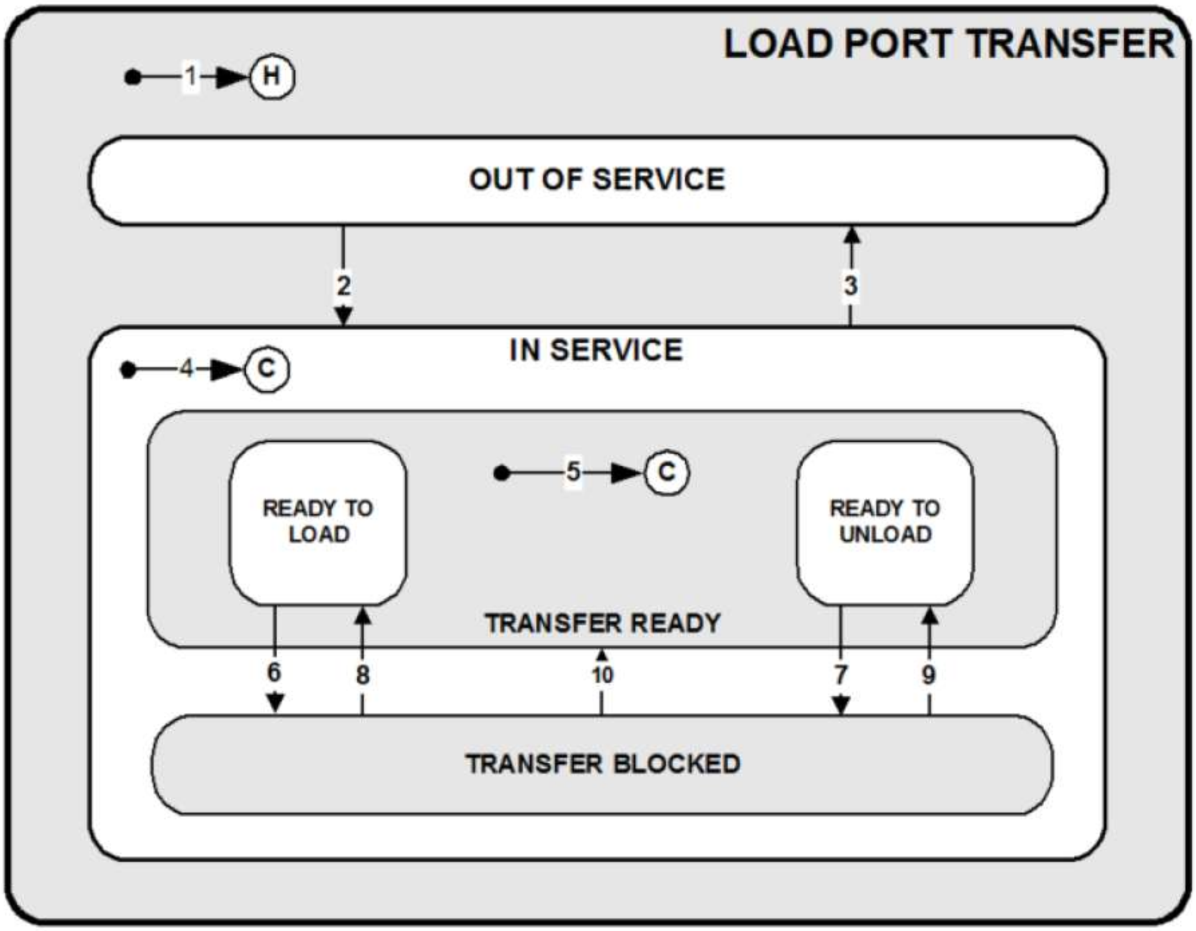


Figure 3.4.1.Load Port Transfer State Model Diagram

Load Port Transfer State Definitions

LOAD PORT TRANSFER
The super state for the IN SERVICE and OUT OF SERVICE states.

OUT OF SERVICE
Transfer to/from this load port is disabled. A transition to IN SERVICE is required to continue using this load port for transfers.

IN SERVICE
Transfer to/from this load port is enabled. A transition to OUT OF SERVICE disables the load port for transfer use.

TRANSFER READY
A substate of IN SERVICE. The load port is available for carrier transfer. The transfer can either be manual or automated, and can be a load or an unload. This state contains two substates, which are used depending on whether or not a carrier is present on the load port (READY TO LOAD and READY TO UNLOAD).

READY TO LOAD
A substate of TRANSFER READY. When transitioning to the TRANSFER READY state, if a carrier is not present on the specified load port, this is the active substate. In this state, the load port is available to be loaded with an external carrier, or with a carrier that is currently located inside the equipment.

READY TO UNLOAD
A substate of TRANSFER READY. When transitioning to the TRANSFER READY state, if a carrier is present on the specified load port, this is the active substate. In this state, the load port is available for unloading of a carrier from the load port to material handling equipment. When the load port is being used by the equipment, the state shall transition to TRANSFER BLOCKED.

TRANSFER BLOCKED
The carrier transfer state is neither READY TO LOAD nor READY TO UNLOAD. Because of load port related activity being performed, transfer is not available to/from this load port at this time.

Load Port Transfer State Transition Table

	Previous State	Trigger	New State	Comments
1	(no state)	System reset.	IN SERVICE	
2	OUT OF SERVICE	The host or an operator has invoked the ChangeServiceStatus service for this load port with a value of IN SERVICE.	IN SERVICE	Load port is now usable for transfer.
3	IN SERVICE	The host or an operator has invoked the ChangeServiceStatus service for this load port with a value of OUT OF SERVICE.	OUT OF SERVICE	Load port is now rendered unusable for transfer.
4	IN SERVICE	*Service:* The host or an operator has invoked the ChangeServiceStatus service for this load port with a value of IN SERVICE. *System Reset:* This transition can be activated by an equipment re-initialization.	TRANSFER READY or TRANSFER BLOCKED	This is the default entry into IN SERVICE. The state is TRANSFER BLOCKED if the carrier, or load port, is not available for carrier transfer. Otherwise, the state is TRANSFER READY.

5	TRANSFER READY	*Service:* The host or an operator has invoked the ChangeServiceStatus service for this load port with a value of IN SERVICE. *System Reset:* This transition can be activated by an equipment re-initialization. *Failed Transfer:* If a transfer fails, this transition is activated by transition #10.	READY TO LOAD or READY TO UNLOAD	When entering the TRANSFER READY state, if a carrier is present, the substate is READY TO UNLOAD, else the substate is READY TO LOAD.
6	READY TO LOAD	*Manual: The equipment recognizes the logical indication of the start of a manual load transfer. This trigger is configurable by the user, examples are included in Table 8.* *Automated: The PIO load transfer is beginning and the PIO 'READY' signal is activated (see SEMI E84).* *Internal Buffer: A CarrierOut service has started for this load port.*	TRANSFER BLOCKED	When a CarrierOut service is queued and the equipment load port is currently in the TRANSFER BLOCKED state, the equipment shall keep the load port in the TRANSFER BLOCKED state.
7	READY TO UNLOAD	*Manual: The equipment recognizes the logical indication of the start of a manual unload transfer. This trigger is configurable by the user, examples are included in Table 8.* *Automated: The PIO unload transfer is beginning and the PIO 'READY' signal is activated (see SEMI E84).* *Internal Buffer: A CarrierIn service has started for this load port.* *By CarrierReCreate Service: A CarrierReCreate service command has been issued by host or operator.*	TRANSFER BLOCKED	When a CarrierOut service is queued and the equipment load port is currently in the TRANSFER BLOCKED state, the equipment shall keep the load port in the TRANSFER BLOCKED state
8	TRANSFER BLOCKED	*Manual: The carrier unload transfer has completed, and the load port is now empty and ready for load transfer. This is indicated when two conditions are met, the presence signal indicates that no carrier is present and the operator has logically indicated that the transfer is complete.* *Automated: The PIO unload transfer ends with the PIO 'COMPT' signal (see SEMI E84).* *Internal Buffer: The carrier has finished its move from the load port into the internal buffer, and no CarrierOut services are queued for this load port.*	READY TO LOAD	A carrier can now be loaded onto the load port from either an external entity, or by the equipment's internal material handling resource.

9	TRANSFER BLOCKED	*Manual: Processing for substrates contained within the carrier has completed, or a CancelCarrier/ CancelCarrierAtPort service has been issued, and the carrier has returned to the load/unload position on the load port.* *Automated: Processing for the substrates belonging to the carrier has completed, or a CancelCarrier/CancelCarrier-AtPort service has been received, and the carrier has returned to the load/unload position.* *Internal Buffer: A carrier has completed its move from the internal buffer to the load port.*	READY TO UNLOAD	The carrier on the load port can now be unloaded from the load port to an external entity.
10	TRANSFER BLOCKED	*The transfer was unsuccessful, and the carrier was not loaded or unloaded.*	TRANSFER READY	The substate of TRANSFER READY which is decided by transition #5.

3.5 Load Port Access Mode State Model

The Access Mode State Model defines the host view of equipment access mode, as well as the host interactions with the equipment necessary to switch the access mode. Each load port has its own Access Mode State Model. There are two access mode states: MANUAL and AUTO. The access mode for a load port may be switched at any time by the host or the operator.

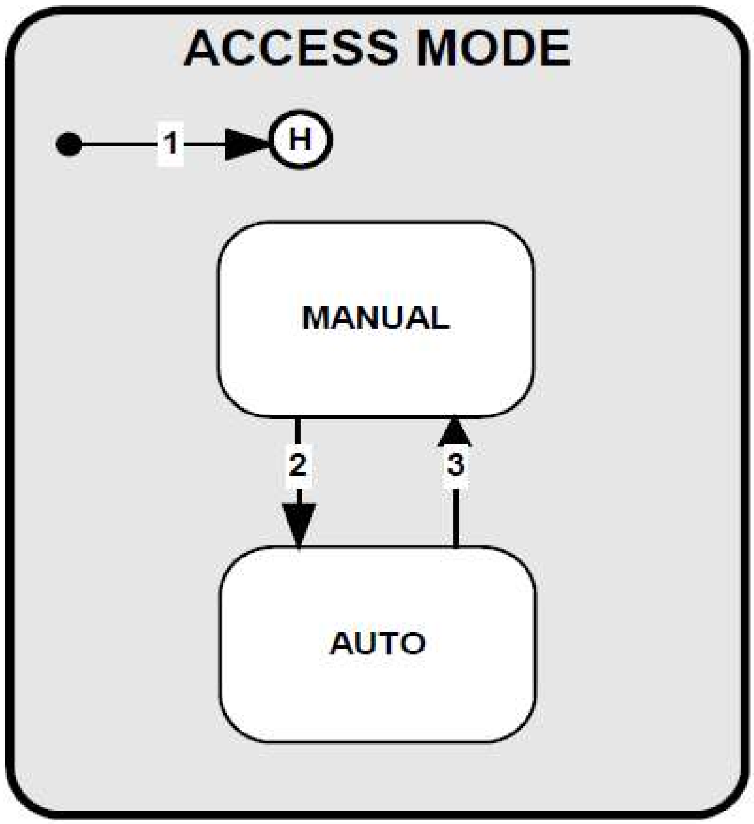


Figure 3.5.1. Access Mode State Model Diagram

Access Mode State Definitions

ACCESS MODE
The parent state for the MANUAL and AUTO substates.

MANUAL
A substate of ACCESS MODE. When the production equipment or specified load ports are in this mode, only manual (non-AMHS) carrier transfers are allowed. If a ChangeAccess service with the value of MANUAL is received in this state, the equipment shall accept the service and no event is sent for this action.

AUTO
A substate of ACCESS MODE. When the production equipment or specified load ports are in this mode, only automated (AMHS) carrier transfers are allowed in normal operation. If a ChangeAccess service with the value of AUTO is received in this state, the equipment shall accept the service and no event is sent for this action.

Access Mode State Transition Table

#	Previous State	Trigger	New state	Actions	Comments
1	(no state)	System restart.	MANUAL		
2	MANUAL	The host or operator has executed a ChangeAccess service with the value of AUTO.	AUTO		The operator may also trigger this transaction from the production equipment console.
3	AUTO	The host or operator has executed a ChangeAccess service with the value of MANUAL.	MANUAL		The operator may also trigger this transaction from the production equipment console or a manual switch at the load port.

3.6 Load Port Reservation State Model

The purpose of the Reservation State Model is to define the host view of future activity at a specific load port. Each load port has its own Load Port Reservation State Model.

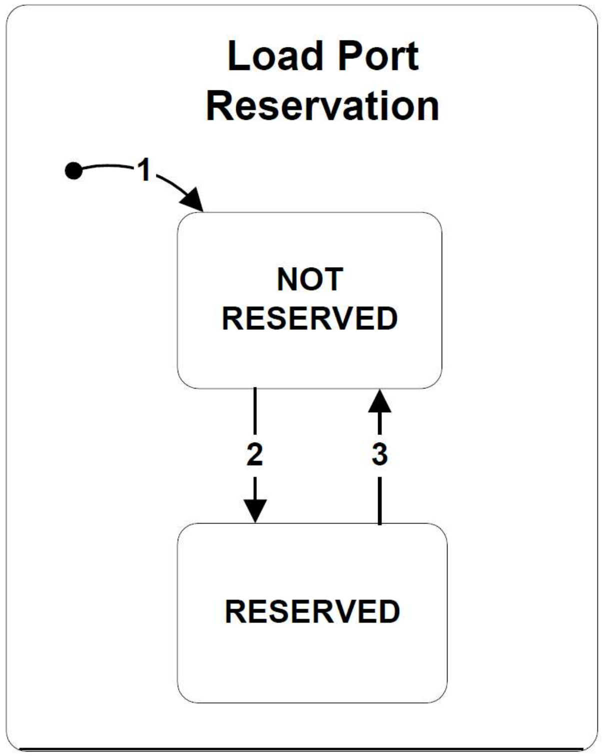


Figure 3.6.1. Load Port Reservation State Model Diagram

Load Port Reservation State Definitions

LOAD PORT RESERVATION
The super state of the substates NOT RESERVED and RESERVED.

NOT RESERVED
A substate of LOAD PORT RESERVATION, this state is active when there is no reservation existing at the load port.

RESERVED
A substate of LOADPORT RESERVATION, this state is active when there is a reservation for future activity at the load port. When in this state, the access mode for a load port may not be changed.

Load Port Reservation State Transition Table

#	Previous State	Trigger	New State	Actions
1	(no state)	System reset.	NOT RESERVED	
2	NOT RESERVED	*Service:* If reserved by service, the host or operator sends a ReserveAtPort or a Bind service to the production equipment. *CarrierOut:* This happens when the equipment physically initiates a CarrierOut operation.	RESERVED	If the user has configured the equipment to use the reservation visible signal indicator, it is activated for this load port.
3	RESERVED	*Service:* If a reservation is cancelled by service, the host or operator sends a CancelBind or a CancelReservationAtPort. *Carrier arrival:* A carrier arrives at the reserved port.	NOT RESERVED	If the user has configured the equipment to use the reservation visible signal, the indicator is deactivated for this load port.

3.7 Load Port/Carrier Association State Model

The purpose of the Carrier Association State Model is to define the host view of carrier to load port association of the production equipment, as well as the host interactions with the production equipment necessary to associate a carrier to load port, and to perform equipment based carrier verification. Each load port shall maintain an independent instance of the Carrier Association State Model. Each instance of this state model must not influence the state of the same state model for a different load port.

This state model provides the ability to perform CarrierID verification with two different methods. If the CarrierID is provided before the equipment reads the CarrierID, the CarrierID that becomes associated with the load port can be used later for equipment based carrier verification. If the association happens by CarrierID read (not by a service execution), then the production equipment shall report the CarrierID information in a data collection event.

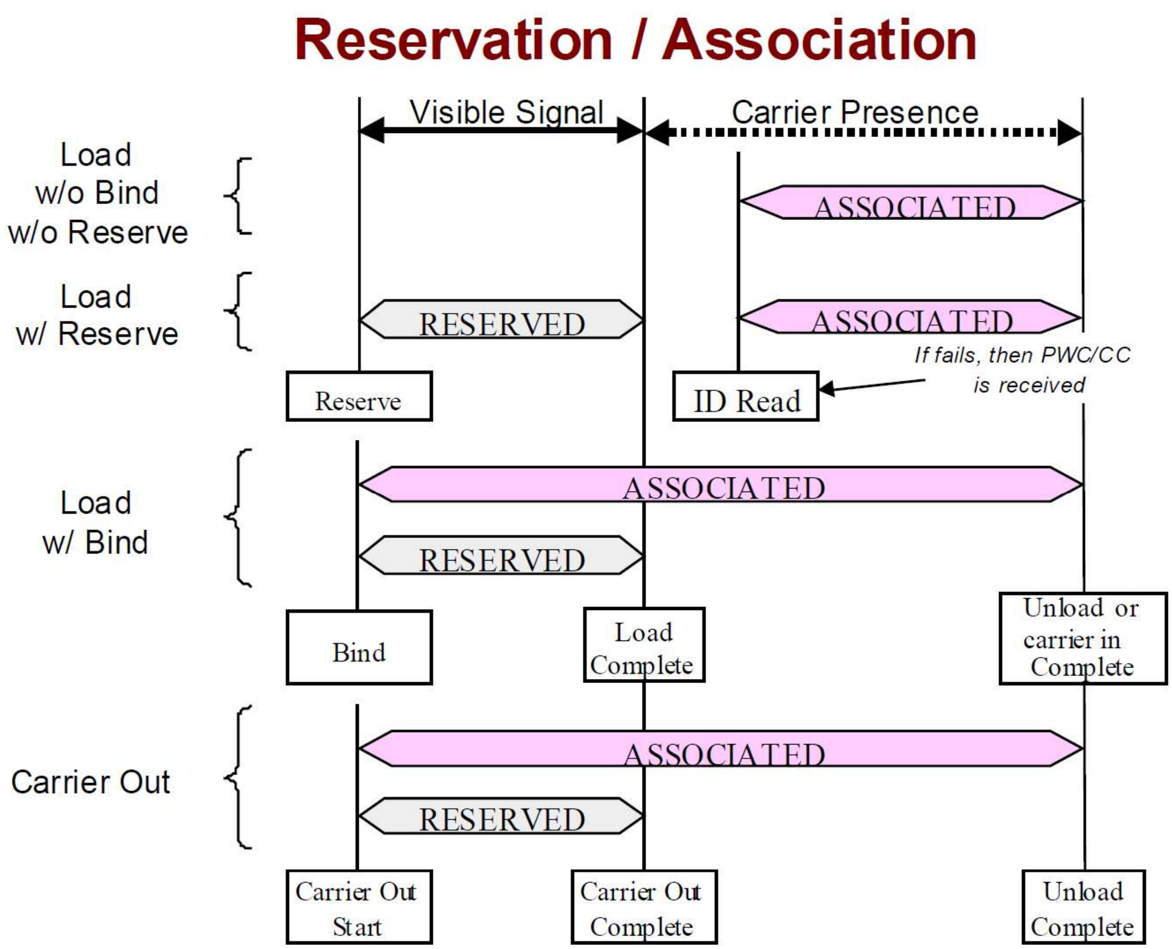


Figure 3.7.1. Relation of Reservation to Association

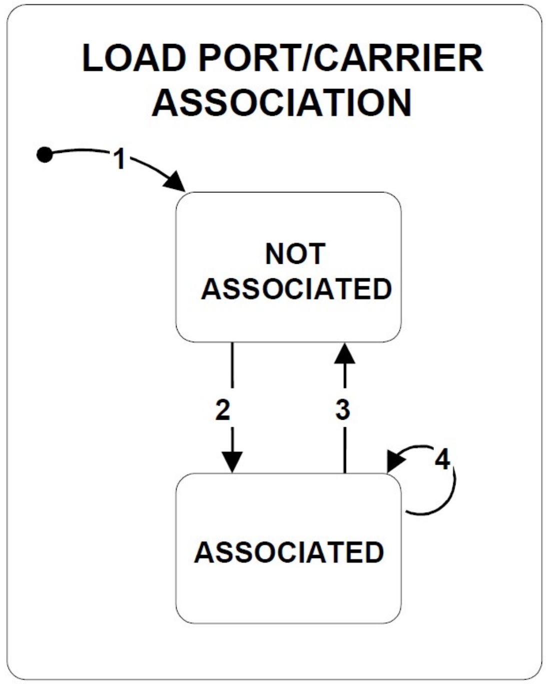


Figure 3.7.2. Load Port/Carrier Association State Model Diagram

Load Port/Carrier Association State Definitions

LOAD PORT/CARRIER ASSOCIATION
The parent state of the NOT ASSOCIATED and ASSOCIATED substates.

NOT ASSOCIATED
A substate of LOAD PORT/CARRIER ASSOCIATION. There is no carrier association present for this load port.

ASSOCIATED
A substate of LOAD PORT/CARRIER ASSOCIATION. A CarrierID has been associated with this load port. The load port is not available for a new carrier association

Load Port/Carrier Association State Transition Table

#	Previous State	Trigger	New State	Comments
1	(no state)	System reset.	NOT ASSOCIATED	
2	NOT ASSOCIATED	Service Normal: The host sends a Bind service to the equipment when the port is unoccupied. Service Abnormal: A carrier is instantiated at the load port because the host sends either a ProceedWithCarrier or CancelCarrier service in response to one of the following events from the equipment: CarrierID Read Fail Event UnknownCarrierID CarrierID Read: A CarrierID not currently existing at the equipment is successfully read. Known Carrier: A known carrier to the equipment is being moved onto the load port from an internal location. This happens when the CarrierOut service is initiated.	ASSOCIATED	Once the CarrierID to load port association is complete, the load port is not available for association until the state returns to NOT ASSOCIATED again.
3	ASSOCIATED	Service: If cancellation of a load port association is required; then, this can be accomplished by sending a CancelBind service to the production equipment before the carrier arrives to the loadport or before a transfer sequence has started. Carrier Unload: An association cancellation may also be performed by removing the carrier from the load port or by the production equipment moving a carrier to an internal buffer position.	NOT ASSOCIATED	A carrier unload, may happen before or after processing occurs. The load port is available for another association once the carrier is removed.
4	ASSOCIATED	Equipment based carrier verification fails, and the carrier assumes the ID value from the carrier that is on the load port. Internal buffer: A carrier is unloaded and a queued CarrierOut service starts.	ASSOCIATED	This transition only occurs when the Bind command has been used. The existing carrierID that was associated by a Bind service is unassociated by the equipment and the new carrierID is now associated to the load port. The equipment shall delay further action until receiving either a CancelCarrier or a ProceedWithCarrier command from the host.

3.8 Carrier State Model

The purpose of the Carrier State Model is to define the host's view of a carrier. The equipment shall maintain a separate and independent state model for each carrier in/at the equipment. The CARRIER state has three fundamental states of CARRIER ID STATUS, CARRIER SLOT MAP STATUS and CARRIER ACCESSING STATUS, and an additional state CARRIER READY TO UNLOAD PREDICTION STATUS.

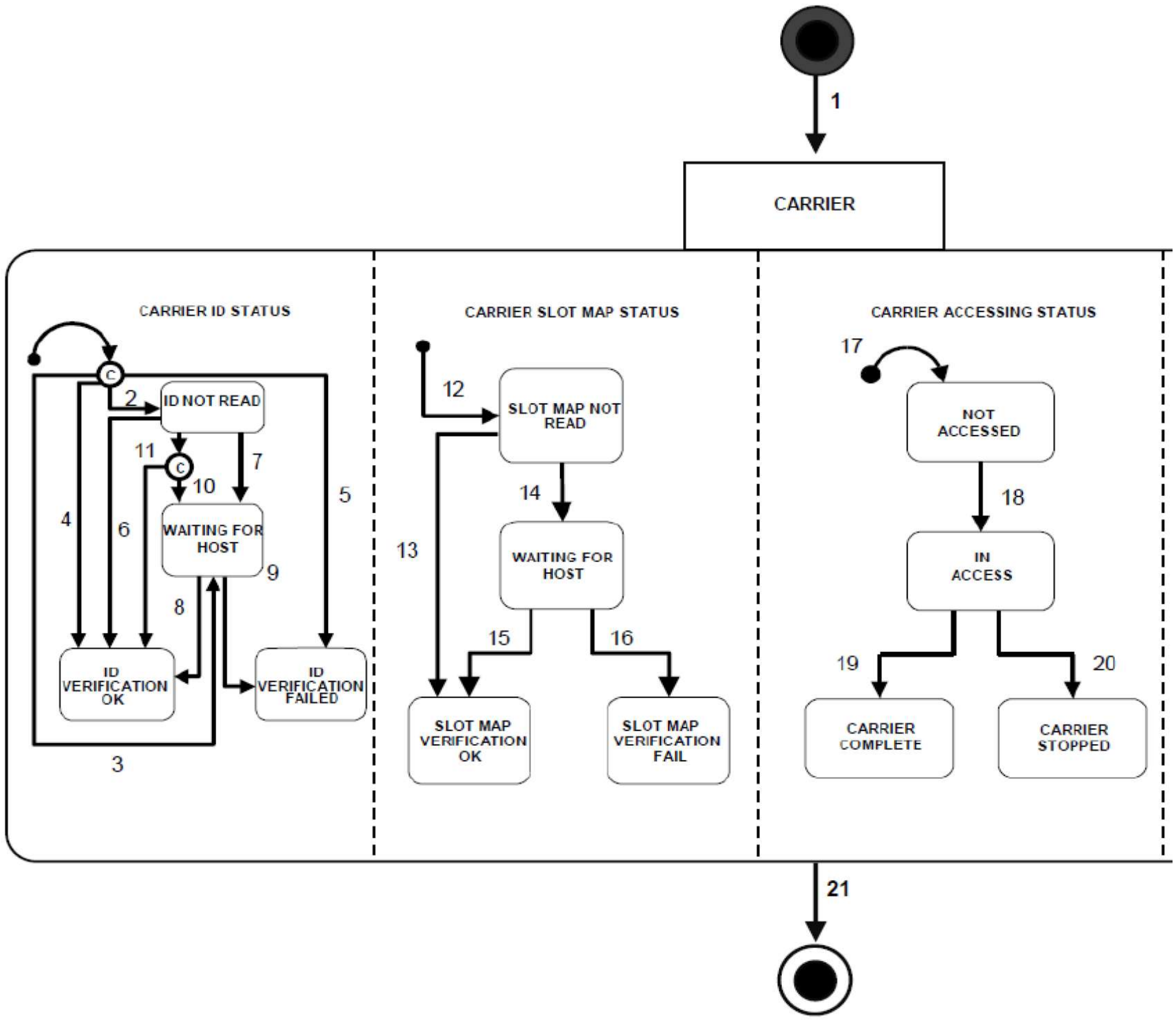


Figure 3.8.1. Carrier State Diagram

Carrier State Definitions

CARRIER ACCESSING STATUS
This is a substate of CARRIER and indicates the current accessing status of the carrier. It has four substates, NOT ACCESSED, IN ACCESS, CARRIER COMPLETE, and CARRIER STOPPED. The initial default entry substate is NOT ACCESSED.

NOT ACCESSED
This is a substate of CARRIER ACCESSING STATUS and is active when access by the equipment to the carrier has not been started. The carrier can be moved out.

IN ACCESS
This is a substate of CARRIER ACCESSING STATUS and is active when access by the equipment to the carrier has been started but has not been finished, and the carrier should not be moved out.

CARRIER COMPLETE
This is a substate of CARRIER ACCESSING STATUS and is active when the access by the equipment to the carrier has been finished, and the carrier should be moved out. This is a final state.

CARRIER STOPPED
This is a substate of CARRIER ACCESSING STATUS and is active when the access by the equipment to the carrier has been stopped abnormally, and the carrier should be moved out. This is a final state.

CARRIER ID STATUS
This is a substate of CARRIER and indicates the current status of the carrier with respect to its identifier. It has four substates, ID NOT READ, WAITING FOR HOST, ID VERIFICATION FAILED, ID VERIFICATION OK.

ID NOT READ
This is a substate of CARRIER ID STATUS. This state is active whenever the CarrierID has not been read by the equipment.

ID VERIFICATION FAILED
This is a substate of CARRIER ID STATUS and is active when the carrierID has failed verification by the host following the CancelCarrier service. This is a final state.

ID VERIFICATION OK
This is a substate of CARRIER ID STATUS and is active as soon as the CarrierID has been accepted. The ID is determined to be accepted by either successful verification by the equipment or the host, or by bypassing ID read because a carrier ID reader is not available and the BypassReadID variable is set to true. This is a final state.

WAITING FOR HOST
This is a substate of CARRIER ID STATUS and is active during the period of time when the CarrierID has been read by the equipment successfully or unsuccessfully and has not yet been verified by the host.

CARRIER SLOT MAP STATUS
This is a substate of CARRIER and indicates the current status of the carrier with respect to its slot map. It has four substates, SLOT MAP NOT READ, WAITING FOR HOST, SLOT MAP VERIFICATION FAILED, SLOT MAP VERIFICATION OK. The initial default entry substate is SLOT MAP NOT READ.

SLOT MAP NOT READ
This is a substate of CARRIER SLOT MAP STATUS and is the default entry state. It is active when the Carrier is first loaded at the equipment until the Slot Map has been read successfully by the equipment at the Substrate Port.

SLOT MAP VERIFICATION FAIL
This is a substate of CARRIER SLOT MAP STATUS and is active when the Slot Map has been read by the equipment and has failed verification by the host. This is a final state.

SLOT MAP VERIFICATION OK
This is a substate of CARRIER SLOT MAP STATUS and is active as soon as the slot map has been verified. This is a final state.

WAITING FOR HOST
This is a substate of CARRIER SLOT MAP STATUS and is active when the equipment is waiting for input from the host

Carrier State Transition Table

#	Previous State	Trigger	New State	Actions
1	(no state)	A carrier is instantiated.	CARRIER	None.
2	(no state)	Normal: A Bind or Carrier Notification service is received.	ID NOT READ	None.
3	(no state)	Normal: A CarrierID not currently existing at the equipment is successfully read. Abnormal: A CarrierID is read successfully but an equipment based verification failed.	WAITING FOR HOST	None.
4	(no state)	ID Read fail or UnknownCarrierID Events: ProceedWithCarrier service is received.	ID VERIFICATION OK	A carrier is instantiated having the CarrierID provided by the Proceed WithCarrier service.
5	(no state)	ID Read fail or UnknownCarrierID Events: A CancelCarrier service is received.	ID VERIFICATION FAIL	A carrier is instantiated having the CarrierID provided by the Cancel Carrier service.
6	ID NOT READ	Normal: CarrierID is read successfully and the equipment has verified the carrierID successfully. Abnormal: The carrier was instantiated via a CarrierNotification service and upon carrier placement the equipment has sent a CarrierID Read Fail Event or UnknownCarrierID Event. Subsequently the host has issued a ProceedWithCarrier service. The ProceedWithCarrier service must provide the PortID parameter as well as the CarrierID parameter in order to unambiguously identify the physical carrier.	ID VERIFICATION OK	None.
7	ID NOT READ	CarrierID is read unsuccessfully.	WAITING FOR HOST	None.
8	WAITING FOR HOST	A ProceedWithCarrier service is received.	ID VERIFICATION OK	None.
9	WAITING FOR HOST	A CancelCarrier service is received.	ID VERIFICATION FAIL	None.
10	ID NOT READ	BypassReadID variable is set to FALSE, and a carrier is received when the ID reader is not in service or not installed.	WAITING FOR HOST	Wait for ProceedWith-Carrier.
11	ID NOT READ	BypassReadID variable is set to TRUE, and a carrier is received when the id reader is not in service or not installed.	ID VERIFICATION OK	None.
12	(no state)	A carrier is instantiated.	SLOT MAP NOT READ	None.
13	SLOT MAP NOT READ	Slot Map is read and verified successfully by the equipment.	SLOT MAP VERIFICATION OK	None.

14	SLOT MAP NOT READ	Normal host based verification: Slot Map is read successfully and the equipment is waiting for host verification. Equipment based verification fail: Slot Map is read successfully but equipment based verification has failed. Slot map read fail: Slot Map cannot be read. Abnormal substrate position within the carrier: The Slot Map read has indicated an abnormal substrate position.	WAITING FOR HOST	Save new slot map in the SlotMap attribute.
15	WAITING FOR HOST	A ProceedWithCarrier service is received.	SLOT MAP VERIFICATION OK	Proceed with the carrier as instructed.
16	WAITING FOR HOST	A CancelCarrier service is received.	SLOT MAP VERIFICATION FAIL	Prepare the carrier for Unload.
17	(no state)	A carrier object is instantiated.	NOT ACCESSED	None.
18	NOT ACCESSED	The equipment starts accessing the carrier.	IN ACCESS	None.
19	IN ACCESS	The equipment finishes accessing the carrier normally.	CARRIER COMPLETE	None.
20	IN ACCESS	The equipment finishes accessing the carrier abnormally.	CARRIER STOPPED	None.
21	CARRIER	Normal: The carrier is unloaded from the equipment. Abnormal by service: CancelBind or CancelCarrierNotification service is received prior to the carrier load. Abnormal by equipment: An equipment based verification fails and the equipment performs a self-initiated CancelBind service.	(no state)	The equipment destroys the instance of this carrier object.

3.9 Process Job State Model

The Process Job is created on request of the supervisor, executes, and then is deleted by the equipment. The job usually spans the time period from shortly before wafers are physically delivered to the equipment, through the processing, and until shortly after finished processing. Process Jobs are not queued by the equipment that supports Control Jobs, rather they are pooled waiting to be scheduled by their respective Control Job

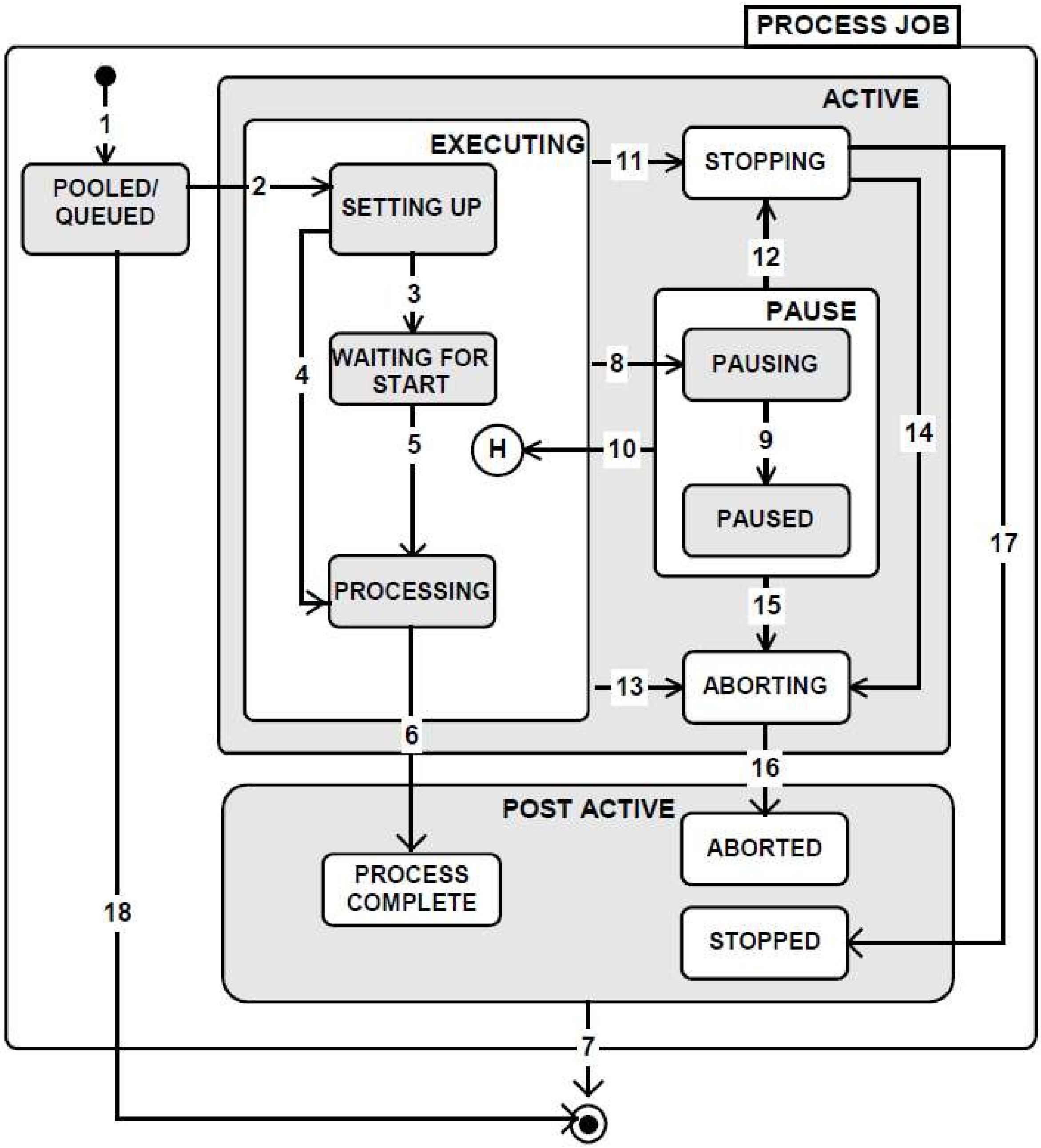


Figure 3.9.1. Process Job State Diagram

Process Job State Descriptions

ABORTING (ACTIVE Substate)
While the PR Job is in the ABORTING substate, the equipment is performing an abort or an optional error recovery procedure. The abort procedure will cause immediate termination of the processing.

ACTIVE
ACTIVE is the parent state of all substates where the context of an active process job execution exists.

EXECUTING (ACTIVE Substate)
EXECUTING is the parent state of those substates that refer to the preparation and execution of a process job.

SETTING UP (EXECUTING Substate)
While the PR Job is in the SETTING UP substate, the processing resource performs pre-conditioning, awaits material arrival, and prepares for material processing. Pre-conditioning includes all operations in the processing resource, which are required by the recipe in advance of material arrival.

PAUSE (ACTIVE Substate)
While the PR Job is in the PAUSE substate the equipment is suspending or has suspended activity. This state is currently not supported by equipment.

PAUSED (PAUSE Substate)
While the PR Job is in the PAUSED substate all equipment activity has ceased. The PR Job is awaiting a RESUME (or STOP or ABORT) command. This state is currently not supported by equipment.

PAUSING (PAUSE Substate)
While the PR Job is in the PAUSING substate, the equipment continues to the first safe, continuable pausing place and then ceases activity. The activity may only cease at points that allow for resumption of the activity such that material integrity is maintained and the processing goals are accomplished. This state is currently not supported by equipment.

PROCESSING (EXECUTING Substate)
While the PR Job is in the PROCESSING substate, the equipment is doing the actual material processing using the equipment recipe(s) specified by the PR Job.

PROCESS COMPLETE (POST ACTIVE Substate)
While the PR Job is in the PROCESS COMPLETE substate the equipment has completed processing all material specified by the PR Job. The job terminates successfully while the material is still present, and the next queued job will transition to execution.

QUEUED/POOLED
While the PR Job is in the QUEUED/POOLED substate, the process job has been accepted by the equipment through a PR Job Create/Acknowledge transaction (such as PRJobCreate, PRJobCreateEnh, and PRJobMultiCreate) and is awaiting execution.

STOPPING (ACTIVE Substate)
While the PR Job is in the STOPPING substate, the equipment is performing a stop procedure to terminate processing in an orderly manner.

WAITING FOR START (EXECUTING Substate)
WAITING FOR START is used only in manual start process jobs. It is entered once SETUP is complete and a PR Job Start Process has not yet been received by the equipment. Manual start

is defined by the supervisor in PR Job Create. The job remains in this state, ready to process the material, until the PR Job Start Process is received or Abort or Stop terminates the job.

POST ACTIVE
This is the parent state of those states that refer to the final state (completion) of the process jobs. Process jobs that have completed, stopped or aborted will immediately transition to extinction and deleted.

STOPPED (POST ACTIVE substate)
This is the final state for those jobs that have been in the STOPPING state.

ABORTED (POST ACTIVE substate)
This is the final state for those jobs that have been in the ABORTING state.

Process Job State Transition Table

#	Current State	Trigger	New State	Action(s)
1	(no state)	The equipment accepts a Process Job create request.	QUEUED/POOLED	The job is placed into queue/pool. Acknowledge the Process Job creation.
2	QUEUED/ POOLED	The equipment has been allocated to the Process Job.	SETTING UP	The job is removed from the queue/pool. PR Job Setup event is triggered.
3	SETTING UP	Job material is present AND the equipment is ready to start the process job AND PRProcessStart attribute is not set.	WAITING FOR START	PR Job Waiting for Start event is triggered.
4	SETTING UP	Material is present and started processing. PRProcessStart attribute is set.	PROCESSING	PR Job Processing event is triggered. Material is processed.
5	WAITING FOR START	Material is present and started processing after Job Start directive.	PROCESSING	PR Job Processing event is triggered. Material is processed.
6	PROCESSING	Material processing completed.	PROCESS COMPLETE	PR Job Processing Complete event is triggered. Await material departure.
7	Any POST ACTIVE substate	Process job becomes extinct after processing completed.	(Extinction)	PR Job Complete event is triggered. The process job is deleted.
8	EXECUTING	Pausing is currently not supported by equipment.	PAUSING	The processing resource pauses at the first convenient time.
9	PAUSING	Pausing is currently not supported by equipment.	PAUSED	None.
10	PAUSE	Pausing is currently not supported by equipment.	EXECUTING	The processing resource resumes the activity that was paused.
11	EXECUTING	The equipment initiated a process stop action. (it received a STOP command or initiated an internal stop)	STOPPING	The equipment equipment stops the current execution activity at the first convenient time.

12	PAUSE	Pausing is currently not supported by equipment.	STOPPING	The processing resource stops the current execution activity at the first convenient time.
13	EXECUTING	The equipment initiated a process abort action. (it received an ABORT command or initiated an internal abort)	ABORTING	The equipment terminates the current execution activity immediately.
14	STOPPING	The equipment initiated a process abort action. (it received an ABORT command or initiated an internal abort)	ABORTING	The equipment terminates the current execution activity immediately.
15	PAUSE	Pausing is currently not supported by equipment.	ABORTING	The processing resource terminates the current execution activity immediately.
16	ABORTING	The abort procedure is completed.	ABORTED	
17	STOPPING	The stop procedure is completed and all material is in a safe condition.	STOPPED	
18	QUEUED/ POOLED	‘CANCEL’, ‘ABORT’, or ‘STOP’ command received.	(no state)	Remove the process job from the queue/pool. PR Job Complete event is triggered. Delete the process job.

3.10 Control Job State Model

The equipment supports the use of Control job to provide a supervisory level of control for process jobs on equipment. They can be used to reduce the amount of host level interaction required for material processing. A factory host is provided with methods for instructing the equipment to provide only significant factory level events, such as, a carrier complete. Control Jobs specifies the order for process jobs and are queued and managed in a FIFO approach.

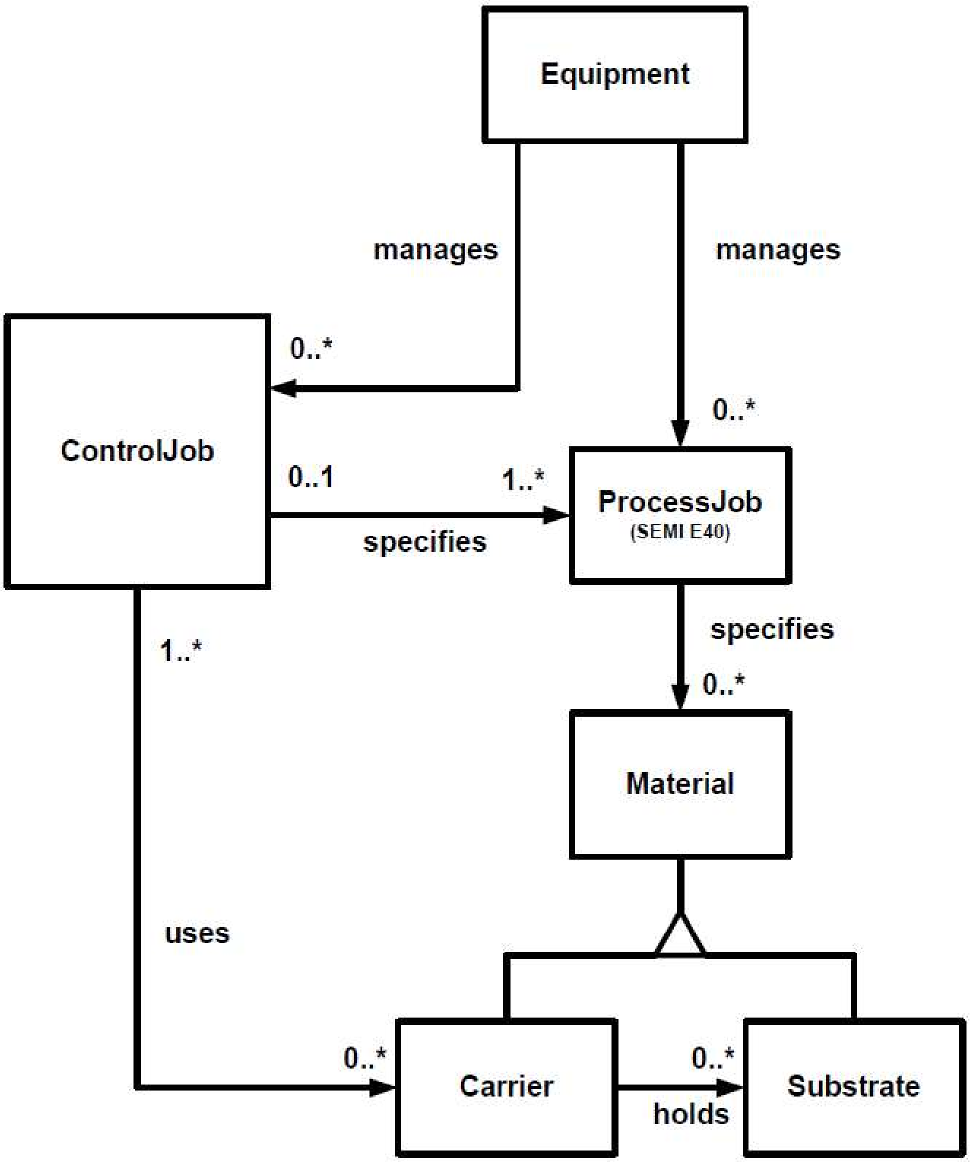


Figure 3.10.1. Control Job Object Model

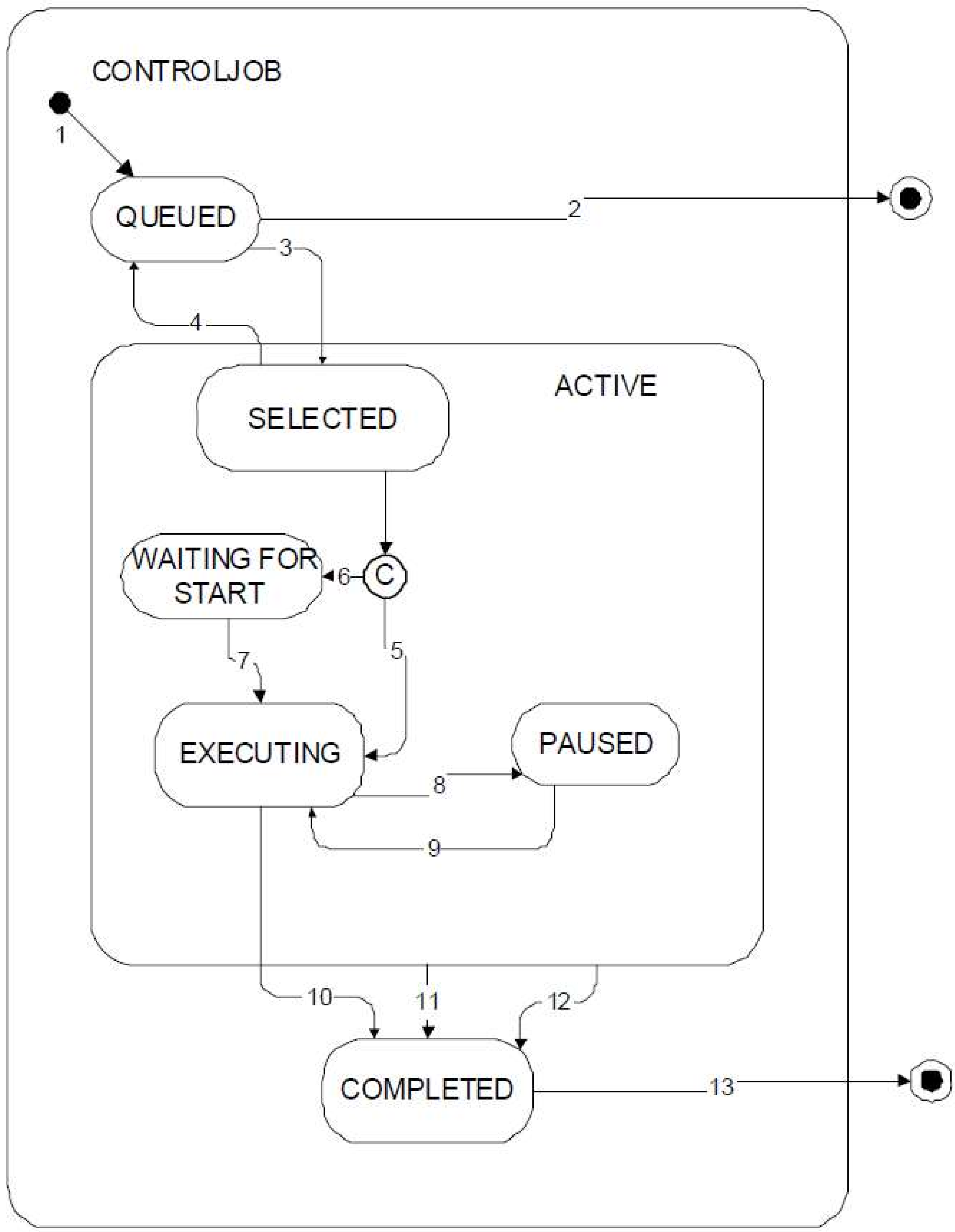


Figure 3.10.2. Control Job State Diagram

Control Job State Descriptions

QUEUED
A ControlJob is queued after its creation or de-selection. A newly created ControlJob is placed at the tail of the queue.

SELECTED
If there is no ControlJob in the ACTIVE state, the first ControlJob in the queue will be dequeued and transition into the SELECTED substate. The equipment is reserved (not available for any other jobs) by the ControlJob in the SELECTED state. If materials, required by the ControlJob, for processing have not arrived at the equipment, the ControlJob will stay in this state until

materials arrive. If the ControlJob or the first process job in the ControlJob does not require material, this state is exited immediately. A SELECTED ControlJob can be de-selected if specified materials have not arrived.

WAITING FOR START
The ControlJob is waiting to receive a start command manually or remotely from the host. The ControlJob transitions to this state only if the StartMethod is set to FALSE (UserStart) and materials have arrived.

EXECUTING
In this state, each ProcessJob in the ProcessingCtrlSpec is initiated in order, based on the value of the ControlJob's ProcessOrderMgmt attribute as required resources become available and material for the job has been verified.

PAUSED
When the ControlJob is paused it shall not commence the initiation of any more Process jobs. The equipment may permit modification of various attributes of the ControlJob while in this state. This is currently not supported by equipment.

COMPLETED
A ControlJob enters this state once all of its process jobs have been completed, stopped or aborted.

Control Job State Transition Table

#	Current State	Trigger	New State	Actions	Comments
1	(No State)	Receive 'Create' command from host or operator through operator console.	QUEUED	Create ControlJob and put it at the tail of a control job queue.	If job queue is full, 'Create' request is rejected.
2	QUEUED	Receive 'Cancel', 'Abort', or 'Stop' command from host or operator through operator console.	(No State)	De-queue and terminate the job.	If other control jobs are waiting behind the canceled job in the queue, they are shifted forward to fill in the gap after the de-queuing of the canceled control job.
3	QUEUED	The processing resource has capacity to begin work on the next ControlJob.	SELECTED	Select and de-queue the ControlJob at the head of the queue.	Materials are not necessarily at the equipment.
4	SELECTED	Receive 'De-select' command from host or operator through operator console and materials for the control job have not arrived yet.	QUEUED	De-selected job moves to the head of the job queue and the job that was at the head becomes the SELECTED job.	The command shall be rejected if the resources for the job at the head of queue are not available. See the Queue Model.
5	SELECTED	Material for the first process job arrives or in the case where the	EXECUTING		Process jobs associated with a carrier will not initiate until the identifier and

		first (or only) process job does not require material, this transition shall be taken as soon as the processing resource for that process job becomes available. 'StartMethod' attribute in the ControlJob is set for Auto.			substrate slot map for the carrier have been verified. Process jobs that don't use material can be initiated immediately.
6	SELECTED	Same as for transition 5 except that the 'StartMethod' attribute in the control job is set for user start.	WAITING FOR START		
7	WAITING FOR START	User START command received.	EXECUTING		Same as for transition 5.
8	EXECUTING	The equipment received a 'Pause' command from the host or operator or a ControlJob PauseEvent has occurred.	PAUSED		Pausing is currently not supported by equipment.
9	PAUSED	The equipment received a 'Resume' command from the host or operator.	EXECUTING	Commence initiating process jobs.	Pausing is currently not supported by equipment.
10	EXECUTING	All the ProcessJobs specified for the ControlJob have completed.	COMPLETED		
11	ACTIVE	The equipment received a CJStop command from the host or operator or generated a CJStop command internally, and all the process jobs under the ControlJob that were ACTIVE when the CJStop command was received have entered the POST ACTIVE state.	COMPLETED		
12	ACTIVE	The equipment received a CJAbort command from the host or operator or generated a CJAbort command internally, and all the process jobs under the ControlJob that were ACTIVE when the CJAbort command was received have entered the POST ACTIVE state.	COMPLETED		
13	COMPLETED	ControlJob entered ACTIVE state.	(No State)	Control Job is deleted.	

4 Message Summary

4.1 Stream 1

Stream 1	Equipment Status	Direction
S1,F1	Are You There Request	H⇔E
S1,F2	On Line Data	H⇔E
S1,F3	Selected Equipment Status Request	H⇨E
S1,F4	Selected Equipment Status Data	H⇦E
S1,F11	Status Variable Namelist Request	H⇨E
S1,F12	Status Variable Namelist Reply	H⇦E
S1,F13	Establish Communications Request	H⇔E
S1,F14	Establish Communications Request Acknowledge	H⇔E
S1,F15	Request OFF-LINE	H⇨E
S1,F16	OFF-LINE Acknowledge	H⇦E
S1,F17	Request ON-LINE	H⇨E
S1,F18	ON-LINE Acknowledge	H⇦E

4.2 Stream 2

Stream 2	Equipment Control and Diagnostics	Direction
S2,F13	Equipment Constant Request	H⇨E
S2,F14	Equipment Constant Data	H⇦E
S2,F15	New Equipment Constant Send	H⇨E
S2,F16	New Equipment Constant Acknowledge	H⇦E
S2,F17	Date and Time Request	H⇔E
S2,F18	Date and Time Data	H⇔E
S2,F23	Trace Initialize Send	H⇨E
S2,F24	Trace Initialize Acknowledge	H⇦E
S2,F29	Equipment Constant Namelist Request	H⇨E
S2,F30	Equipment Constant Namelist	H⇦E
S2,F31	Date and Time Set Request	H⇨E
S2,F32	Date and Time Set Acknowledge	H⇦E
S2,F33	Define Report	H⇨E
S2,F34	Define Report Acknowledge	H⇦E
S2,F35	Link Event Report	H⇨E
S2,F36	Link Event Report Acknowledge	H⇦E
S2,F37	Enable/Disable Event Report	H⇨E
S2,F38	Enable/Disable Event Report Acknowledge	H⇦E
S2,F39	Multi-block Inquire	H⇨E
S2,F40	Multi-block Grant	H⇦E
S2,F41	Host Command Send	H⇨E
S2,F42	Host Command Acknowledge	H⇦E

4.3 Stream 3

Stream 3	Materials Status	Direction
S3,F17	Carrier Action Request	H⇨E
S3,F18	Carrier Action Acknowledge	H⇦E
S3,F25	Port Action Request	H⇨E

S3,F26	Port Action Acknowledge	H⇦E
S3,F27	Change Access	H⇨E
S3,F28	Change Access Acknowledge	H⇦E

4.4 Stream 5

Stream 5	Exception Handling	Direction
S5,F1	Alarm Report Send	H⇨E
S5,F2	Alarm Report Acknowledge	H⇦E
S5,F3	Enable/Disable Alarm Send	H⇨E
S5,F4	Enable/Disable Alarm Acknowledge	H⇦E
S5,F5	List Alarms Request	H⇨E
S5,F6	List Alarm Data	H⇦E

4.5 Stream 6

Stream 6	Data Collection	Direction
S6,F1	Trace Data Send	H⇦E
S6,F2	Trace Data Acknowledge	H⇨E
S6,F11	Event Report Send	H⇦E
S6,F12	Event Report Acknowledge	H⇨E
S6,F13	Annotated Event Report Send	H⇦E
S6,F14	Annotated Event Report Acknowledge	H⇨E
S6,F15	Event Report Request	H⇨E
S6,F16	Event Report Send	H⇦E
S6,F17	Annotated Event Report Request	H⇨E
S6,F18	Annotated Event Report Data	H⇦E
S6,F19	Individual Report Request	H⇨E
S6,F20	Individual Report Data	H⇦E

4.6 Stream 7

Stream 7	Process Program Management	Direction
S7,F1	Process Program Load Inquire	H⇔E
S7,F2	Process Program Load Grant	H⇔E
S7,F3	Process Program Send	H⇔E
S7,F4	Process Program Acknowledge	H⇔E
S7,F5	Process Program Request	H⇔E
S7,F6	Process Program Data	H⇔E
S7,F17	Delete Process Program Send	H⇨E
S7,F18	Delete Process Program Acknowledge	H⇦E
S7,F19	Current EPPD Request	H⇨E
S7,F20	Current EPPD Data	H⇦E
S7,F23	Formatted Process Program Send	H⇔E
S7,F24	Formatted Process Program Acknowledge	H⇔E
S7,F25	Formatted Process Program Request	H⇔E
S7,F26	Formatted Process Program Data	H⇔E

4.7 Stream 9

Stream 9	System Errors	Direction
S9,F1	Unrecognized Device ID	H⇦E
S9,F3	Unrecognized Stream Type	H⇦E
S9,F5	Unrecognized Function Type	H⇦E
S9,F7	Illegal Data	H⇦E
S9,F9	Transaction Timer Timeout	H⇦E
S9,F11	Data Too Long	H⇦E
S9,F13	Conversation Timeout	H⇦E

4.8 Stream 10

Stream 10	Terminal Services	Direction
S10,F1	Terminal Request	H⇦E
S10,F2	Terminal Request Acknowledge	H⇨E
S10,F3	Terminal Display, Single	H⇨E
S10,F4	Terminal Display, Single Acknowledge	H⇦E
S10,F5	Terminal Display, Multi-Block	H⇨E
S10,F6	Terminal Display, Multi-Block Acknowledge	H⇦E
S10,F7	Multi-Block Not Allowed	H⇦E

4.9 Stream 14

Stream 14	Object Services	Direction
S14,F9	Create Object Request	H⬄E
S14,F10	Create Object Acknowledge	H⬄E
S14,F11	Delete Object Request	H⬄E
S14,F12	Delete Object Acknowledge	H⬄E

4.10 Stream 16

Stream 16	Processing Management	Direction
S16,F5	Process Job Command Request	H⇨E
S16,F6	Process Job Command Acknowledge	H⇦E
S16,F11	PRJobCreateEnh	H⇨E
S16,F12	PRJobCreateEnh Acknowledge	H⇦E
S16,F15	PRJobMultiCreate Request	H⇨E
S16,F16	PRJobMultiCreate Acknowledge	H⇦E
S16,F17	PRJobDequeue	H⇨E
S16,F18	PRJobDequeue Acknowledge	H⇦E
S16,F19	PRGetAllJobs	H⇨E
S16,F20	PRGetAllJobs Send	H⇦E
S16,F21	PRGetSpace	H⇨E
S16,F22	PRGetSpace Send	H⇦E
S16,F27	Control Job Command Request	H⇨E
S16,F28	Control Job Command Acknowledge	H⇦E

5 Data Item Dictionary

5.1 Item Format Codes

The following table is provided for reference:

Format	Format Code Binary	Octal	Meaning
LIST	000 000	00	List (length in elements)
B	001 000	10	Binary
BOOL	001 001	11	Boolean
A	010 000	20	ASCII[1]
J	010 001	21	JIS-8 (not supported)
I8	011 000	30	8-byte integer (signed)
I1	011 001	31	1-byte integer (signed)
I2	011 010	32	2-byte integer (signed)
I4	011 100	34	4-byte integer (signed)
F8	100 000	40	8-byte floating point
F4	100 100	44	4-byte floating point
U8	101 000	50	8-byte integer (unsigned)
U1	101 001	51	1-byte integer (unsigned)
U2	101 010	52	2-byte integer (unsigned)
U4	101 100	54	4-byte integer (unsigned)

5.2 Data Item Dictionary

ACKC5	**Format: B**
Acknowledge Code, 1 byte 0 = Accepted >0 = Error, not accepted 1 – 63 = Reserved Where used: S5F2, F4	

ACKC6	**Format: B**
Acknowledge Code, 1 byte 0 = Accepted >0 = Error, not accepted 1 – 63 = Reserved Where used: S6F2, F4, F10, F12, F14	

ACKC7	**Format: B**
Acknowledge Code, 1 byte 0 = Accepted 1 = Permission not granted 2 = Length error 3 = Matrix overflow	

n = PPID not found 5 = Mode not supported >5 = Other error 6 – 63 = Reserved Where used: S7F4, F18	

ACKC7A	**Format: B**
Acknowledge Code, 1 byte 0 = Accepted 1 = MDLN is inconsistent 2 = SOFTREV 3 = Invalid CCODE n = PPARM value n = Other error (described by ERRW7) n – 63 = Reserved Where used: S7F27	

ACKC10	**Format: B**
Acknowledge Code, 1 byte 0 = Accepted for display 1 = Message will not be displayed 2 = Terminal not available 3 – 63 = Reserved Where used: S10F2, F4, F6	

ALCD	**Format: B**
Alarm code byte bit 8 = 1 means alarm set bit 8 = 0 means alarm cleared bit 7-1 is alarm category 0 = Not used 1 = Personal safety 2 = Equipment safety 3 = Parameter control warning n = Parameter control error n = Irrecoverable error n = Equipment status warning 7 = Attention flags 8 = Data integrity >8 = Other categories 9-63 = Reserved Where used: S5F1, F6	

ALED	**Format: B**
Alarm Enable/Disable Code, 1 byte bit 8 = 1 means enable alarm bit 8 = 0 means disable alarm Where used: S5F3	

ALID	Format: U4
Alarm Identification Where used: S5F1, F3, F5, F6	

ALTX	Format: A
Alarm Text; limited to 40 characters Where used: S5F1,F6	

CAACK	Format: U1
Carrier Action Acknowledge Code, 1 byte. 0 = Acknowledge, command has been performed. 1 = Invalid command 2 = Cannot perform now 3 = Invalid data or argument n = Acknowledge, request will be performed with completion 49ignalled later by an event. n = Rejected. Invalid state. n = Command performed with errors. 7-63 = Reserved Where used: S3F18, F20, F22, F24, F26, F30, F32	

CARRIERACTION	Format: A
Specifies the action requested for a carrier. Where used: S3F17	

CARRIERID	Format: A
The identifier of a carrier. Where used: S3F17	

CARRIERSPEC	Format: A
The object specifier for a carrier. Conforms to OBJSPEC. Where used: S3F29, F31	

CATTRDATA	Format: LIST,B,A,I(),U(),F(),BOOL
The value of a carrier attribute. Where used: S3F17	

CEED	Format: BOOL
Collection Event Enable/Disable Code, 1 byte FALSE = Disable TRUE = Enable Where used: S2F37	

CEID	Format: U4
Collection Event ID Where used: S2F35, F37, S6F11	

COMMACK	Format: B
Establish Communications Acknowledge Code, 1 byte	

0 = Accepted 1 = Denied, Try Again 2 – 63 = Reserved Where used: S1F13, F14

CPACK	Format:B
Command Parameter Acknowledge Code 1 = Parameter Name (CPNAME) does not exist 2 = Illegal value specified for CPVAL 3 = Illegal format specified for CPVAL >3 = Other error 4 – 63 = Reserved Where used: S2F42	

CPNAME	Format: A
Command Parameter Name; limited to 40 characters Where used: S2F41, F42	

CPVAL	Format: A,U(),I()
Command Parameter Value Where used: S2F41	

DATAID	Format: U2
Data ID Where used: S2F33, F35, F39, S6F5, F11	

DATALENGTH	Format: U2
Total bytes to be sent Where used: S2F39, S6F5	

DRACK	Format: B
Define Report Acknowledge Code, 1 byte 0 = Accepted 1 = Denied. Insufficient space 2 = Denied. Invalid format 3 = Denied. At least one RPTID already defined 4 = Denied. At least one VID does not exist >4 = Other errors 5 – 63 = Reserved Where used: S2F34	

DSPER	Format: A
Data Sample Period hhmmss, 6 bytes Where used: S2F23	

EAC	Format: B
Equipment Acknowledge Code, 1 byte 0 = Acknowledge	

1 = Denied. At least one constant does not exist
2 = Denied. Busy
3 = Denied. At least one constant out of range
n = Denied. At least one ECV has an improper format.
n – 63 = Reserved
Where used: S2F16

ECDEF	Format: A,U(),F(),I()

Equipment Constant Default Value
Where used: S2F30

ECID	Format: U4

Equipment Constant ID
Where used: S2F15

ECMAX	Format: A,U(),F(),I()

Equipment Constant Maximum Value
Where used: S2F30

ECMIN	Format: A,U(),F(),I()

Equipment Constant Minimum Value
Where used: S2F30

ECNAME	Format: A

Equipment Constant Name
Where used: S2F30

ECV	Format: A,U(),F(),I()

Equipment Constant Value
Where used: S2F15

ERACK	Format: B

Enable/Disable Event Report
Where used: S2F38

ERRW7	Format: A

Text string describing error found in process program
Where used: S7F27

GRANT	Format: B

Grant Code, 1 byte
0 = Positive response, load OK
1 = Busy, try again
2 = No space
3 = Duplicate name
n – 63 = Reserved
Where used: S2F40

GRANT6	Format: B

Permission to Send, 1 byte 0 = Permission granted 1 = Busy, try again 2 = Not interested 3 – 63 = Reserved Where used: S6F6	

HCACK	**Format: B**
Host Command Parameter Acknowledge Code, 1 byte 0 = Acknowledge, command will be performed 1 = Command does not exist 2 = Cannot perform now 3 = At least one parameter is invalid 4 = Acknowledge, command will be performed with completion signaled later by an event 5 = Rejected, tool is not ready to perform the command Where used: S2F42	

LENGTH	**Format: U2**
Length of the service program or process program in bytes Where used: S7F1	

LRACK	**Format: B**
Link Report Acknowledge Code, 1 byte 0 = Accepted 1 = Denied. Insufficient space 2 = Denied. Invalid format 3 = Denied. At least one CEID link already defined n = Denied. At least one CEID does not exist n = Denied. At least one RPTID does not exist n – 63 = Reserved Where used: S2F36	

MDLN	**Format: A**
Equipment Model Type, 6 bytes Where used: S1F2, F13, F14	

MHEAD	**Format: B**
SECS message block header associated with message block in error. Where used: S9F1, F3, F5, F7, F11	

OFLACK	**Format: B**
Acknowledge Code for OFF-LINE Request 0 = OFF-LINE Acknowledge 1 – 63 = Reserved Where used: S1F16	

ONLACK	**Format: B**
Acknowledge Code for ON-LINE Request	

0 = ON-LINE Accepted 1 = ON-LINE not allowed 2 = Equipment already ON-LINE 3 – 63 = Reserved Where used: S1F18

PPARM	Format: A,I(),U()
Process Parameter Where used: S7F23, S7F26	

PPBODY	Format: B
Process Program Body Where used: S7F3, F6	

PPGNT	Format: B
Process Program Grant Status, 1 byte 0 = OK 1 = Already have 2 = No space 3 = Invalid PPID n = Busy, try later n = Will not accept n – 63 = Reserved Where used: S7F2	

PPID	Format: A
Process Program ID; limited to 12 characters Where used: S2F27; S7F1, F3, F5, F6	

RCMD	Format: A
Remote Command String; limited to 20 characters Where used: S2F41	

REPGSZ	Format: U2
Report Group Size Where used: S2F23	

RPTID	Format: U4
Report ID Where used: S2F33, F35; S6F11	

SHEAD	Format: B
Stored header related to the transaction timer. Where used: S9F9	

SMPLN	Format: U2
Sample Number Where used: S6F1	

STIME	Format: A
Sample Time, 12 bytes yymmddhhmmss or 16 bytes YYYYMMDDhhmmsscc Where used: S6F1 Support for the 12-byte format shall be supported as a configurable option using the equipment constant Time Format. This is a format requirement only and does not imply either precision or accuracy.	

SOFTREV	Format: A
Software Revision Code, 6 bytes maximum Where used: S1F2, F13, F14	

SV	Format: A,I,F,U
Status Variable Value Where used: S1F4; S6F1, F11, F16, F20	

SVID	Format: U4
Status Variable ID Where used: S1F3, F11, F12; S2F23, F33	

SVNAME	Format: A
Status Variable Name Where used: S1F12	

TEXT	Format: A
A single line of characters. Where Used: S10F1, F3, F5	

TIAACK	Format: B
Equipment Acknowledgement Code, 1 byte 0 = Everything correct 1 = Too many SVIDs 2 = No more traces allowed 3 = Invalid period n = Invalid SVID n – 63 = Reserved Where used: S2F24	

TIACK	Format: B
Time Acknowledge Code, 1 byte 0 = OK 1 = Error not done 2 – 63 = Reserved Where used: S2F32	

TID	Format: B
Terminal Number, 1 byte 0 = OK >0 = Additional terminals at the same equipment Where Used: S10F1, F3, F5	

TIME	Format: A
Time of Day, 12 bytes yymmddhhmmss or 16 bytes YYYYMMDDhhmmsscc Where Used: S2F18 Support for the 12-byte format shall be supported as a configurable option using the equipment constant Time Format. This is a format requirement only and does not imply either precision or accuracy.	

TIMESTAMP	Format: A
Current value of system's internal clock, 16 bytes YYYYMMDDhhmmsscc Where Used: S5F73	

TOTSMP	Format: U2
Total samples to be made Where used: S2F23	

TRID	Format: U2
Trace Request ID Where used: S1F4	

UNITS	Format: A
Units Identifier Where used: S1F12	

V	Format: A,I,F,U
Variable Data Where Used: S1F4; S6F1, F11, F16, F20	

VID	Format: U4
Variable ID Where Used: S1F3, F11, F12; S2F23, F33	

6 SECS Message Details

6.1 Stream 1: Equipment Status

S1,F1	Are You There Request	S,H<->E,reply
Description	Establishes if the equipment is on-line. A function 0 response to this message means the communication is inoperative. In the equipment, a function 0 is equivalent to a timeout on the receive timer after issuing S1,F1 to the host.	
Structure	`Header only`	
Exception	None	

S1,F2	On Line Data	S,H<->E
Description	Data signifying that the equipment is alive.	
Structure	L,2 <MDLN> <SOFTREV>	
Exception	The host sends a zero-length list to the equipment.	

S1,F3	Selected Equipment Status Request	S,H->E,reply
Description	A request to the equipment to report selected values of its status.	
Structure	L,n n. <$SVID_1$> . . n. <$SVID_n$>	
Exception	A zero-length list means report all SVIDs.	

S1,F4	Selected Equipment Status Data	M,H<-E
Description	The equipment reports the value of each SVID requested in the order requested. The host remembers the names of values requested.	
Structure	L,n n. <SV_1> . . n. <SV_n>	
Exception	A zero-length list item for SV_i means that $SVID_i$ does not exist.	

S1,F11	Status Variable Namelist Request	S,H->E,reply
Description	A request to the equipment to identify certain status variables.	
Structure	L,n n. <$SVID_1$> . . n. <$SVID_n$>	
Exception	A zero length means report all SVIDs.	

S1,F12	**Status Variable Namelist Reply** **M,H<-E**
Description	The equipment reports to the host the name and units of the requested SVs.
Structure	L,n 1. L,3 1. <SVID$_1$> 2. <SVNAME$_1$> 3. <UNITS$_1$> 2. L,3 . . n. L,3 1. <SVID$_n$> 2. <SVNAME$_n$> 3. <UNITS$_n$>
Exception	Zero-length ASCII items for both SVNAME$_i$ and UNITS$_i$ indicates that the SVID does not exist.

S1,F13	**Establish Communications Request** **S,H<->E,reply**
Description	The purpose of this message is to provide a formal means of initializing communications at a logical application level both on power-up and following a break in communications. It should be the following any period where host and Equipment SECS applications are unable to communicate. An attempt to send an Establish Communications Request (S1,F13) should be repeated at programmable intervals until an Establish Communications Acknowledge (S1,F14) is received within the transaction timeout period with an acknowledgement code accepting the establishment.
Structure	L,2 1. <MDLN> 2. <SOFTREV>
Exception	The host sends a zero-length list to the equipment.

S1,F14	**Establish Communications Request Acknowledge** **S,H<->E**
Description	Accept or deny Establish Communications Request (S1,F13). MDLN and SOFTREV are on-line data and are valid only if COMMACK = 0.
Structure	L,2 1. <COMMACK> 2. L,2 1. <MDLN> 2. <SOFTREV>
Exception	The host sends a zero-length list for item 2 to the equipment.

S1,F15	**Request OFF-LINE** **S,H->E,reply**
Description	The host requests that the equipment transition to the OFF-LINE state.
Structure	Header only
Exception	None

S1,F16	**OFF-LINE Acknowledge** **S,H<-E**
Description	Acknowledge or error
Structure	<OFLACK>
Exception	None

S1,F17	Request ON-LINE S,H->E,reply
Description	The host requests that the equipment transition to the ON-LINE state
Structure	`Header only`
Exception	None

S1,F18	ON-LINE Acknowledge S,H<-E
Description	Acknowledge or error
Structure	`<ONLACK>`
Exception	None

6.2 Stream 2: Equipment Control and Diagnostics

S2,F13	**Equipment Constant Request**	**S,H->E,reply**
Description	Constants such as for calibration, servo gain, alarm limits, data collection mode, and other values that are changed infrequently can be obtained using this message.	
Structure	L,n n. <ECID$_1$> . . n. <ECID$_n$>	
Exception	A zero-length list means report all ECVs according to a predefined order.	

S2,F14	**Equipment Constant Data**	**M,H<-E**
Description	Data Response to S2,F13 in the order requested.	
Structure	L,n 1. <ECV$_1$> 2. <ECV$_2$> . . n. <ECV$_n$>	
Exception	A zero-length list item for ECV$_i$ means that ECID$_i$ does not exist. The list format for this data item is not allowed, except in this case.	

S2,F15	**New Equipment Constant Send**	**S,H->E,reply**
Description	Change one or more equipment constants.	

Structure	L,n
	1. L,2
	    1. <ECID$_1$>
	    2. <ECV$_1$>
	2. L,2
	.
	.
	n. L,2
	    1. <ECID$_n$>
	    2. <ECV$_n$>	

Exception	None	

S2,F16	**New Equipment Constant Acknowledge**	**S,H<-E**
Description	Acknowledge or error if EAC contains a non-zero error code, the equipment should not change any of the ECIDs specified in S2F15.	
Structure	<EAC>	
Exception	None	

S2,F17	**Date and Time Request**	**S,H<->E,reply**
Description	Useful to check equipment time base or for equipment to synchronize with the host time base.	
Structure	Header only	
Exception	None	

S2,F18	**Date and Time Data**	**S,H<->E**
Description	Actual time data.	
Structure	<TIME>	
Exception	A zero-length item means no time exists.	

S2,F23	Trace Initialize Send	M,H->E,reply
Description	Status variables exist at all times. This function provides a way to sample a subset of those status variables as a function of time. The trace data is returned on S6,F1 and is related to the original request by the TRID Multiple trace requests may be made to that equipment allowing it. If equipment receives S2,F23 with the same TRID as a trace function that is currently in progress, the equipment should terminate the old trace and then initiate the new trace. A trace function currently in progress may be terminated by S2,F23 with TRID of that trace and TOTSMP = 0. If S2,F23 is multi-block, it must be preceded by the S2,F39/S2,F40 Inquire/Grant transaction. Some equipment may support only single-Block S6,F1, and may refuse a S2,F23 message which would cause a multi-block S6,F1. Each equipment shall document its trace performance limits. The Host Computer shall not send an S2,F23 which exceeds the equipment's performance limits, or the equipment may operate incorrectly.	

Structure	`L,5`
	`1. <TRID>`
	`2. <DSPER>`
	`3. <TOTSMP>`
	`4. <REPGSZ>`
	`5. L,n`
	`1. <`$SVID_1$`>`
	`.`
	`.`
	`n. <`$SVID_n$`>`	

Exception	None	

S2,F24	Trace Initialize Acknowledge	S,H<-E
Description	Acknowledge or error.	
Structure	`<TIAACK>`	
Exception	None	

S2,F29	Equipment Constant Namelist Request	S,H->E,reply
Description	This function allows the host to retrieve basic information about what equipment constants are available in the equipment.	

Structure	`L,n`
	`n. <`$ECID_1$`>`
	`.`
	`.`
	`n. <`$ECID_n$`>`	

Exception	A zero-length list means send information for all ECIDs.	

S2,F30	**Equipment Constant Namelist**	**M,H<-E**
Description	Data Response.	

Structure	`L,n (number of equipment constants)`
	1. L,6
	    1. <ECID$_1$>
	    2. <ECNAME$_1$>
	    3. <ECMIN$_1$>
	    4. <ECMAX$_1$>
	    5. <ECDEF$_1$>
	    6. <UNITS$_1$>
	2. L,6
	.
	.
	n. L,6
	    1. <ECID$_n$>
	    2. <ECNAME$_n$>
	    3. <ECMIN$_n$>
	    4. <ECMAX$_n$>
	    5. <ECDEF$_n$>
	    6. <UNITS$_n$>	

Exception	Zero-length ASCII items for ECNAME$_i$, ECMIN$_i$, ECMAX$_i$, ECDEF$_i$, and UNITS$_i$ indicates that the ECID does not exist.	

S2,F31	**Date and Time Set Request**	**S,H->E,reply**
Description	Useful to synchronize the equipment time with the host time base.	
Structure	<TIME>	
Exception	None	

S2,F32	**Date and Time Set Acknowledge**	**S,H<-E**
Description	Acknowledge the receipt of time and date.	
Structure	<TIACK>	
Exception	None	

S2,F33	Define Report	M,H->E,reply
Description	The purpose of this message is for the host to define a group of reports for the equipment. The type of report to be transmitted is designated by a Boolean “Equipment Constant”. An “Equipment Constant Value” of “False” means that an “Event Report” (S6,F11) will be sent, and a value of “True” means that an “Annotated Event Report” (S6,F13) will be sent. If S2,F33 is Multi-block, it must be preceded by the S2,F39/S2,F40 Inquire/Grant transaction.	

Structure	`L,2`
	`1. <DATAID>`
	`2. L,a                          # reports`
	`     1. L,2                     report 1`
	`          1. <RPTID`$_1$`>`
	`          2. L,b                # VIDs this report`
	`               1. <VID`$_1$`>`
	`               .`
	`               .`
	`               b.<VID`$_b$`>`
	`     a. L,2                     report a`
	`          1. <RPTID`$_a$`>`
	`          2. L,c                # VIDs this report`
	`               1. <VID`$_1$`>`
	`               .`
	`               .`
	`               c. <VID`$_c$`>`	

Exception	1. A list of zero-length following <DATAID> deletes all report definitions and associated links. See S2,F35 (Link Event/Report). 2. A list of zero-length following <RPTID> deletes report type RPTID. All CEID links to this RPTID are also deleted.	

S2,F34	Define Report Acknowledge	S,H<-E
Description	Acknowledge or error if an error condition is detected the entire message is rejected (i.e., partial changes are not allowed).	
Structure	`<DRACK>`	
Exception	None	

S2,F35	**Link Event Report**	**M,H->E,reply**
Description	The purpose of this message is for the host to link n reports to an event (CEID). These linked event reports will default to ‘disabled’ upon linking. That is, the occurrence of an event would not cause the report to be sent until enabled. See S2,F37 for enabling reports. If S2,F35 is Multi-block, it must be preceded by the S2,F39/S2,F40 Inquire/Grant transaction.	
Structure	L,2 1. <DATAID> 2. L,a # events 1. L,2 event 1 1. <$CEID_1$> 2. L,b 1. <$RPTID_1$> . . b. <$RPTID_b$> . . a. L,2 event a 1. <$CEID_a$> # RPTIDS this event 2.L,c 1.<$RPTID_1$> . . c. <$RPTID_c$>	
Exception	A list of zero length following CEID deletes all report links to that event.	

S2,F36	**Link Event Report Acknowledge**	**S,H<-E**
Description	Acknowledge or error If an error condition is detected the entire message is rejected (i.e., partial changes are not allowed).	
Structure	<LRACK>	
Exception	None	

S2,F37	**Enable/Disable Event Report**	**S,H->E,reply**
Description	The purpose of this message is for the host to enable or disable reporting for a group of events (CEIDs).	
Structure	L,2 1. <CEED> enable/disable 2. L,n #CEIDs 1. <CEID1> . . n. <CEIDn>	
Exception	A list of zero length following <CEED> means all CEIDs.	

S2,F38	**Enable/Disable Event Report Acknowledge**	**S,H<-E**
Description	Acknowledge or error if an error condition is detected the entire message is rejected, i.e., partial changes are not allowed.	
Structure	<ERACK>	
Exception	None	

S2,F39	**Multi-block Inquire**	**S,H->E,reply**
Description	If a S2,F23, S2,F33, S2,F35, S2,F45, or S2,F49 message is more than one block, this transaction must precede the message.	
Structure	L,2 1. <DATAID> 2. <DATALENGTH>	
Exception	None	

S2,F40	**Multi-block Grant**	**S,H<-E**
Description	Grant permission to send multi-block message.	
Structure	<GRANT>	
Exception	None	

S2,F41	**Host Command Send**	**S,H->E,reply**
Description	The Host requests the Equipment perform the specified remote command with the associated parameters.	
Structure	L,2 1. <RCMD> 2. L,n 1. L,2 1. <CPNAME$_1$> 2. <CPVAL$_1$> . . n. L,2 1. <CPNAME$_n$> 2. <CPVAL$_n$>	 # of parameters parameter 1 name parameter 1 value parameter n name parameter n value
Exception	None	

S2,F42	**Host Command Acknowledge**	**S,H<-E**
Description	Acknowledge Host command or error. If command is not accepted due to one or more invalid parameters (i.e., HCACK = 3), then a list of invalid parameters will be returned containing the parameter name and reason for being invalid.	
Structure	L,2 1. <HCACK> 2. L,n 1. L,2 1. <CPNAME$_1$> 2. <CPACK$_1$> . . n. L,2 1. <CPNAME$_n$> 2. <CPACK$_n$>	 # of parameters parameter 1 name parameter 1 reason parameter n name parameter n reason
Exception	If there are no invalid parameters, then a list of zero length will be sent for item 2.	

6.3 Stream 3: Materials Status

S3,F17	Carrier Action Request	M,H->E,reply
Description	This message requests an action to be performed for a specified carrier. If using the SECS-I protocol and multi-block, this message must be preceded by the S3,F15/F16 transaction. When using other protocols like HSMS this preceding transaction is optional.	
Structure	L,5 1. <DATAID> 2. <CARRIERACTION> 3. <CARRIERID> 4. <PTN> 5. L,n n = number of carrier attributes 1. L,2 1. <CATTRID1> 2. <CATTRDATA1> . . n. L,2 1. <CATTRIDn> 2. <CATTRDATAn>	
Exception	If n = 0, then no carrier attributes are included. If CARRIERID is not a zero-length item, then PTN may be omitted (a zero-length item). ATTRID and ATTRDATA may be substituted for CATTRID and CATTRDATA respectively. ReticlePodLocationID may be used as one of <CATTRID> when the CARRIERACTION is PodRelease and the carrier is not at a Load Port.	

S3,F18	Carrier Action Acknowledge	S,H<-E
Description	This message acknowledges the carrier action request.	
Structure	L,2 1. <CAACK> 2. L,n 1. L,2 1. <ERRCODE1> 2. <ERRTEXT1> . . n. L,2 1. <ERRCODEn> 2. <ERRTEXTn>	
Exception	If n = 0, no errors exist.	

Carrier Attribute Definition Table

CATTRID	Definition	CATTRDATA Form
Capacity	Maximum number of substrates a carrier can hold.	Format Code 51 Capacity Capacity Range: 1 to 225 Capacity Examples: 1, 13, 25
ContentMap	Ordered list of lot and substrate identifiers corresponding to slot 1,2,3,…n.	L,n where n = Capacity 1. L,2 1. Format Code 20 LotID 2. Format Code 20 SubstID … n. L,2 1. Format Code 20 LotID 2. Format Code 20 SubstID SubstID conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6. When no Lot ID is provided by the host, the LotID value should be null.

		When a slot has no substrate or the host does not know substrate identifier, the LotID and SubstrateID value should be null.
SlotMap	Ordered list of slot status as provided by the host and corresponding to slot 1,2,3…n until a successful slot map read, then as read by the equipment.	L,n where n = Capacity 1. Format Code 51 enumerated … n. Format Code 51 enumerated Enumerated per variable SlotMap following enumeration: UNDEFINED, EMPTY, NOT EMPTY, CORRECTLY OCCUPIED, DOUBLESLOTTED, CROSS SLOTTED. (NOT EMPTY provided for equipment that cannot detect incorrectly slotted substrates).
SubstrateCount	The number of substrates currently in the carrier.	Format Code 51 SubstrateCount SubstrateCount Range: 0 to Capacity SubstrateCount Examples: 1, 3, 21, 25
Usage	The type of material contained in the carrier (i.e., TEST, DUMMY, PRODUCT, FILLER, etc.).	Format Code 20 Usage Usage is equipment defined, examples: ‘TEST’, ‘DUMMY’, ‘PRODUCT’

S3,F25	**Port Action Request**	**S,H->E,reply**
Description	This message requests an action be performed for a port.	

Structure	`L,3`
	`  1. <PORTACTION>`
	`  2. <PTN>`
	`  3. L,m`
	`       1. L,2`
	`            1. <PARAMNAME1>`
	`            2. <PARAMVAL1>`
	`       .`
	`       .`
	`       m. L,2`
	`            1. <PARAMNAMEm>`
	`            2. <PARAMVALm>`	

Exception	If m = 0, then no parameters are provided.	

S2,F26	**Port Action Acknowledge**	**S,H<-E**
Description	This message acknowledges the port action request.	

Structure	`L,2`
	`  1. <CAACK>`
	`  2. L,n`
	`       1. L,2`
	`            1. <ERRCODE1>`
	`            2. <ERRTEXT1>`
	`       .`
	`       .`
	`       n. L,2`
	`            1. <ERRCODEn>`
	`            2. <ERRTEXTn>`	

Exception	If n = 0, no errors exist.	

<table>
<tr><th>S3,F27</th><th>Change Access</th><th>S,H->E,reply</th></tr>
<tr><td>Description</td><td colspan="2">The Host requests the Equipment to change the Access Mode for the specified Load Ports. ACCESSMODE specifies the desired Access Mode. PTN specifies a desired Load Port Number.</td></tr>
<tr><td>Structure</td><td colspan="2"><pre>L,2
  1. <ACCESSMODE> (0: MANUAL, 1: AUTO)
  2. L,n
       1. <PTN1>
       .
       .
       n. <PTNn></pre></td></tr>
<tr><td>Exception</td><td colspan="2">If n = 0, then the command applies to all Load Ports on the equipment. If any specified port is already in the specified Access Mode, then the Equipment shall accept the command, and toggle all loadports to specified mode. If the Equipment is unable to change one or more of specified Port(s) to the specified Access Mode, then the Equipment shall accept the command (with appropriate response acknowledgement), and shall change only the Access Mode of those Port(s) allowed by the equipment, supplying the host with an indication that not all ports were successfully changed.</td></tr>
</table>

<table>
<tr><th>S2,F28</th><th>Change Access Acknowledge</th><th>S,H<-E</th></tr>
<tr><td>Description</td><td colspan="2">This message is used to acknowledge the Change Access request.</td></tr>
<tr><td>Structure</td><td colspan="2"><pre>L,2
  1. <CAACK>
  2. L,n
       1. L,3
            1. <PTN1>
            2. <ERRCODE1>
            3. <ERRTEXT1>
       .
       .
       n. L,3
            1. <PTNn>
            2. <ERRCODEn>
            3. <ERRTEXTn></pre></td></tr>
<tr><td>Exception</td><td colspan="2">If the command is successful, CAACK = 0, and n = 0. If the command was successful for some ports, CAACK = 6, and n > 0.</td></tr>
</table>

Services Message Mapping Table

Service Name	Stream,Function	Parameter
Bind	S3,F17/18	CARRIERACTION
CancelBind	S3,F17/18	CARRIERACTION
CancelCarrier	S3,F17/18	CARRIERACTION
CancelCarrierAtPort	S3,F17/18	CARRIERACTION
CancelCarrierNotification	S3,F17/18	CARRIERACTION
CancelCarrierOut	S3,F17/18	CARRIERACTION
CancelReservationAtPort	S3,F25/26	PORTACTION
CarrierIn	S3,F17/18	CARRIERACTION
CarrierNotification	S3,F17/18	CARRIERACTION
CarrierOut	S3,F17/18	CARRIERACTION
CarrierReCreate	S3,F17/18	CARRIERACTION
CarrierRelease	S3,F17/18	CARRIERACTION
ChangeServiceStatus	S3,F25/26	PORTACTION
ProceedWithCarrier	S3,F17/18	CARRIERACTION
ReserveAtPort	S3,F25/26	PORTACTION

6.4 Stream 5: Exception Handling

S5,F1	**Alarm Report Send**	**S,H<-E,[reply]**
Description	This message reports a change in or presence of an alarm condition. One message will be issued when the alarm is set and one message will be issued when the alarm is cleared. Irrecoverable errors and attention flags may not have a corresponding clear message.	

Structure	`L,3`
	`1. <ALCD>`
	`2. <ALID>`
	`3. <ALTX>`	

Exception	None	

S5,F2	**Alarm Report Acknowledge**	**S,H->E**
Description	Acknowledge or error.	
Structure	`<ACKC5>`	
Exception	None	

S5,F3	**Enable/Disable Alarm Send**	**S,H->E,[reply]**
Description	This message will change the state of the enable bit in the equipment. The enable bit determines if the alarm will be sent to the host. Alarms which are not controllable in this way are unaffected by this message.	

Structure	`L,2`
	`1. <ALED>`
	`2. <ALID>`	

Exception	A zero-length item for ALID means all alarms.	

S5,F4	**Enable/Disable Alarm Acknowledge (EAA)**	**S,H<-E**
Description	Acknowledge or error.	
Structure	`<ACKC5>`	
Exception	None	

S5,F5	**List Alarms Request**	**S,H->E,reply**
Description	This message requests the equipment to send binary and analog alarm information to the host.	
Structure	<$ALID_1$, . . . ,$ALID_n$>	
Exception	A zero-length item means send all possible alarms regardless of the state of ALED.	

S5,F6	**List Alarm Data**	**M,H<-E**
Description	This message contains the alarm data known to the equipment. There are “m” alarms in the list.	

Structure	L,m
	1. L,3
	    1. <ALCD$_1$>
	    2. <ALID$_1$>
	    3. <ALTX$_1$>
	2. L,3
	.
	.
	m. L,3
	    1. <ALCD$_m$>
	    2. <ALID$_m$>
	    3. <ALTX$_m$>	

Exception	If m = 0, no response can be made. A zero-length item returned for ALCD$_i$ or ALTX$_i$ means that value does not exist.	

S5,F7	**List Enabled Alarm Request**	**S,H->E,[reply]**
Description	List alarms wich are enabled.	
Structure	`Header Only`	
Exception	None	

S5,F8	**List Enabled Alarm Data**	**M,H<-E**
Description	This message is similar to S5,F6 except that it lists only alarms wich are enabled.	
Structure	`Same as S5,F6`	
Exception	None	

6.5 Stream 6: Data Collection

S6,F1	Trace Data Send	M,H<-E,[reply]
Description	This function sends samples to the host according to the trace setup done by S2,F23. Trace is a time-driven form of equipment status. Even if S6,F1 is multi-block, it is not preceded by an Inquire/Grant transaction, because the Host S2,F23 is an implicit grant. Some equipment may support only single-block S6,F1, and may refuse an S2,F23 (Trace Initiate Send) message which would cause a multi-block S6,F1.	
Structure	L,4 1. <TRID> 2. <SMPLN> 3. <STIME> 4. L,n 1. <SV1> 2. <SV2> . . n. <SVn>	
Exception	A zero-length STIME means no value is given and that the time is to be derived from SMPLN along with knowledge of the request.	

S6,F2	Trace Data Acknowledge	S,H->E
Description	Acknowledge or error.	
Structure	<ACKC6>	
Exception	None	

S6,F11	Event Report Send	M,H<-E, reply
Description	The purpose of this message is for the equipment to send a defined, linked, and enabled group of reports to the host upon the occurrence of an event (CEID). If S6,F11 is Multi-block, it must be preceded by the S6,F5/S6,F6 Inquire/Grant transaction.	
Structure	L,3 1. <DATAID> 2. <CEID> 3. L,a 1. L,2 1. <RPTID$_1$> 2. L,b 1. <V$_1$> . b. <V$_b$> . . a. L,2 report a 1. <RPTID$_a$> 2. L,c #Vs this report 1. <V$_1$> . . c. <V$_c$>	
Exception	If there are no reports linked to the event a 'null' report is assumed. A zero-length list for # of reports means there are no reports linked to the given CEID.	

<table>
<tr><th>S6,F13</th><th>Annotated Event Report Send</th><th>M,H<-E,reply</th></tr>
<tr><td>Description</td><td colspan="2">This message is the same as S6F11 with the exception that VID’s are sent with data. If S6,F13 is Multi-block, it must be preceded by the S6,F5/S6,F6 Inquire/Grant transaction.</td></tr>
<tr><td>Structure</td><td colspan="2">L,3
 1. <DATAID>
 2. <CEID>
 3. L,a
     1. L,2
         1. <RPTID$_{1}$>
         2. L,b
             1. L,2
                 1. <VID$_{1}$>
             .
             .
             b. L,2
                 1. <VID$_{b}$>
                 b. <V$_{b}$>
     .
     .
     a. L,2
         1. <RPTID$_{a}$>
         2. L,c
             1. L,2
                 1. <VID$_{1}$>
                 2. <V$_{1}$>
             .
             .
             c. L,2
                 1. <VID$_{c}$>
                 2. <V$_{c}$></td></tr>
<tr><td>Exception</td><td colspan="2">If there are no reports linked to the event a ‘null’ report is assumed. A zero-length list for # of reports means there are no reports linked to the given CEID.</td></tr>
</table>

S6,F14	**Annotated Event Report Acknowledge**	**S,H->E**
Description	Acknowledge or error.	
Structure	<ACKC6>	
Exception	None	

S6,F15	**Event Report Request**	**S,H->E, reply**
Description	The purpose of this message is for the host to demand a given report group from the equipment.	
Structure	<CEID>	
Exception	None	

S6,F16	**Event Report Data**	**M,H<-E**
Description	Equipment sends reports linked to given CEID to host.	
Structure	Identical to structure of S6,F11.	
Exception	A zero-length item means there are no reports linked to the given CEID.	

S6,F17	**Annotated Event Report Request**	**S,H->E,reply**
Description	Same as S6,F15, but requests annotated reports.	
Structure	<CEID>	
Exception	None	

S6,F18	Annotated Event Report Data M,H<-E
Description	Equipment sends annotated reports linked to given CEID.
Structure	`Same as S6,F13.`
Exception	A zero-length item means there are no reports linked to the given CEID.

S6,F19	Individual Report Request S,H->E,reply
Description	The purpose of this message is for the host to request a defined report from the equipment.
Structure	`<RPTID>`
Exception	None

S6,F20	Individual Report Data M,H<-E
Description	Equipment sends variable data defined for the given RPTID to the host.

Structure	`L,n # of variable data items`
	` n. <V1>`
	` .`
	` .`
	` n. <Vn>`

Exception	None

6.6 Stream 7: Process Program Management

Formated Process Programm Management is not supported by the equipment.

S7,F1	Process Program Load Inquire	S,H<->E,reply
Description	This message is used to initiate the transfer of a process program or to select from stored programs. The message may be used to initiate the transfer of an unformatted process program (S7,F3/S7,F4) .	

Structure	`L,2`
	`1. <PPID>`
	`2. <LENGTH>`	

Exception	None	

S7,F2	Process Program Load Grant	S,H<->E
Description	This message gives permission for the process program to be loaded.	
Structure	`<PPGNT>`	
Exception	None	

S7,F3	Process Program Send	S,H<->E,reply
Description	The program is sent. If S7,F3 is multi-block, it must be preceded by the S7,F1/S7,F2 Inquire/Grant transaction.	

Structure	`L,2`
	`1. <PPID>`
	`2. <PPBODY>`	

Exception	None	

S7,F4	Process Program Acknowledge	S,H<->E
Description	Acknowledge or error.	
Structure	`<ACKC7>`	
Exception	None	

S7,F5	Process Program Request	S,H<->E,reply
Description	This message is used to request the transfer of a process program.	
Structure	`<PPID>`	
Exception	None	

S7,F6	Process Program Data	M,H<->E
Description	This message is used to transfer a process program. PPBODY contains the binary data of a **.gz** file containing the process program **.xml** file. (The **.gz** file needs to be unzipped to extract the process program **.xml** file)	

Structure	`L,2`
	`1. <PPID>`
	`2. <PPBODY>`	

Exception	A zero-length list means request denied.	

S7,F17	Delete Process Program Send	S,H->E,reply
Description	This message is used by the host to request the equipment to delete process programs from equipment storage.	
Structure	`L,n (Number of process programs to be deleted)` n. <$PPID_1$> . . n. <$PPID_n$>	
Exception	If n = 0, then delete all.	

S7,F18	Delete Process Program Acknowledge	S,H<-E
Description	Acknowledge or error.	
Structure	`<ACKC7>`	
Exception	None	

S7,F19	Current EPPD Request	S,H->E,reply
Description	This message is used to request the transmission of the current equipment process program directory (EPPD). This is a list of all the PPIDs of the process programs stored in the equipment.	
Structure	`Header only`	
Exception	None	

S7,F20	Current EPPD Data	M,H<-E
Description	This message is used to transmit the current EPPD.	
Structure	L,n (number of process programs in the directory) n. <PPID$_1$> . . n. <PPID$_n$>	
Exception	None	

S7,F22	Equipment Process Capability Data	M,H<->E
Description	Not supported	
S7,F23	**Formatted Process Program Send**	**M,H<->E, reply**
Description	Not supported as our Process Program is already formatted in .xml format.	
S7,F24	**Formatted Process Acknowledge**	**M,H<->E**
Description	Not supported as our Process Program is already formatted in .xml format.	
S7,F25	**Formatted Process Program Request**	**M,H<->E, reply**
Description	Not supported as our Process Program is already formatted in .xml format.	
S7,F26	**Formatted Process Program Data**	**M,H<->E**
Description	Not supported as our Process Program is already formatted in .xml format.	
S7,F27	**Formatted Program Verification**	**M,H<->E, reply**
Description	Not supported as our Process Program is already formatted in .xml format.	
S7,F28	**Formatted Program Verification Acknowledge**	**M,H<->E**
Description	Not supported as our Process Program is already formatted in .xml format.	
S7,F29	**Formatted Program Verification Inquire**	**M,H<->E, reply**
Description	Not supported as our Process Program is already formatted in .xml format.	
S7,F30	**Formatted Program Verification Grant**	**M,H<->E**
Description	Not supported as our Process Program is already formatted in .xml format.	

6.7 Stream 9: System Errors

S9,F1	Unrecognized Device ID	S,H<-E
Description	The device ID in the message block header did not correspond to any known device ID in the node detecting the error.	
Structure	`<MHEAD>`	
Exception	None	

S9,F3	Unrecognized Stream Type	S,H<-E
Description	The equipment does not recognize the stream type in the message block header.	
Structure	`<MHEAD>`	
Exception	None	

S9,F5	Unrecognized Function Type	S,H<-E
Description	This message indicates that the function in the message ID is not recognized by the receiver.	
Structure	`<MHEAD>`	
Exception	None	

S9,F7	Illegal Data	S,H<-E
Description	This message indicates that the stream and function were recognized, but the message format was incorrect. This message may be used to indicate that a message received by the equipment did not conform to § 10.3.1.3 “Compliance to Message Definitions” or that data within the received message was not an allowed data type. If the Message Definition allows for multiple formats for items or for multiple data types, this message may be used to indicate that the equipment does not support a format or type that was used in the instance of the message. Equipment suppliers are encouraged to document which data formats and data types are supported when multiple formats or types are allowed for items within a message. Similarly, if the message for a custom stream and function is not formatted according to the rules in § 10.3.1.3 for the Message Definition provided by the equipment supplier, this message may be used to indicate the message format was incorrect.	
Structure	`<MHEAD>`	
Exception	None	

S9,F9	Transaction Timer Timeout	S,H<-E
Description	This message indicates that a transaction (receive) timer has timed out and that the corresponding transaction has been aborted. It is up to the host to respond to this error in an appropriate manner to keep the system operational.	
Structure	`<SHEAD>`	
Exception	None	

S9,F11	Data Too Long S,H<-E
Description	This message to the host indicates that the equipment has been sent more data than it can handle.
Structure	`<MHEAD>`
Exception	None

S9,F13	Conversation Timeout S,H<-E
Description	Data were expected but none were received within a reasonable length of time. Resources have been cleared.

Structure	`L,2`
	` 1. <MEXP>`
	` 2. <EDID>`

Exception	None

6.8 Stream 10: Terminal Services

S10,F1	Terminal Request	S,H<-E,[reply]
Description	A terminal text message to the host.	

Structure	`L,2`
	` 1. <TID>`
	` 2. <TEXT>`	

Exception	None	

S10,F2	Terminal Request Acknowledge	S,H->E
Description	Acknowledge or error.	
Structure	`<ACKC10>`	
Exception	None	

S10,F3	Terminal Display, Single	S,H->E, [reply]
Description	Data to be displayed.	

Structure	`L,2`
	` 1. <TID>`
	` 2. <TEXT>`	

Exception	None	

S10,F4	Terminal Display, Single Acknowledge	S,H<-E
Description	Acknowledge or error.	
Structure	`<ACKC10>`	
Exception	None	

S10,F5	Terminal Display, Multi-Block	M,H->E,[reply]
Description	Data to be displayed on the equipment's terminal.	

Structure	`L,2`
	` 1. <TID>`
	` 2. L,n`
	`      1. <TEXT1>`
	`      .`
	`      n.<TEXTn>`	

Exception	None	

S10,F6	Terminal Display, Multi-block Acknowledge	S,H<-E
Description	Acknowledge or error.	
Structure	`<ACKC10>`	
Exception	None	

S10,F7	Multi-block Not Allowed	S,H<-E
Description	An error message from a terminal that cannot handle a multi-block message from S10,F5.	
Structure	`<TID>`	
Exception	None	

6.9 Stream 14: Object Services

S14,F9	**Create Object Request**	**M,H->E,[reply]**
Description	This message is used to request an object owner to create an object instance. OBJSPEC specifies the object owner.	
Structure	L,3 1. <OBJSPEC> 2. <OBJTYPE> 3. L,a a = # attributes requested 1. L,2 1. <ATTRID$_1$> 2. <ATTRDATA$_1$> . . a. L,2 1. <ATTRID$_a$> 2. <ATTRDATA$_a$>	
Exception	If OBJSPEC is a null-length item, no object specifier is provided. If a = 0, no specific attribute settings are requested for the new object.	

S14,F10	**Create Object Acknowledge**	**M,H<->E**
Description	This message is used to acknowledge the success or failure of creating the new object specified. If successful, OBJSPEC is the object specifier of the new object. The list of attributes returned is dependent upon the type of object specified.	
Structure	L,3 1. <OBJSPEC> 2. L,b b = number of attributes returned 1. L,2 1. <ATTRID$_1$> 2. <ATTRDATA$_1$> . . b. L,2 1. <ATTRID$_b$> 2. <ATTRDATA$_b$> 3. L,2 1. <OBJACK> 2. L,p p = number of errors reported 1. L,2 1. <ERRCODE$_1$> 2. <ERRTEXT$_1$> . . p. L,2 1. <ERRCODE$_p$> 2. <ERRTEXT$_p$>	
Exception	If OBJSPEC is a null-length item, no object was created. If b = 0, no attributes of the new object are returned. If p = 0, no errors were detected.	

ControlJob Object Attributes

ATTRID	Definition	ATTRDATA Form
ObjID	Host defined identifier of the control job.	Text
ObjType	Object type.	Text = ‘ControlJob’
CarrierInputSpec	A list of carrierID for material that will be used by the ControlJob. An empty list is allowed.	List of CarrierID
MtrlOutSpec	Maps material from source to destination after processing. In Slot Integrity Mode the list shall be empty.	List of Structure: SourceMap DestinationMap
ProcessingCtrlSpec	A list of structures that defines the process jobs and rules for running each that will be run within this ControlJob.	(list of) Structure: PRJobID ControlRule OutputRule
ProcessOrderMgmt	Define the method for the order in which process jobs are initiated.	Enumeration: LIST ARRIVAL OPTIMIZE
StartMethod	A logical flag that determines if the ControlJob can start automatically. A user start may come through either the host connection or the operator console.	Boolean: TRUE – Auto FALSE – UserStart

S14,F11	Delete Object Request S,H->E,[reply]
Description	This message is used to request that the object specified in OBJSPEC be deleted. The list of attribute settings depends upon the type of object to be deleted.

Structure	`L,2`
	`1. <OBJSPEC>`
	`2. L,a          a = # attributes requested`
	`1. L,2`
	`1. <ATTRID1>`
	`2. <ATTRDATA1>`
	`.`
	`.`
	`a. L,2`
	`1. <ATTRIDa>`
	`2. <ATTRDATAa>`

Exception	If n = 0, no attribute settings are provided.

S14,F12	Delete Object Acknowledge	M,H<->E
Description	This message is used to acknowledge the success or failure of deleting the object specified. The list of attributes returned is dependent upon the type of object to be deleted.	
Structure	L,2 1. L,b b = number of attributes returned 1. L,2 1. <ATTRID$_1$> 2. <ATTRDATA$_1$> . . b. L,2 1. <ATTRID$_b$> 2. <ATTRDATA$_b$> 2. L,2 1. <OBJACK> 2. L,p p = number of errors reported 1. L,2 1. <ERRCODE$_1$> 2. <ERRTEXT$_1$> . . p. L,2 1. <ERRCODE$_p$> 2. <ERRTEXT$_p$>	
Exception	If n = 0, no attribute values are returned. If p = 0, no errors were detected.	

6.10 Stream 16: Processing Management

<table>
<tr><th>S16,F5</th><th>Process Job Command Request</th><th>M,H->E,[reply]</th></tr>
<tr><td>Description</td><td colspan="2">Send a job control command to a processing job.</td></tr>
<tr><td>Structure</td><td colspan="2"><pre>L,4
  1. <DATAID>
  2. <PRJOBID>
  3. <PRCMDNAME>   ABORT, STOP, CANCEL, PAUSE,
                     RESUME, STARTPROCESS
  4. L,n
       1. L,2
            1. <CPNAME1>
            2. <CPVAL1>
       .
       .
       n. L,2
            1. <CPNAMEn>
            2. <CPVALn></pre></td></tr>
<tr><td>Exception</td><td colspan="2">The CPNAME, CPVAL pairs are command parameter identifiers and values; n = 0 is valid for some commands (PRCMDNAME).</td></tr>
</table>

<table>
<tr><th>S16,F6</th><th>Process Job Command Acknowledge</th><th>S,H<-E</th></tr>
<tr><td>Description</td><td colspan="2">The processing service sends its confirmation for receipt of a command request.</td></tr>
<tr><td>Structure</td><td colspan="2"><pre>L,2
  1. <PRJOBID>
  2. L,2
       1. <ACKA>
       2. L,n   (n = {0,n})
            1. L,2
                 1. <ERRCODE1>
                 2. <ERRTEXT1>
            .
            .
            n. L,2
                 1. <ERRCODEn>
                 2. <ERRTEXTn></pre></td></tr>
<tr><td>Exception</td><td colspan="2">This list n may be zero length.</td></tr>
</table>

Process Command Parameter Table

Parameter	Form
DATAID	Numeric Format.
PRJOBID	ASCII Format. Process Job ID
PRCMDNAME	ASCII Format. ABORT, STOP, CANCEL, PAUSE, RESUME, STARTPROCESS
(List of) CmdParameter	(No required parameters)

<table>
<tr><th>S16,F11</th><th>PRJobCreateEnh</th><th>M,H->E,[reply]</th></tr>
<tr><td>Description</td><td colspan="2">Request equipment to create a Process Job with the given PRJOBID. If multi-block, this message must be preceded by the S16,F1/F2 transaction.</td></tr>
<tr><td>Structure</td><td colspan="2"><pre>L,7
  1. <DATAID>
  2. <PRJOBID>
  3. <MF>
  4a. L,n   [MF = carrier, n = # of carriers]
        1. L,2
             1. <CARRIERID1>
             2. L,j [j = # of slots, may be implemented
as an array]
                  1. <SLOTID1>
                  2. <SLOTID2>
                  .
                  .
                  j. <SLOTIDj>
        .
        .
        n. L,2
             1. <CARRIERIDn>
             2. L,j [j = # of slots, may be implemented
as an array]
                  1. <SLOTID1>
                  2. <SLOTID2>
                  .
                  .
                  j. <SLOTIDj>
  4b. L,n   [MF = substrate]
        1. <MID1>
        .
        . n. <MIDn>
  5. L,3
       1. <PRRECIPEMETHOD>
       2. <RCPSPEC> or <PPID>
       3. L,m [m = # recipe parameters]
            1. L,2
                 1. <RCPPARNM1>
                 2. <RCPPARVAL1>
                 .
                 m. L,2
                      1. <RCPPARNMm>
                      2. <RCPPARVALm>
  6. <PRPROCESSSTART>
  7. <PRPAUSEEVENT></pre></td></tr>
<tr><td>Exception</td><td colspan="2">The list for specifying material (item 4a and 4b) is empty (L,0instead of L,n), when no material is specified for the process job. The form of data item 4 (a or b) depends on the value in MF.If an implementer used PPID for this message, the format of PPID shall be ASCII because RecID is defined as the text in SEMI E40.</td></tr>
</table>

S16,F12	PRJobCreateEnh Acknowledge	S,H<-E
Description	This message acknowledges the request and reports any errors in the creation of a process job.	

Structure

```
L,2
  1. <PRJOBID>
  2. L,2
       1. <ACKA>
       2. L,n
            1. L,2
                 1. <ERRCODE1>
                 2. <ERRTEXT1>
            .
            .
            n. L,2
                 1. <ERRCODEn>
                 2. <ERRTEXTn>
```

Exception	If n = 0, no errors exist.

S16,F15	PRJobMultiCreate	M,H->E,[reply]
Description	Use this single message to Create Multiple Process Jobs, each of which may be unique in its association of material to process recipe. If multi-block, this message must be preceded by the S16,F1/F2 transaction.	

Structure

```
L,2
  1. <DATAID>
  2. L,p [p = # of process jobs being created]
       1. L,6
            1. <PRJOBID1>
            2. <MF1>
            3a. L,n [MF = carrier, n = # of carriers]
                  1. L,2
                       1. <CARRIERID1>
                       2. L,j [j = # of slots, may be
implemented as an array]
                            1. <SLOTID1>
                            2. <SLOTID2>
                       .
                       .
                       j. <SLOTIDj>
                  .
                  .
                  n. L,2
                       1. <CARRIERID1>
                       2. L,j [j = # of slots, may be
implemented as an array]
                            1. <SLOTID1>
                            2. <SLOTID2>
                       .
                       .
                       j. <SLOTIDj>
            3b. L,n [MF = substrate, n = # of MID]
                  1. <MID1>
                  .
                  .
                  n. <MIDn>
            4.  L,3
```

```
                  1. <PRRECIPEMETHOD1>
                  2. <RCPSPEC1> or <PPID1>
                  3. L,m [m = # recipe parameters]
                       1. L,2
                            1. <RCPPARNM1>
                            2. <RCPPARVAL1>
                       .
                       .
                       m. L,2
                            1. <RCPPARNMm>
                            2. <RCPPARVALm>
            5. <PRPROCESSSTART1>
            6. <PRPAUSEEVENT1>
  .
  .
  p. L,6
       1. <PRJOBIDp>
       2. <MFp>
       3a. L,n [MF = carrier, n = # of carriers]
             1. L,2
                  1. <CARRIERID1>
                  2. L,j [j = # of slots, may be
implemented as an array]
                      1. <SLOTID1>
                      2. <SLOTID2>
                      .
                      .
                      j. <SLOTIDj>
             .
             .
             n. L,2
                  1. <CARRIERIDn>
                  2. L,j [j = # of slots, may be
implemented as an array]
                      1. <SLOTID1>
                      2. <SLOTID2>
                      .
                      .
                      j. <SLOTIDj>
       3b. L,n [MF = substrate, n = # of MID]
             1. <MID1>
             .
             .
             n. <MIDn>
       4.  L,3
             1. <PRRECIPEMETHODp>
             2. <RCPSPECp> or <PPIDp>
             3. L,m [m = # recipe parameters]
                  1. L,2
                       1. <RCPPARNM1>
                       2. <RCPPARVAL1>
                  .
                  .
                  m. L,2
                       1. <RCPPARNM1>
```

	2. <RCPPARVAL$_1$> 5. <PRPROCESSSTART$_p$> 6. <PRPAUSEEVENT$_p$>
Exception	The list for specifying material (item 3a and 3b) is empty (L,0instead of L,n), when no material is specified for the process job. The form of data item 3 (a or b) depends on the value in MF.If an implementer uses PPIDi for this message, the format of PPIDi shall be ASCII because RecID is defined as the text in SEMI E40.

S16,F16	**PRJobMultiCreate Acknowledge S,H<-E**
Description	This message acknowledges the request and reports any errors in the creation of a process job. ERRTEXT contains the identifier of process jobs that were not created.
Structure	L,2 1. L,m [m = # jobs created] 1. <PRJOBID$_1$> . . m. <PRJOBID$_m$> 2. L,2 1. <ACKA> 2. L,n 1. L,2 1. <ERRCODE$_1$> 2. <ERRTEXT$_1$> . . n. L,2 1. <ERRCODE$_n$> 2. <ERRTEXT$_n$>
Exception	If n = 0, no errors exist.

S16,F17	**PRJobDequeue S,H->E,[reply]**
Description	Used to remove Process Jobs from the equipment for jobs that have not begun processing.
Structure	1. L,m [m = # jobs created] 1. <PRJOBID$_1$> . . m. <PRJOBID$_m$>
Exception	If m = 0, then de-queue all.

S16,F18	PRJobDequeue Acknowledge	S,H<-E
Description	Acknowledge the request to de-queue and report any errors. ERRTEXT will contain the identifier of any jobs that were not de-queued.	
Structure	L,2 1. L,m [m = # jobs removed] 1. <PRJOBID$_1$> . . m. <PRJOBID$_m$> 2. L,2 1. <ACKA> 2. L,n 1. L,2 1. <ERRCODE$_1$> 2. <ERRTEXT$_1$> . . n. L,2 1. <ERRCODE$_n$> 2. <ERRTEXT$_n$>	
Exception	If n = 0, no errors exist.	

S16,F19	PRGetAllJobs	S,H->E,[reply]
Description	Requests the equipment to return a list of Process Jobs which have not completed. They may be running or waiting to run.	
Structure	Header only	
Exception	None.	

S16,F20	PRGetAllJobs Send	S,H<-E
Description	Returns the requested list of Process Jobs.	
Structure	L,m [m = # jobs in the list] 1. L,2 1. <PRJOBID1> 2. <PRSTATE1> . . m. L,2 1. <PRJOBIDm> 2. <PRSTATEm>	
Exception	If m = 0, then no Process Jobs are running or waiting to run. PRSTATE: QUEUED/POOLED – 0 SETTING UP – 1 WAITING FOR START – 2 PROCESSING – 3 PROCESS COMPLETE – 4 PAUSING – 5 PAUSED – 6 STOPPING – 7 ABORTING – 8 STOPPED – 9 ABORTED – 10	

S16,F21	**PRGetSpace**	**S,H->E,[reply]**
Description	Requests the equipment to return the number of Process Jobs it has space to create.	
Structure	`Header only`	
Exception	None.	

S16,F22	**PRGetSpace Send**	**S,H<-E**
Description	Sends the host the number of Process Jobs which can be created.	
Structure	<PRJOBSPACE>	
Exception	None.	

S16,F27	**Control Job Command Request**	**S,H->E,[reply]**
Description	Send a control job command to a control job.	
Structure	L,3 1. <CTLJOBID> 2. <CTLJOBCMD> 1=CJStart, 2=CJPause, 3=CJResume, 4=CJCancel, 5=CJDeselect, 6=CJStop, 7=CJAbort, 8=CJHOQ 3. L,2 1. <CPNAME> 2. <CPVAL>	
Exception	3. L,2 IS L,0 for commands that do not need parameters.	

S16,F28	**Control Job Command Acknowledge**	**S,H<-E**
Description	Indicates success or failure of command request to a control job. If applicable ERRTEXT shall contain information on specificcommand parameter names or values that caused the error.	
Structure	L,2 1. <ACKA> 2. L,2 1. <ERRCODE> 2. <ERRTEXT>	
Exception	2. L,2 IS L,0 if no errors.	

7 SECS Scenarios

7.1 Establish Communications

Host attempts to Establish Communications

Comment	Host	Equipment	Comment
Communications state is Enabled (any substate)			
Establish Communications Request	S1F13 ⇨		
		⇦ S1F14	Reply COMMACK = Accept and Communications state = COMMUNICATING

Equipment Receives S1,F14 From Host Before Sending S1,F14

Comment	Host	Equipment	Comment
Communications state = NOT COMMUNICATING			
		⇦ S1F13	Establish Communications Request
Establish Communications Request	S1F13 ⇨		
Reply COMMACK = Accept	S1F14 ⇨		
		⇦ S1F14	Reply COMMACK = Accept

Equipment attempts to Establish Communications and Host Acknowledges

Comment	Host	Equipment	Comment
Communications state = NOT COMMUNICATING			
		⇦ S1F13	[LOOP] [LOOP] –SEND Establish Communications Request
Establish Communications Acknowledge	S1F14 ⇨		[IF] S1,F14 received w/o timeouts [THEN] exit_loop–SEND [ELSE] Delay for interval in Establish-Communications-Timeout [ENDIF] [END_LOOP]–SEND [IF] COMMACK = Accept [THEN] Communications state = Communicating exit_loop– [ELSE] Reset timer for delay, and delay for interval specified in Establish-Communications-Timeout [ENDIF]

			[END_LOOP]

7.2 Data Collection

Event Data Collection

Collection Event Reporting Set-Up

Comment	Host	Equipment	Comment
Send report definitions	S2F33 ⇨		DATAIDs, RTPIDs and VIDs received
		⇦ S2F34	DRACK = 0 reports are OK

Collection Event Reporting Set-Up

Comment	Host	Equipment	Comment
Link reports to events	S2F35 ⇨		CEIDs and corresponding RPTIDs are received.
		⇦ S2F36	LRACK = 0 the event linkages are acceptable.
Enable specific collection events	S2F37 ⇨		
		⇦ S2F38	ERACK = 0 OK, will generate the specified reports when the appropriate collection events happen

Collection Event Occurs on the Equipment

Comment	Host	Equipment	Comment
		⇦ S6F11	Equipment sends Event Report
Host acknowledges Event Report	S6F12 ⇨		

Variable Data Collection

Host Requests Report

Comment	Host	Equipment	Comment
Host requests data variables contained in report RPTID	S6F19 ⇨		
		⇦ S2F34	Equipment responds with a list of variable data for the given RPTID.

Trace Data Collection

Host Initiates Trace Report

Comment	Host	Equipment	Comment
Trace Data initialization Requested	S2F23 ⇨		
		⇦ S2F34	Acknowledge, trace initiated [DO] TOTSMP + REPGSZ times [DO] REPGSZ many times; collect SVID1,..,SVIDn data, delay time by DSPER [END_DO] Send SV1,..,SVn
		⇦ S6F1	
Acknowledge receipt	S6F2 ⇨		[END_DO]
Optional:			
Request trace termination prior to completion (TOTSMP = 0)	S2F23 ⇨		
		⇦ S2F23	Acknowledge premature termination

Status Data Collection

Request Equipment Status Report

Comment	Host	Equipment	Comment
Host requests report of selected status variable values	S1F3 ⇨		
		⇦ S1F4	Equipment responds with the requested status variable data.

Request Equipment Status Variable Namelist

Comment	Host	Equipment	Comment
Host requests equipment to identify status variables	S1F11 ⇨		
		⇦ S1F12	Equipment responds with the requested status variable descriptions

On-line Identification

Host Initiated

Comment	Host	Equipment	Comment

Are you there?	S1F1 ⇨		
		⇦ S1F2	Equipment replies with MDLN and SOFTREV.

7.3 Alarm Management

Enable/Disable Alarms

Comment	Host	Equipment	Comment
Enable/disable Alarm	S5F3 ⇨		
		⇦ S5F4	Acknowledge

Upload Alarm Information

Comment	Host	Equipment	Comment
Request alarm data/text	S5F5 ⇨		
		⇦ S5F6	Send alarm data/text

Send Alarm Report

Comment	Host	Equipment	Comment
Alarm occurrence detected by the equipment			
		⇦ S5F1	Send alarm report (if enabled)
Acknowledge	S5F2 ⇨		
		⇦ S6F11	Send event report (if enabled)
Acknowledge	S6F12 ⇨		

7.4 Remote Control

Comment	Host	Equipment	Comment
Host command send	S2F41 ⇨		
		⇦ S2F42	HCACK – Host Command Acknowledge

7.5 Equipment Constants

Host Sends Equipment Constants

Comment	Host	Equipment	Comment
Host sends equipment constants	S2F15 ⇨		
		⇦ S2F16	EAC = 0 sets constants

Host Equipment Constants Request

Comment	Host	Equipment	Comment

Host constant request	S2F13 ⇨		
		⇦ S2F14	Equipment constant data

Host Equipment Constant Namelist Request

Comment	Host	Equipment	Comment
Host constant namelist request	S2F29 ⇨		
		⇦ S2F30	Equipment constant namelist

7.6 Process Program Management

Process Program Created, Edited, or Deleted by Operator

Comment	Host	Equipment	Comment
			New process program created, edited, or deleted by operator at equipment. PPChangeName = PPID PPChangeStatus = 1 (Created) = 2 (Edited) = 3 (Deleted) [IF] CEID for Process Program Change Event enabled [THEN]
		⇦ S6F11	Send Event Report
Event Report Acknowledge	S6F12 ⇨		
			[END_IF]

Process Program Deletion by the Host

Comment	Host	Equipment	Comment
Delete Process Program Send	S7F17 ⇨		
		⇦ S7F18	Delete Process Program Acknowledge. The process program is removed from non-volatile storage.

Process Program Directory Request

Comment	Host	Equipment	Comment
Current EPPD Request	S7F19 ⇨		
		⇦ S7F20	Current EPPD Data

Host-Initiated Process Program Upload – Unformatted

Comment	Host	Equipment	Comment
Process Program Request	S7F5 ⇨		
		⇦ S7F6	Process Program Data (PPBODY contains the

			binary data of a **.gz** file containing the process program **.xml** file. The **.gz** file needs to be unzipped to extract the process program **.xml** file)

Equipment-Initiated Process Program Upload – Unformatted

Comment	Host	Equipment	Comment
		⇦ S7F3	Process Program Send
Process Program Acknowledge	S7F4 ⇨		

Host-Initiated Process Program Download – Unformatted

Comment	Host	Equipment	Comment
Process Program Send	S7F3 ⇨		
		⇦ S7F4	Process Program Acknowledge

Equipment-Initiated Process Program Download – Unformatted

Comment	Host	Equipment	Comment
		⇦ S7F5	Process Program Request
Process Program Data	S7F6 ⇨		

7.7 Error Messages

Message Fault Due to Unrecognized Device ID

Comment	Host	Equipment	Comment
Host sends message	SxFy ⇨		Equipment detects an unrecognized device ID within the message from the host.
		⇦ S9F1	Unrecognized Device ID

Message Fault Due to Unrecognized Stream Type

Comment	Host	Equipment	Comment
Host sends message	SxFy ⇨		Equipment detects an unrecognized stream type within the message from the host.
		⇦ S9F3	Unrecognized Stream Type

Message Fault Due to Unrecognized Function Type

Comment	Host	Equipment	Comment
Host sends message	SxFy ⇨		Equipment detects an unrecognized function type within the message from the host.

		⇦ S9F5	Unrecognized Function Type

Message Fault Due to Illegal Data Format

Comment	Host	Equipment	Comment
Host sends message	SxFy ⇨		Equipment detects illegal data format within the message from the host.
		⇦ S9F7	Illegal Data Format

Message Fault Due to Transaction Timer Timeout

Comment	Host	Equipment	Comment
			Equipment does not receive an expected reply message from the host and a transaction timer timeout occurs.
		⇦ S9F9	Transaction Timer Timeout

7.8 Clock

Host Instructs Equipment to Set Time

Comment	Host	Equipment	Comment
Host instructs equipment to set its time	S2F31 ⇨		
		⇦ S2F32	Equipment sets its internal time

Host Requests Equipment's Current Time Value

Comment	Host	Equipment	Comment
Host requests equipment time.	S2F17 ⇨		
		⇦ S2F18	Equipment returns its internal time

7.9 Processing Management

Create a Process Job

Comment	Host	Equipment	Comment
Host sends message with Process Job ID and parameters	S16F11 ⇨		
		⇦ S16F12	Equipment returns result of Process Job creation

Create multiple Process Jobs

Comment	Host	Equipment	Comment
Host sends message with Process Jobs ID and parameters	S16F15 ⇨		
		⇦ S16F16	Equipment returns result of each Process Job creation

Create a Control Job

Comment	Host	Equipment	Comment
Host sends message with Control Job ID and parameters	S14F9 ⇨	⇦ S14F10	Equipment returns result of Control Job creation

Send Process Job Command Request

Comment	Host	Equipment	Comment
Host sends message with PRCMDNAME (ABORT, CANCEL, STARTPROCESS)	S16F5 ⇨	⇦ S16F6	Equipment returns result of Process Job creation

Send Control Job Command Request

Comment	Host	Equipment	Comment
Host sends message with CTLJOBCMD (1=CJStart, 2=CJPause, 3=CJResume, 4=CJCancel, 5=CJDeselect, 6=CJStop, 7=CJAbort, 8=CJHOQ)	S16F27 ⇨	⇦ S16F28	Equipment returns result of Process Job creation

8 Appendix

8.1 Collection Events (CEIDs)

The following collection events are maintained and supported by the equipment.

CEID	Collection Event	Collection Event Description
0	EquipmentOffline	EquipmentOffline
1	ControlStateLocal	ControlStateLocal
2	ControlStateRemote	ControlStateRemote
3	OperatorCommandIssued	OperatorCommandIssued
4	ProcessingStarted	ProcessingStarted
5	ProcessingCompleted	ProcessingCompleted
6	ProcessingStopped	ProcessingStopped
7	ProcessingStateChange	ProcessingStateChange
8	AlarmDetected	AlarmDetected
9	AlarmCleared	AlarmCleared
10	OperatorEquipmentConstantChange	OperatorEquipmentConstantChange
11	LimitZoneTransition	LimitZoneTransition
12	ProcessProgramChange	ProcessProgramChange
13	processRecipeSelected	processRecipeSelected
14	MaterialReceived	MaterialReceived
15	MaterialRemoved	MaterialRemoved
16	SpoolingActivated	SpoolingActivated
17	SpoolingDeactivated	SpoolingDeactivated
18	SpoolTransmitFailure	SpoolTransmitFailure
19	MessageRecognition	MessageRecognition
100	initCompleted	initCompleted
101	processStateSetup	processStateSetup
102	setupCompleted	setupCompleted
103	readyForProcess	readyForProcess
110	buzzerStateChanged	buzzerStateChanged
111	signalTowerStateChanged	signalTowerStateChanged
120	port1ReadyToLoad	port1ReadyToLoad
121	port2ReadyToLoad	port2ReadyToLoad
122	port3ReadyToLoad	port3ReadyToLoad
123	port4ReadyToLoad	port4ReadyToLoad
124	port1ReadyToUnload	port1ReadyToUnload
125	port2ReadyToUnload	port2ReadyToUnload
126	port3ReadyToUnload	port3ReadyToUnload
127	port4ReadyToUnload	port4ReadyToUnload
130	port1CasPlaced	port1CasPlaced
131	port2CasPlaced	port2CasPlaced
132	port3CasPlaced	port3CasPlaced

133	port4CasPlaced	port4CasPlaced
134	port1CasRemoved	port1CasRemoved
135	port2CasRemoved	port2CasRemoved
136	port3CasRemoved	port3CasRemoved
137	port4CasRemoved	port4CasRemoved
140	port1CasMapped	port1CasMapped
141	port2CasMapped	port2CasMapped
142	port3CasMapped	port3CasMapped
143	port4CasMapped	port4CasMapped
145	cassetteMapped	A cassette has been mapped on any of the load ports
150	processingStartedPort1	ProcessingStarted
151	processingStartedPort2	ProcessingStarted
152	processingStartedPort3	ProcessingStarted
153	processingStartedPort4	ProcessingStarted
200	pm1StatusChanged	pm1StatusChanged
210	pm1Occupied	pm1Occupied
211	pm1Unoccupied	pm1Unoccupied
212	pm1WaferStarted	pm1WaferStarted
213	pm1WaferFinished	pm1WaferFinished
214	pm1ProcessStopping	processStopping
215	pm1ProcessAborting	processAborting
216	pm1ProcessAborted	processAborted
220	pm1StepStarted	pm1StepStarted
221	pm1StepFinished	pm1StepFinished
222	pm1MediumStepStarted	pm1MediumStepStarted
223	pm1MediumStepFinished	pm1MediumStepFinished
224	pm1DiStepStarted	pm1DiStepStarted
225	pm1DiStepFinished	pm1DiStepFinished
226	pm1N2DryStepStarted	pm1N2DryStepStarted
227	pm1N2DryStepFinished	pm1N2DryStepFinished
228	pm1DiwO3StepStarted	pm1DiwO3StepStarted
229	pm1DiwO3StepFinished	pm1DiwO3StepFinished
230	pm1MediumOffStepStarted	pm1MediumOffStepStarted
231	pm1MediumOffStepFinished	pm1MediumOffStepFinished
300	pm2StatusChanged	pm2StatusChanged
310	pm2Occupied	pm2Occupied
311	pm2Unoccupied	pm2Unoccupied
312	pm2WaferStarted	pm2WaferStarted
313	pm2WaferFinished	pm2WaferFinished
314	pm2ProcessStopping	pm2ProcessStopping
315	pm2ProcessAborting	pm2ProcessAborting
316	pm2ProcessAborted	pm2ProcessAborted
320	pm2StepStarted	pm2StepStarted
321	pm2StepFinished	pm2StepFinished
322	pm2MediumStepStarted	pm2MediumStepStarted
323	pm2MediumStepFinished	pm2MediumStepFinished

324	pm2DiStepStarted	pm2DiStepStarted
325	pm2DiStepFinished	pm2DiStepFinished
326	pm2N2DryStepStarted	pm2N2DryStepStarted
327	pm2N2DryStepFinished	pm2N2DryStepFinished
328	pm2DiwO3StepStarted	pm2DiwO3StepStarted
329	pm2DiwO3StepFinished	pm2DiwO3StepFinished
330	pm2MediumOffStepStarted	pm2MediumOffStepStarted
331	pm2MediumOffStepFinished	pm2MediumOffStepFinished
400	chc1StateChanged	chc1StateChanged
401	chc2StateChanged	chc2StateChanged
402	chc3StateChanged	chc3StateChanged
410	chc1Ready	chc1Ready
411	chc2Ready	chc2Ready
412	chc3Ready	chc3Ready
415	med1NotReady	med1NotReady
416	med2NotReady	med2NotReady
417	med3NotReady	med3NotReady
420	med1RefillStarted	med1RefillStarted
421	med1RefillFinished	med1RefillFinished
422	med2RefillStarted	med2RefillStarted
423	med2RefillFinished	med2RefillFinished
424	med3RefillStarted	med3RefillStarted
425	med3RefillFinished	med3RefillFinished
450	med1Comp1ConcHighLimit	med1Comp1ConcHighLimit
451	med1Comp2ConcHighLimit	med1Comp2ConcHighLimit
452	med1Comp3ConcHighLimit	med1Comp3ConcHighLimit
453	med1Comp4ConcHighLimit	med1Comp4ConcHighLimit
454	med1Comp5ConcHighLimit	med1Comp5ConcHighLimit
455	med2Comp1ConcHighLimit	med2Comp1ConcHighLimit
456	med2Comp2ConcHighLimit	med2Comp2ConcHighLimit
457	med2Comp3ConcHighLimit	med2Comp3ConcHighLimit
458	med2Comp4ConcHighLimit	med2Comp4ConcHighLimit
459	med2Comp5ConcHighLimit	med2Comp5ConcHighLimit
460	med3Comp1ConcHighLimit	med3Comp1ConcHighLimit
461	med3Comp2ConcHighLimit	med3Comp2ConcHighLimit
462	med3Comp3ConcHighLimit	med3Comp3ConcHighLimit
463	med3Comp4ConcHighLimit	med3Comp4ConcHighLimit
464	med3Comp5ConcHighLimit	med3Comp5ConcHighLimit
465	med1Comp1ConcLowLimit	med1Comp1ConcLowLimit
466	med1Comp2ConcLowLimit	med1Comp2ConcLowLimit
467	med1Comp3ConcLowLimit	med1Comp3ConcLowLimit
468	med1Comp4ConcLowLimit	med1Comp4ConcLowLimit
469	med1Comp5ConcLowLimit	med1Comp5ConcLowLimit
470	med2Comp1ConcLowLimit	med2Comp1ConcLowLimit
471	med2Comp2ConcLowLimit	med2Comp2ConcLowLimit
472	med2Comp3ConcLowLimit	med2Comp3ConcLowLimit

473	med2Comp4ConcLowLimit	med2Comp4ConcLowLimit
474	med2Comp5ConcLowLimit	med2Comp5ConcLowLimit
475	med3Comp1ConcLowLimit	med3Comp1ConcLowLimit
476	med3Comp2ConcLowLimit	med3Comp2ConcLowLimit
477	med3Comp3ConcLowLimit	med3Comp3ConcLowLimit
478	med3Comp4ConcLowLimit	med3Comp4ConcLowLimit
479	med3Comp5ConcLowLimit	med3Comp5ConcLowLimit
510	Atmsi1MeasFinished	Atmsi1MeasFinished
511	Atmsi2MeasFinished	Atmsi2MeasFinished
514	Pm1HpcStepStarted	Pm1HpcStepStarted
515	Pm1HpcStepFinished	Pm1HpcStepFinished
516	Pm2HpcStepStarted	Pm2HpcStepStarted
517	Pm2HpcStepFinished	Pm2HpcStepFinished
518	Pm1BemStepStarted	Pm1BemStepStarted
519	Pm1BemStepFinished	Pm1BemStepFinished
520	Pm2BemStepStarted	Pm2BemStepStarted
521	Pm2BemStepFinished	Pm2BemStepFinished
522	Pm1LowFlowStepStarted	Pm1LowFlowStepStarted
523	Pm1LowFlowStepFinished	Pm1LowFlowStepFinished
524	Pm2LowFlowStepStarted	Pm2LowFlowStepStarted
525	Pm2LowFlowStepFinished	Pm2LowFlowStepFinished
531	Roughness1MeasFinished	Roughness 1 Measurement Finished
532	Roughness2MeasFinished	Roughness 2 Measurement Finished
533	Pm1EpdStepStarted	Pm1EpdStepStarted
534	Pm1EpdStepFinished	Pm1EpdStepFinished
535	Pm1EpdEndpointDetected	Pm1EpdEndpointDetected
536	Pm2EpdStepStarted	Pm2EpdStepStarted
537	Pm2EpdStepFinished	Pm2EpdStepFinished
538	Pm2EpdEndpointDetected	Pm2EpdEndpointDetected
600	WaferAlignmentStatus	PreAligner finished an alignment
700	PjStateChanged	PjStateChanged
701	PjNoStateToQueued	PjNoStateToQueued
702	PjQueuedToSettingUp	PjQueuedToSettingUp
703	PjSettingUpToWaitingForStart	PjSettingUpToWaitingForStart
704	PjSettingUpToProcessing	PjSettingUpToProcessing
705	PjWaitingForStartToProcessing	PjWaitingForStartToProcessing
706	PjProcessingToProcessComplete	PjProcessingToProcessComplete
707	PjPostActiveToNoState	PjPostActiveToNoState
708	PjExecutingToPausing	PjExecutingToPausing
709	PjPausingToPaused	PjPausingToPaused
710	PjPauseToExecuting	PjPauseToExecuting
711	PjExecutingToStopping	PjExecutingToStopping
712	PjPauseToStopping	PjPauseToStopping
713	PjExecutingToAborting	PjExecutingToAborting
714	PjStoppingToAborting	PjStoppingToAborting
715	PjPauseToAborting	PjPauseToAborting

716	PjAbortingToAborted	PjAbortingToAborted
717	PjStoppingToStopped	PjStoppingToStopped
718	PjQueuedToNoState	PjQueuedToNoState
720	CjStateChanged	CjStateChanged
721	CjNoStateToQueued	CjNoStateToQueued
722	CjQueuedToNoState	CjQueuedToNoState
723	CjQueuedToSelected	CjQueuedToSelected
724	CjSelectedToQueued	CjSelectedToQueued
725	CjSelectedToExecuting	CjSelectedToExecuting
726	CjSelectedToWaitingForStart	CjSelectedToWaitingForStart
727	CjWaitingForStartToExecuting	CjWaitingForStartToExecuting
728	CjExecutingToPaused	CjExecutingToPaused
729	CjPausedToExecuting	CjPausedToExecuting
730	CjExecutingToCompleted	CjExecutingToCompleted
731	CjAbortedToCompleted	CjAbortedToCompleted
732	CjStoppedToCompleted	CjStoppedToCompleted
733	CjCompletedToNoState	CjCompletedToNoState
751	LpNoStateToServiceState	LpNoStateToServiceState
752	LpOutOfServiceToInService	LpOutOfServiceToInService
753	LpInServiceToOutOfService	LpInServiceToOutOfService
754	LpInServiceToTransferState	LpInServiceToTransferState
755	LpTransferReadyToReadyState	LpTransferReadyToReadyState
756	LpReadyToLoadToTransferBlocked	LpReadyToLoadToTransferBlocked
757	LpReadyToUnloadToTransferBlocked	LpReadyToUnloadToTransferBlocked
758	LpTransferBlockedToReadyToLoad	LpTransferBlockedToReadyToLoad
759	LpTransferBlockedToReadyToUnload	LpTransferBlockedToReadyToUnload
760	LpTransferBlockedToTransferReady	LpTransferBlockedToTransferReady
772	CarrierIdNoStateToNotRead	CarrierIdNoStateToNotRead
773	CarrierIdNoStateToWaitingForHost	CarrierIdNoStateToWaitingForHost
774	CarrierIdNoStateToVerificationOk	CarrierIdNoStateToVerificationOk
775	CarrierIdNoStateToVerificationFail	CarrierIdNoStateToVerificationFail
776	CarrierIdNotReadToVerificationOk	CarrierIdNotReadToVerificationOk
777	CarrierIdNotReadToWaitingForHost	CarrierIdNotReadToWaitingForHost
778	CarrierIdWaitingForHostToVerificationOk	CarrierIdWaitingForHostToVerificationOk
779	CarrierIdWaitingForHostToVerificationFail	CarrierIdWaitingForHostToVerificationFail
783	CarrierSlotMapNotReadToVerificationOk	CarrierSlotMapNotReadToVerificationOk
784	CarrierSlotMapNotReadToWaitingForHost	CarrierSlotMapNotReadToWaitingForHost
785	CarrierSlotMapWaitingForHostToVerificationOk	CarrierSlotMapWaitingForHost ToVerificationOk
786	CarrierSlotMapWaitingForHostToVerificationFail	CarrierSlotMapWaitingForHost ToVerificationFail
787	CarrierNotAccessedToInAccess	CarrierNotAccessedToInAccess
788	CarrierInAccessToCarrierComplete	CarrierInAccessToCarrierComplete
789	CarrierInAccessToCarrierStopped	CarrierInAccessToCarrierStopped
790	CarrierToNoState	CarrierToNoState
801	LpAccessModeNoStateToManualOrAuto	LpAccessModeNoStateToManualOrAuto
802	LpAccessModeManualToAuto	LpAccessModeManualToAuto

803	LpAccessModeAutoToManual	LpAccessModeAutoToManual
804	LpNotReservedToReserved	LpNotReservedToReserved
805	LpReservedToNotReserved	LpReservedToNotReserved
806	LpNotAssociatedToAssociated	LpNotAssociatedToAssociated
807	LpAssociatedToNotAssociated	LpAssociatedToNotAssociated
808	LpAssociatedToAssociated	LpAssociatedToAssociated
850	SubstNoStateToAtSource	SubstNoStateToAtSource
851	SubstAtSourceToAtWork	SubstAtSourceToAtWork
852	SubstAtWorkToAtSource	SubstAtWorkToAtSource
853	SubstAtWorkToAtWork	SubstAtWorkToAtWork
854	SubstAtWorkToAtDestination	SubstAtWorkToAtDestination
855	SubstAtDestinationToNoState	SubstAtDestinationToNoState
856	SubstNoStateToNeedsProcessing	SubstNoStateToNeedsProcessing
857	SubstNeedsProcessingToInProcess	SubstNeedsProcessingToInProcess
858	SubstInProcessToComplete	SubstInProcessToComplete
859	SubstNeedsProcessingToComplete	SubstNeedsProcessingToComplete
860	SubstLocUnoccupiedToOccupied	SubstLocUnoccupiedToOccupied
861	SubstLocOccupiedToUnoccupied	SubstLocOccupiedToUnoccupied
862	SubstToNoState	SubstToNoState
880	chc1CanisterFillPaused	chc1CanisterFillPaused
881	chc2CanisterFillPaused	chc2CanisterFillPaused
882	chc3CanisterFillPaused	chc3CanisterFillPaused
883	chc4CanisterFillPaused	chc4CanisterFillPaused

8.2 Variables

This chapter lists all the avaliable variables within the machine. This includes Status Variables (SVs) and their identifiers (SVIDs), Equipment Constants (ECVs) and their identifiers (ECIDs) and Data Values (DVVALs) and their identifiers (DVNAMEs).
The term VID (Variable Identifier) encompasses all SVIDs, ECIDs and DVNAMEs.

Data Variables (DVVALs)

Data variables/values may only be valid upon the occurence of a particular event. Each VID has one or more Collection Events (CEID) that indicates the validity of the variable.

VID	Data Variable	Format	Description	CEID
0	AlarmId	A	Not supported	-
1	EventLimit	A	Not supported	-
2	LimitVariable	A	Not supported	-
3	TransitionType	A	Not supported	-
100	port1N2ChuckFlowMinLot	F4	Minimum N2 chuck flow previous lot	5, 6, 216, 316
101	port1N2ChuckFlowMaxLot	F4	Maximum N2 chuck flow previous lot	5, 6, 216, 316
102	port1N2ChuckFlowAvrLot	F4	Average N2 chuck flow previous lot	5, 6, 216, 316
103	port1N2DryFlowMinLot	F4	Minimum N2 dry flow previous lot	5, 6, 216,

				316
104	port1N2DryFlowMaxLot	F4	Maximum N2 dry flow previous lot	5, 6, 216, 316
105	port1N2DryFlowAvrLot	F4	Average N2 dry flow previous lot	5, 6, 216, 316
110	port1Med1TempMinLot	F4	Minimum temperature of medium 1 previous lot	5, 6, 216, 316
111	port1Med2TempMinLot	F4	Minimum temperature of medium 2 previous lot	5, 6, 216, 316
112	port1Med3TempMinLot	F4	Minimum temperature of medium 3 previous lot	5, 6, 216, 316
113	port1Med1TempMaxLot	F4	Maximum temperature of medium 1 previous lot	5, 6, 216, 316
114	port1Med2TempMaxLot	F4	Maximum temperature of medium 2 previous lot	5, 6, 216, 316
115	port1Med3TempMaxLot	F4	Maximum temperature of medium 3 previous lot	5, 6, 216, 316
116	port1Med1TempAvrLot	F4	Average temperature of medium 1 previous lot	5, 6, 216, 316
117	port1Med2TempAvrLot	F4	Average temperature of medium 2 previous lot	5, 6, 216, 316
118	port1Med3TempAvrLot	F4	Average temperature of medium 3 previous lot	5, 6, 216, 316
130	port1Med1FlowMinLot	F4	Minimum flow of medium 1 previous lot	5, 6, 216, 316
131	port1Med2FlowMinLot	F4	Minimum flow of medium 2 previous lot	5, 6, 216, 316
132	port1Med3FlowMinLot	F4	Minimum flow of medium 3 previous lot	5, 6, 216, 316
133	port1Med1FlowMaxLot	F4	Maximum flow of medium 1 previous lot	5, 6, 216, 316
134	port1Med2FlowMaxLot	F4	Maximum flow of medium 2 previous lot	5, 6, 216, 316
135	port1Med3FlowMaxLot	F4	Maximum flow of medium 3 previous lot	5, 6, 216, 316
136	port1Med1FlowAvrLot	F4	Average flow of medium 1 previous lot	5, 6, 216, 316
137	port1Med2FlowAvrLot	F4	Average flow of medium 2 previous lot	5, 6, 216, 316
138	port1Med3FlowAvrLot	F4	Average flow of medium 3 previous lot	5, 6, 216, 316
150	port1DiFlowMinLot	F4	Minimum di flow previous lot	5, 6, 216, 316
151	port1DiFlowMaxLot	F4	Maximum di flow previous lot	5, 6, 216, 316
152	port1DiFlowAvrLot	F4	Average di flow previous lot	5, 6, 216, 316
160	port1ChuckSpeedMinLot	F4	Minimum chuck speed previous lot	5, 6, 216, 316
161	port1ChuckSpeedMaxLot	F4	Maximum chuck speed previous lot	5, 6, 216, 316
162	port1ChuckSpeedAvrLot	F4	Average chuck speed previous lot	5, 6, 216, 316
200	port1wafersFinished	U4	Port 1 wafers finished current lot	124
201	port1wafersToProcess	U4	Port 1 wafers to process current lot	150

202	port1wafersInCassette	U4	Port 1 wafers in cassette current lot	140
210	port1OutputPort	U4	Port 1 output port current lot	150
211	port1StartProcessDate	A	Port 1 date when process started current lot	150
212	port1StartProcessTime	A	Port 1 time when process started current lot	150
213	port1TotalLotTime	U4	Port 1 total time for current lot	124
214	port1TotalProcessTime	U4	Port 1 total time for process current lot	124
300	port2wafersFinished	U4	Port 2 wafers finished current lot	125
301	port2wafersToProcess	U4	Port 2 wafers to process current lot	151
302	port2wafersInCassette	U4	Port 2 wafers in cassette current lot	141
310	port2OutputPort	U4	Port 2 output port current lot	151
311	port2StartProcessDate	A	Port 2 date when process started current lot	151
312	port2StartProcessTime	A	Port 2 time when process started current lot	151
313	port2TotalLotTime	U4	Port 2 total time for current lot	125
314	port2TotalProcessTime	U4	Port 2 total time for process current lot	125
400	port3wafersFinished	U4	Port 3 wafers finished current lot	126
401	port3wafersToProcess	U4	Port 3 wafers to process current lot	152
402	port3wafersInCassette	U4	Port 3 wafers in cassette current lot	142
410	port3OutputPort	U4	Port 3 output port current lot	152
411	port3StartProcessDate	A	Port 3 date when process started current lot	152
412	port3StartProcessTime	A	Port 3 time when process started current lot	152
413	port3TotalLotTime	U4	Port 3 total time for current lot	126
414	port3TotalProcessTime	U4	Port 3 total time for process current lot	126
500	port4wafersFinished	U4	Port 4 wafers finished current lot	127
501	port4wafersToProcess	U4	Port 4 wafers to process current lot	153
502	port4wafersInCassette	U4	Port 4 wafers in cassette current lot	143
510	port4OutputPort	U4	Port 4 output port current lot	153
511	port4StartProcessDate	A	Port 4 date when process started current lot	153
512	port4StartProcessTime	A	Port 4 time when process started current lot	153
513	port4TotalLotTime	U4	Port 4 total time for current lot	127
514	port4TotalProcessTime	U4	Port 4 total time for process current lot	127
1000	pm1N2ChuckFlowMinWafer	F4	Process module 1 minimum N2 chuck flow last wafer	213
1001	pm1N2ChuckFlowMaxWafer	F4	Process module 1 maximum N2 chuck flow last wafer	213
1002	pm1N2ChuckFlowAvrWafer	F4	Process module 1 average N2	213

			chuck flow last wafer	
1003	pm1N2DryFlowMinWafer	F4	Process module 1 minimum N2 dry flow last wafer	213
1004	pm1N2DryFlowMaxWafer	F4	Process module 1 maximum N2 dry flow last wafer	213
1005	pm1N2DryFlowAvrWafer	F4	Process module 1 average N2 dry flow last wafer	213
1010	pm1Med1TempAvrWafer	F4	Process module 1 average medium 1 temperature last wafer	213
1011	pm1Med2TempAvrWafer	F4	Process module 1 average medium 2 temperature last wafer	213
1012	pm1Med3TempAvrWafer	F4	Process module 1 average medium 3 temperature last wafer	213
1013	pm1Med1TempMaxWafer	F4	Process module 1 maximum medium 1 temperature last wafer	213
1014	pm1Med2TempMaxWafer	F4	Process module 1 maximum medium 2 temperature last wafer	213
1015	pm1Med3TempMaxWafer	F4	Process module 1 maximum medium 3 temperature last wafer	213
1016	pm1Med1TempMinWafer	F4	Process module 1 minimum medium 1 temperature last wafer	213
1017	pm1Med2TempMinWafer	F4	Process module 1 minimum medium 2 temperature last wafer	213
1018	pm1Med3TempMinWafer	F4	Process module 1 minimum medium 3 temperature last wafer	213
1030	pm1Med1FlowMaxWafer	F4	Process module 1 maximum medium 1 flow last wafer	213
1031	pm1Med2FlowMaxWafer	F4	Process module 2 maximum medium 1 flow last wafer	213
1032	pm1Med3FlowMaxWafer	F4	Process module 3 maximum medium 1 flow last wafer	213
1033	pm1Med1FlowAvrWafer	F4	Process module 1 average medium 1 flow last wafer	213
1034	pm1Med2FlowAvrWafer	F4	Process module 2 average medium 1 flow last wafer	213
1035	pm1Med3FlowAvrWafer	F4	Process module 3 average medium 1 flow last wafer	213
1036	pm1Med1FlowMinWafer	F4	Process module 1 minimum medium 1 flow last wafer	213
1037	pm1Med2FlowMinWafer	F4	Process module 2 minimum medium 1 flow last wafer	213
1038	pm1Med3FlowMinWafer	F4	Process module 3 minimum medium 1 flow last wafer	213
1050	pm1DiFlowMaxWafer	F4	Process module 1 maximum di flow last wafer	213
1051	pm1DiFlowAvrWafer	F4	Process module 1 average di flow last wafer	213
1052	pm1DiFlowMinWafer	F4	Process module 1 minimum di flow last wafer	213
1060	pm1ChuckSpeedMaxWafer	F4	Process module 1 maximum chuck speed last wafer	213
1061	pm1ChuckSpeedAvrWafer	F4	Process module 1 average chuck speed last wafer	213
1062	pm1ChuckSpeedMinWafer	F4	Process module 1 minimum chuck speed last wafer	213
1100	pm1N2ChuckFlowMinPrevStep	F4	Process module 1 minimum N2	223,225,

			chuck flow of previous step	227,229
1101	pm1N2ChuckFlowMaxPrevStep	F4	Process module 1 maximum N2 chuck flow of previous step	223,225, 227,229
1102	pm1N2ChuckFlowAvrPrevStep	F4	Process module 1 average N2 chuck flow of previous step	223,225, 227,229
1103	pm1N2DryFlowMinPrevStep	F4	Process module 1 minimum N2 dry flow of previous step	227
1104	pm1N2DryFlowMaxPrevStep	F4	Process module 1 maximum N2 dry flow of previous step	227
1105	pm1N2DryFlowAvrPrevStep	F4	Process module 1 average N2 dry flow of previous step	227
1130	pm1MedTempMinPrevStep	F4	Process module 1 minimum medium temperature of previous step	223
1131	pm1MedTempMaxPrevStep	F4	Process module 1 maximum medium temperature of previous step	223
1132	pm1MedTempAvrPrevStep	F4	Process module 1 average medium temperature of previous step	223
1150	pm1MedFlowMinPrevStep	F4	Process module 1 minimum medium flow of previous step	223
1151	pm1MedFlowMaxPrevStep	F4	Process module 1 maximum medium flow of previous step	223
1152	pm1MedFlowAvrPrevStep	F4	Process module 1 average medium flow of previous step	223
1160	pm1DiFlowMinPrevStep	F4	Process module 1 minimum di flow of previous step	225
1161	pm1DiFlowMaxPrevStep	F4	Process module 1 maximum di flow of previous step	225
1162	pm1DiFlowAvrPrevStep	F4	Process module 1 average di flow of previous step	225
1170	pm1ChuckSpeedMinPrevStep	F4	Process module 1 minimum chuck speed of previous step	223,225, 227,229
1171	pm1ChuckSpeedMaxPrevStep	F4	Process module 1 maximum chuck speed of previous step	223,225, 227,229
1172	pm1ChuckSpeedAvrPrevStep	F4	Process module 1 average chuck speed of previous step	223,225, 227,229
1180	pm1StepTypePrevStep	A	Process module 1 type of previous step	221,223, 225,227, 229
1181	pm1StepTypeCurrentStep	A	Process module 1 type of current step	220,222, 224,226, 228
1182	pm1PrevStepElapsedTime	U4	Process module 1 elapsed time of previous step	223,225, 227,229
1183	pm1TotalProcessTimeWafer	U4	Process module 1 total process time last wafer	213
1184	pm1StepElapsedTime	U4	Process module 1 elapsed time last step	223,225, 227,229
1187	pm1CurrRecpNumOfSteps	U4	Process module 1 number of steps in current recipe	212
1188	pm1CurrStepNumber	U4	Process module 1 current recipe step number	220,222 224,226 228
1189	pm1PrevStepNumber	U4	Process module 1 previous recipe	221,223

			step number	225,227 229
1200	pm2N2ChuckFlowMinWafer	F4	Process module 2 minimum N2 chuck flow last wafer	313
1201	pm2N2ChuckFlowMaxWafer	F4	Process module 2 maximum N2 chuck flow last wafer	313
1202	pm2N2ChuckFlowAvrWafer	F4	Process module 2 average N2 chuck flow last wafer	313
1203	pm2N2DryFlowMinWafer	F4	Process module 2 minimum N2 dry flow last wafer	313
1204	pm2N2DryFlowMaxWafer	F4	Process module 2 maximum N2 dry flow last wafer	313
1205	pm2N2DryFlowAvrWafer	F4	Process module 2 average N2 dry flow last wafer	313
1210	pm2Med1TempAvrWafer	F4	Process module 2 average medium 1 temperature last wafer	313
1211	pm2Med2TempAvrWafer	F4	Process module 2 average medium 2 temperature last wafer	313
1212	pm2Med3TempAvrWafer	F4	Process module 2 average medium 3 temperature last wafer	313
1213	pm2Med1TempMaxWafer	F4	Process module 2 maximum medium 1 temperature last wafer	313
1214	pm2Med2TempMaxWafer	F4	Process module 2 maximum medium 2 temperature last wafer	313
1215	pm2Med3TempMaxWafer	F4	Process module 2 maximum medium 3 temperature last wafer	313
1216	pm2Med1TempMinWafer	F4	Process module 2 minimum medium 1 temperature last wafer	313
1217	pm2Med2TempMinWafer	F4	Process module 2 minimum medium 2 temperature last wafer	313
1218	pm2Med3TempMinWafer	F4	Process module 2 minimum medium 3 temperature last wafer	313
1230	pm2Med1FlowMaxWafer	F4	Process module 2 maximum medium 1 flow last wafer	313
1231	pm2Med2FlowMaxWafer	F4	Process module 2 maximum medium 2 flow last wafer	313
1232	pm2Med3FlowMaxWafer	F4	Process module 2 maximum medium 3 flow last wafer	313
1233	pm2Med1FlowAvrWafer	F4	Process module 2 average medium 1 flow last wafer	313
1234	pm2Med2FlowAvrWafer	F4	Process module 2 average medium 2 flow last wafer	313
1235	pm2Med3FlowAvrWafer	F4	Process module 2 average medium 3 flow last wafer	313
1236	pm2Med1FlowMinWafer	F4	Process module 2 minimum medium 1 flow last wafer	313
1237	pm2Med2FlowMinWafer	F4	Process module 2 minimum medium 2 flow last wafer	313
1238	pm2Med3FlowMinWafer	F4	Process module 2 minimum medium 3 flow last wafer	313
1250	pm2DiFlowMaxWafer	F4	Process module 2 maximum di flow last wafer	313
1251	pm2DiFlowAvrWafer	F4	Process module 2 average di flow last wafer	313
1252	pm2DiFlowMinWafer	F4	Process module 2 minimum di flow last wafer	313

1260	pm2ChuckSpeedMaxWafer	F4	Process module 2 maximum chuck speed last wafer	313
1261	pm2ChuckSpeedAvrWafer	F4	Process module 2 average chuck speed last wafer	313
1262	pm2ChuckSpeedMinWafer	F4	Process module 2 minimum chuck speed last wafer	313
1300	pm2N2ChuckFlowMinPrevStep	F4	Process module 2 minimum N2 chuck flow of previous step	323,325, 327,329
1301	pm2N2ChuckFlowMaxPrevStep	F4	Process module 2 maximum N2 chuck flow of previous step	323,325, 327,329
1302	pm2N2ChuckFlowAvrPrevStep	F4	Process module 2 average N2 chuck flow of previous step	323,325, 327,329
1303	pm2N2DryFlowMinPrevStep	F4	Process module 2 minimum N2 dry flow of previous step	327
1304	pm2N2DryFlowMaxPrevStep	F4	Process module 2 maximum N2 dry flow of previous step	327
1305	pm2N2DryFlowAvrPrevStep	F4	Process module 2 average N2 dry flow of previous step	327
1330	pm2MedTempMinPrevStep	F4	Process module 2 minimum medium temperature of previous step	323
1331	pm2MedTempMaxPrevStep	F4	Process module 2 maximum medium temperature of previous step	323
1332	pm2MedTempAvrPrevStep	F4	Process module 2 average medium temperature of previous step	323
1350	pm2MedFlowMinPrevStep	F4	Process module 2 minimum medium flow of previous step	323
1351	pm2MedFlowMaxPrevStep	F4	Process module 2 maximum medium flow of previous step	323
1352	pm2MedFlowAvrPrevStep	F4	Process module 2 average medium flow of previous step	323
1360	pm2DiFlowMinPrevStep	F4	Process module 2 minimum di flow of previous step	325
1361	pm2DiFlowMaxPrevStep	F4	Process module 2 maximum di flow of previous step	325
1362	pm2DiFlowAvrPrevStep	F4	Process module 2 average di flow of previous step	325
1370	pm2ChuckSpeedMinPrevStep	F4	Process module 2 minimum chuck speed of previous step	323,325, 327,329
1371	pm2ChuckSpeedMaxPrevStep	F4	Process module 2 maximum chuck speed of previous step	323,325, 327,329
1372	pm2ChuckSpeedAvrPrevStep	F4	Process module 2 average chuck speed of previous step	323,325, 327,329
1380	pm2StepTypePrevStep	A	Process module 2 type of previous step	321,323 325,327 329
1381	pm2StepTypeCurrentStep	A	Process module 2 type of current step	320,322 324,326 328
1382	pm2PrevStepElapsedTime	U4	Process module 2 elapsed time of previous step	323,325 327,329
1383	pm2TotalProcessTimeWafer	U4	Process module 2 total process time last wafer	313
1384	pm2StepElapsedTime	U4	Process module 2 elapsed time	323,325

			last step	327,329
1387	pm2CurrRecpNumOfSteps	U4	Process module 2 number of steps in current recipe	312
1388	pm2CurrStepNumber	U4	Process module 2 current step number	320,322 324,326 328
1389	pm2PrevStepNumber	U4	Process module 2 previous step number	321,323 325,327 329
1800	portIdMaterialReceived	U4	Port id of received material	14
1801	portIdMaterialRemoved	U4	Port id of removed material	15
1810	lastStartedWaferPmId	U4	Process module id of last started wafer	213,313
1811	lastStartedWaferJobId	A	Job id of last started wafer	213,313
1812	lastStartedWaferLotId	A	Lot id of last started wafer	213,313
1813	lastStartedWaferLoadSlot	U4	Load slot of last started wafer	213,313
1814	lastStartedWaferUnloadSlot	U4	Unload slot of last started wafer	213,313
1815	lastStartedWaferLoadPort	U4	Load port of last started wafer	213,313
1816	lastStartedWaferUnloadPort	U4	Unload port of last started wafer	213,313
1817	lastStartedWaferCid	A	Cid of last started wafer	213,313
1820	lastFinishedWaferPmId	U4	Process module id of last finished wafer	213,313
1821	lastFinishedWaferJobId	A	Job id of last finished wafer	213,313
1822	lastFinishedWaferLotId	A	Lot id of last finished wafer	213,313
1823	lastFinishedWaferLoadSlot	U4	Load slot of last finished wafer	213,313
1824	lastFinishedWaferUnloadSlot	U4	Unload slot of last finished wafer	213,313
1830	jobIdLastFinishedLot	A	Job Id of Last Finished Lot	5,6
1831	lotIdLastFinishedLot	A	Lot Id of Last Finished Lot	5,6
1832	ppLastFinishedLot	A	PPID of Last Finished Lot	5,6
1833	portIdLastFinishedLot	A	Port Id of Last Finished Lot	5,6
1834	cidLastFinishedLot	A	CID of Last Finished Lot	5,6
1835	jobIdLastStartedLot	A	Job Id of Last Started Lot	4
1836	lotIdLastStartedLot	A	Lot Id of Last Started Lot	4
1837	ppLastStartedLot	A	PPID of Last Started Lot	4
1838	portIdLastStartedLot	A	Port Id of Last Started Lot	4
1839	cidLastStartedLot	A	CID of Last Started Lot	4
1850	ppSelectedName	A	Name of the last selected recipe	13
1851	ppSelectedPortId	A	Port id of the last selected recipe	13
1855	ppChangeName	A	Name of the last changed recipe	12
1856	ppChangeStatus	U4	Type of last recipe change	12
1900	pm1CurrWaferJobId	A	Process module 1 job id of current wafer	212-216,221
1901	pm1CurrWaferLotId	A	Process module 1 lot id of current wafer	212-216,221
1902	pm1CurrWaferCId	A	Process module 1 cid of current wafer	212-216,221
1903	pm1CurrWaferPPId	A	Process module 1 PP id of current wafer	212-216,221
1904	pm1CurrWaferLoadSlot	U4	Process module 1 load slot of current wafer	212-216,221

1905	pm1CurrWaferUnloadSlot	U4	Process module 1 unload slot of current wafer	212-216,221
1906	pm1CurrWaferLoadPort	U4	Process module 1 load port of current wafer	212-216,221
1907	pm1CurrWaferUnloadPort	U4	Process module 1 unload port of current wafer	212-216,221
1908	pm1DiwO3FlowMaxWafer	F4	Process module 1 DI Water O3 maximum flow last wafer	213
1909	pm1DiwO3FlowAvrWafer	F4	Process module 1 DI Water O3 average flow last wafer	213
1910	pm1DiwO3FlowMinWafer	F4	Process module 1 DI Water O3 minimum flow last wafer	213
1911	pm1DiwO3FlowMaxPrevStep	F4	Process module 1 DI Water O3 maximum flow previous step	229
1912	pm1DiwO3FlowAvrPrevStep	F4	Process module 1 DI Water O3 average flow previous step	229
1913	pm1DiwO3FlowMinPrevStep	F4	Process module 1 DI Water O3 minimum flow previous step	229
1914	pm1DiwO3ConcentrationMaxPrevStep	F4	Process module 1 DI Water O3 maximum concentration previous step	229
1915	pm1DiwO3ConcentrationAvrPrevStep	F4	Process module 1 DI Water O3 average concentration previous step	229
1916	pm1DiwO3ConcentrationMinPrevStep	F4	Process module 1 DI Water O3 minimum concentration previous step	229
1917	pm1CO2FlowMaxPrevStep	F4	pm1CO2FlowMaxPrevStep	225,515
1918	pm1CO2FlowAvrPrevStep	F4	pm1CO2FlowAvrPrevStep	225,515
1919	pm1CO2FlowMinPrevStep	F4	pm1CO2FlowMinPrevStep	225,515
1920	pm1CO2CondMaxPrevStep	F4	pm1CO2CondMaxPrevStep	225,515
1921	pm1CO2CondAvrPrevStep	F4	pm1CO2CondAvrPrevStep	225,515
1922	pm1CO2CondMinPrevStep	F4	pm1CO2CondMinPrevStep	225,515
1923	pm1CO2TempMaxPrevStep	F4	pm1CO2TempMaxPrevStep	225,515
1924	pm1CO2TempAvrPrevStep	F4	pm1CO2TempAvrPrevStep	225,515
1925	pm1CO2TempMinPrevStep	F4	pm1CO2TempMinPrevStep	225,515
1926	pm1CO2PressMaxPrevStep	F4	pm1CO2PressMaxPrevStep	225,515
1927	pm1CO2PressAvrPrevStep	F4	pm1CO2PressAvrPrevStep	225,515
1928	pm1CO2PressMinPrevStep	F4	pm1CO2PressMinPrevStep	225,515
1929	pm1CO2FlowMaxWafer	F4	pm1CO2FlowMaxWafer	225,515
1930	pm1CO2FlowAvrWafer	F4	pm1CO2FlowAvrWafer	225,515
1931	pm1CO2FlowMinWafer	F4	pm1CO2FlowMinWafer	225,515
1932	pm1CO2CondMaxWafer	F4	pm1CO2CondMaxWafer	225,515
1933	pm1CO2CondAvrWafer	F4	pm1CO2CondAvrWafer	225,515
1934	pm1CO2CondMinWafer	F4	pm1CO2CondMinWafer	225,515
1935	pm1CO2TempMaxWafer	F4	pm1CO2TempMaxWafer	225,515
1936	pm1CO2TempAvrWafer	F4	pm1CO2TempAvrWafer	225,515
1937	pm1CO2TempMinWafer	F4	pm1CO2TempMinWafer	225,515
1938	pm1CO2PressMaxWafer	F4	pm1CO2PressMaxWafer	225,515
1939	pm1CO2PressAvrWafer	F4	pm1CO2PressAvrWafer	225,515
1940	pm1CO2PressMinWafer	F4	pm1CO2PressMinWafer	225,515

2000	pm2CurrWaferJobId	A	Process module 2 job id of current wafer	312-316,321
2001	pm2CurrWaferLotId	A	Process module 2 lot id of current wafer	312-316,321
2002	pm2CurrWaferCId	A	Process module 2 cid of current wafer	312-316,321
2003	pm2CurrWaferPPId	A	Process module 2 PP id of current wafer	312-316,321
2004	pm2CurrWaferLoadSlot	U4	Process module 2 load slot of current wafer	312-316,321
2005	pm2CurrWaferUnloadSlot	U4	Process module 2 unload slot of current wafer	312-316,321
2006	pm2CurrWaferLoadPort	U4	Process module 2 load port of current wafer	312-316,321
2007	pm2CurrWaferUnloadPort	U4	Process module 2 unload port of current wafer	312-316,321
2008	pm2DiwO3FlowMaxWafer	F4	Process module 2 DI Water O3 maxiumum flow last wafer	313
2009	pm2DiwO3FlowAvrWafer	F4	Process module 2 DI Water O3 average flow last wafer	313
2010	pm2DiwO3FlowMinWafer	F4	Process module 2 DI Water O3 minimum flow last wafer	313
2011	pm2DiwO3FlowMaxPrevStep	F4	Process module 2 DI Water O3 maximum flow previous step	329
2012	pm2DiwO3FlowAvrPrevStep	F4	Process module 2 DI Water O3 average flow previous step	329
2013	pm2DiwO3FlowMinPrevStep	F4	Process module 2 DI Water O3 minimum flow previous step	329
2014	pm2DiwO3ConcentrationMaxPrevStep	F4	Process module 2 DI Water O3 maximum concentration previous step	329
2015	pm2DiwO3ConcentrationAvrPrevStep	F4	Process module 2 DI Water O3 average concentration previous step	329
2016	pm2DiwO3ConcentrationMinPrevStep	F4	Process module 2 DI Water O3 minimum concentration previous step	329
2017	pm2CO2FlowMaxPrevStep	F4	pm2CO2FlowMaxPrevStep	325,517
2018	pm2CO2FlowAvrPrevStep	F4	pm2CO2FlowAvrPrevStep	325,517
2019	pm2CO2FlowMinPrevStep	F4	pm2CO2FlowMinPrevStep	325,517
2020	pm2CO2CondMaxPrevStep	F4	pm2CO2CondMaxPrevStep	325,517
2021	pm2CO2CondAvrPrevStep	F4	pm2CO2CondAvrPrevStep	325,517
2022	pm2CO2CondMinPrevStep	F4	pm2CO2CondMinPrevStep	325,517
2023	pm2CO2TempMaxPrevStep	F4	pm2CO2TempMaxPrevStep	325,517
2024	pm2CO2TempAvrPrevStep	F4	pm2CO2TempAvrPrevStep	325,517
2025	pm2CO2TempMinPrevStep	F4	pm2CO2TempMinPrevStep	325,517
2026	pm2CO2PressMaxPrevStep	F4	pm2CO2PressMaxPrevStep	325,517
2027	pm2CO2PressAvrPrevStep	F4	pm2CO2PressAvrPrevStep	325,517
2028	pm2CO2PressMinPrevStep	F4	pm2CO2PressMinPrevStep	325,517
2029	pm2CO2FlowMaxWafer	F4	pm2CO2FlowMaxWafer	325,517
2030	pm2CO2FlowAvrWafer	F4	pm2CO2FlowAvrWafer	325,517
2031	pm2CO2FlowMinWafer	F4	pm2CO2FlowMinWafer	325,517
2032	pm2CO2CondMaxWafer	F4	pm2CO2CondMaxWafer	325,517

2033	pm2CO2CondAvrWafer	F4	pm2CO2CondAvrWafer	325,517
2034	pm2CO2CondMinWafer	F4	pm2CO2CondMinWafer	325,517
2035	pm2CO2TempMaxWafer	F4	pm2CO2TempMaxWafer	325,517
2036	pm2CO2TempAvrWafer	F4	pm2CO2TempAvrWafer	325,517
2037	pm2CO2TempMinWafer	F4	pm2CO2TempMinWafer	325,517
2038	pm2CO2PressMaxWafer	F4	pm2CO2PressMaxWafer	325,517
2039	pm2CO2PressAvrWafer	F4	pm2CO2PressAvrWafer	325,517
2040	pm2CO2PressMinWafer	F4	pm2CO2PressMinWafer	325,517
2080	PortID	U1	PortID	751-760, 773,776-779,783-786,801-808
2081	CarrierID	A	CarrierID	755,759, 772-779, 783-790, 806-808
2082	LocationID	A	Carrier LocationID	783-786
2083	PortTransferState	U1	PortTransferState	751-760
2084	CarrierIDStatus	U1	CarrierIDStatus	772-779
2085	CarrierAccessingStatus	U1	CarrierAccessingStatus	783,787-789
2086	SlotMapStatus	U1	SlotMapStatus	783-786
2087	LpAccessMode	U1	LpAccessMode	801-803
2088	LpReservationState	U1	LpReservationState	804-805
2089	LpAssociationState	U1	LpAssociationState	806-808
2090	Capacity	U1	Capacity	772
2091	ContentMap	L	ContentMap	772
2092	Reason	A	Reason	784
2093	SlotMap	L	SlotMap	772
2094	SubstrateCount	U1	SubstrateCount	772
2095	Usage	A	Usage	772
2100	PjID	A	ID of Process Job with state change	700-718
2101	PjNewState	U1	New State of Process Job	700-718
2102	PjPrevState	U1	Previous State of Process Job	700-718
2103	PjPauseEvent	L	PJ PauseEvent	701
2104	PRJobState	U1	PJ PRJobState	701-718
2105	PRMtlNameList	L	PJ PRMtlNameList	701
2106	PRMtlType	B	PJ PRMtlType	701
2107	PRRecipeMethod	U1	PJ PRRecipeMethod	701
2108	RecID	A	PJ RecID	701
2110	CjID	A	ID of Control Job with state change	720-733
2111	CjNewState	U1	New State of Control Job	720-733
2112	CjPrevState	U1	Previous State of Control Job	720-733
2113	CarrierInputSpec	L	CJ CarrierInputSpec	721
2114	CurrentPRJob	L	CJ CurrentPRJob	721
2115	CjPauseEvent	L	CJ PauseEvent	721

2116	ProcessingCtrlSpec	L	CJ ProcessingCtrlSpec	721
2117	ProcessOrderMgmt	U1	CJ ProcessOrderMgmt	721
2118	StartMethod	BO OL	CJ StartMethod	721
2120	pm1HpcDiwFlowMaxPrevStep	F4	Process module 1 HPC Diw Flow maximum previous step	515
2121	pm1HpcDiwFlowAvrgPrevStep	F4	Process module 1 HPC Diw Flow average previous step	515
2122	pm1HpcDiwFlowMinPrevStep	F4	Process module 1 HPC Diw Flow minimum previous step	515
2123	pm1HpcDiwFlowMaxWafer	F4	Process module 1 HPC Diw Flow maximum last wafer	213
2124	pm1HpcDiwFlowAvrgWafer	F4	Process module 1 HPC Diw Flow average last wafer	213
2125	pm1HpcDiwFlowMinWafer	F4	Process module 1 HPC Diw Flow minimum last wafer	213
2126	pm2HpcDiwFlowMaxPrevStep	F4	Process module 2 HPC Diw Flow maximum previous step	517
2127	pm2HpcDiwFlowAvrgPrevStep	F4	Process module 2 HPC Diw Flow average previous step	517
2128	pm2HpcDiwFlowMinPrevStep	F4	Process module 2 HPC Diw Flow minimum previous step	517
2129	pm2HpcDiwFlowMaxWafer	F4	Process module 2 HPC Diw Flow maximum last wafer	313
2130	pm2HpcDiwFlowAvrgWafer	F4	Process module 2 HPC Diw Flow average last wafer	313
2131	pm2HpcDiwFlowMinWafer	F4	Process module 2 HPC Diw Flow minimum last wafer	313
2132	pm1HpcN2FlowMaxPrevStep	F4	Process module 1 HPC N2 Flow maximum previous step	515
2133	pm1HpcN2FlowAvrgPrevStep	F4	Process module 1 HPC N2 Flow average previous step	515
2134	pm1HpcN2FlowMinPrevStep	F4	Process module 1 HPC N2 Flow minimum previous step	515
2135	pm1HpcN2FlowMaxWafer	F4	Process module 1 HPC N2 Flow maximum last wafer	213
2136	pm1HpcN2FlowAvrgWafer	F4	Process module 1 HPC N2 Flow average last wafer	213
2137	pm1HpcN2FlowMinWafer	F4	Process module 1 HPC N2 Flow minimum last wafer	213
2138	pm2HpcN2FlowMaxPrevStep	F4	Process module 2 HPC N2 Flow maximum previous step	517
2139	pm2HpcN2FlowAvrgPrevStep	F4	Process module 2 HPC N2 Flow average previous step	517
2140	pm2HpcN2FlowMinPrevStep	F4	Process module 2 HPC N2 Flow minimum previous step	517
2141	pm2HpcN2FlowMaxWafer	F4	Process module 2 HPC N2 Flow maximum last wafer	313
2142	pm2HpcN2FlowAvrgWafer	F4	Process module 2 HPC N2 Flow average last wafer	313
2143	pm2HpcN2FlowMinWafer	F4	Process module 2 HPC N2 Flow minimum last wafer	313
2144	pm1BemFlowMaxPrevStep	F4	Process module 1 Bevel Etch Module medium flow maximum previous step	519

2145	pm1BemFlowAvrgPrevStep	F4	Process module 1 Bevel Etch Module medium flow average previous step	519
2146	pm1BemFlowMinPrevStep	F4	Process module 1 Bevel Etch Module medium flow minimum previous step	519
2147	pm1BemMed1FlowMaxWafer	F4	Process module 1 Bevel Etch Module medium 1 flow maximum last wafer	213
2148	pm1BemMed1FlowAvrgWafer	F4	Process module 1 Bevel Etch Module medium 1 flow average last wafer	213
2149	pm1BemMed1FlowMinWafer	F4	Process module 1 Bevel Etch Module medium 1 flow minimum last wafer	213
2150	pm1BemMed2FlowMaxWafer	F4	Process module 1 Bevel Etch Module medium 2 flow maximum last wafer	213
2151	pm1BemMed2FlowAvrgWafer	F4	Process module 1 Bevel Etch Module medium 2 flow average last wafer	213
2152	pm1Bem Med2FlowMinWafer	F4	Process module 1 Bevel Etch Module medium 2 flow minimum last wafer	213
2153	pm1BemMed3FlowMaxWafer	F4	Process module 1 Bevel Etch Module medium 3 flow maximum last wafer	213
2154	pm1BemMed3FlowAvrgWafer	F4	Process module 1 Bevel Etch Module medium 3 flow average last wafer	213
2155	pm1Bem Med3FlowMinWafer	F4	Process module 1 Bevel Etch Module medium 3 flow minimum last wafer	213
2156	pm1BemMed4FlowMaxWafer	F4	Process module 1 Bevel Etch Module medium 4 flow maximum last wafer	213
2157	pm1Bem Med4FlowAvrgWafer	F4	Process module 1 Bevel Etch Module medium 4 flow average last wafer	213
2158	pm1Bem Med4FlowMinWafer	F4	Process module 1 Bevel Etch Module medium 4 flow minimum last wafer	213
2159	pm1BemFlowMaxPrevStep	F4	Process module 1 Bevel Etch Module medium flow maximum previous step	521
2160	pm1BemFlowAvrgPrevStep	F4	Process module 1 Bevel Etch Module medium flow average previous step	521
2161	pm1BemFlowMinPrevStep	F4	Process module 1 Bevel Etch Module medium flow minimum previous step	521
2162	pm2BemMed1FlowMaxWafer	F4	Process module 2 Bevel Etch Module medium 1 flow maximum last wafer	313
2163	pm2BemMed1FlowAvrgWafer	F4	Process module 2 Bevel Etch Module medium 1 flow average	313

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2164	pm2BemMed1FlowMinWafer	F4	Process module 2 Bevel Etch Module medium 1 flow minimum last wafer	313
2165	pm2BemMed2FlowMaxWafer	F4	Process module 2 Bevel Etch Module medium flow maximum last wafer	313
2166	pm2Bem Med2FlowAvrgWafer	F4	Process module 2 Bevel Etch Module medium 2 flow average last wafer	313
2167	pm2BemMed2FlowMinWafer	F4	Process module 2 Bevel Etch Module medium 2 flow minimum last wafer	313
2168	pm2BemMed3FlowMaxWafer	F4	Process module 2 Bevel Etch Module medium 3 flow maximum last wafer	313
2169	pm2BemMed3FlowAvrgWafer	F4	Process module 2 Bevel Etch Module medium 3 flow average last wafer	313
2170	pm2BemMed3FlowMinWafer	F4	Process module 2 Bevel Etch Module medium 3 flow minimum last wafer	313
2171	pm2BemMed4FlowMaxWafer	F4	Process module 2 Bevel Etch Module medium 4 flow maximum last wafer	313
2172	pm2BemMed4FlowAvrgWafer	F4	Process module 2 Bevel Etch Module medium 4 flow average last wafer	313
2173	pm2BemMed4FlowMinWafer	F4	Process module 2 Bevel Etch Module medium 4 flow minimum last wafer	313
2174	pm1LowFlowMedFlowMaxPrevStep	F4	Process module 1 LowFlow Module medium x maximum previous step	523
2175	pm1LowFlowMedFlowAvrPrevStep	F4	Process module 1 LowFlow Module medium x average previous step	523
2176	pm1LowFlowMedFlowMinPrevStep	F4	Process module 1 LowFlow Module medium x minimum previous step	523
2177	pm2LowFlowMedFlowMaxPrevStep	F4	Process module 2 LowFlow Module medium x maximum previous step	525
2178	pm2LowFlowMedFlowAvrPrevStep	F4	Process module 2 LowFlow Module medium x average previous step	525
2179	pm2LowFlowMedFlowMinPrevStep	F4	Process module 2 LowFlow Module medium x minimum previous step	525
2180	pm1LowFlowMedFlowMaxWafer	F4	Process module 1 LowFlow Module medium x flow maximum last wafer	523
2181	pm1LowFlowMedFlowAvrWafer	F4	Process module 1 LowFlow Module medium x flow average last wafer	523
2182	pm1LowFlowMedFlowMinWafer	F4	Process module 1 LowFlow	523

			Module medium x flow minimum last wafer	
2183	pm2LowFlowMedFlowMaxWafer	F4	Process module 2 LowFlow Module medium x flow maximum last wafer	525
2184	pm2LowFlowMedFlowAvrWafer	F4	Process module 2 LowFlow Module medium x flow average last wafer	525
2185	pm2LowFlowMedFlowMinWafer	F4	Process module 2 LowFlow Module medium x flow minimum last wafer	525
2200	SubstDestination	A	Destination SubstLocID	850-859
2201	SubstID	A	Substrate ID of wafer/slot	850-859
2202	SubstSubstLocID	A	Current SubstLocID	850-859
2203	SubsLotID	A	Lot ID of wafer/slot	850-859
2204	SubsMtrlStatus	U1	Material status	850-859
2205	SubstProcState	U1	Process state	850-859
2206	SubstSource	A	Source SubstLocID	850-859
2207	SubstState	U1	Substrate state	850-859
2208	SubstLocID	A	Substrate location	860-861
2209	SubstLocState	U1	Substate state	860-861
2210	SubstLocSubstID	A	Substate ID of wafer/slot at location	860-861
2211	SubstHistory	L	Substrate history	850,860-861
2212	pm1CurrWaferSubstId	A	Process module 1 current Substrate ID	212,213
2213	pm2CurrWaferSubstId	A	Process module 2 current Substrate ID	312,313
2301	atms1PreThickness	F4	atms1PreThickness	510
2302	atms1PostThickness	F4	atms1PostThickness	510
2303	atms1EtchedAmount	F4	atms1EtchedAmount	510
2304	atms1ChcEtchRate	F4	atms1ChcEtchRate	510
2305	atms1CalcEtchRate	F4	atms1CalcEtchRate	510
2306	atms1CalcDispTime	F4	atms1CalcDispTime	510
2307	atms1Slot	U1	atms1Slot	510
2308	atms1CarrierId	A	atms1CarrierId	510
2309	atms1MeasurementType	A	atms1MeasurementType	510
2310	atms1NumOfDatapoints	U8	atms1NumOfDatapoints	510
2311	atms1MeanThickness	F4	atms1MeanThickness	510
2312	atms1MinThickness	F4	atms1MinThickness	510
2313	atms1MaxThickness	F4	atms1MaxThickness	510
2314	atms1StandardDeviation	F4	atms1StandardDeviation	510
2315	atms1RecipeName	A	atms1RecipeName	510
2316	atms1LotId	A	atms1LotId	510
2317	atms1WaferId	A	atms1WaferId	510
2318	atms1PointPreThickness	L	atms1PointPreThickness	510
2319	atms1PointPreEncPosition	L	atms1PointPreEncPosition	510
2320	atms1PointPreValueMax	L	atms1PointPreValueMax	510
2321	atms1PointPreValueMean	L	atms1PointPreValueMean	510

2322	atms1PointPreValueMin	L	atms1PointPreValueMin	510
2323	atms1PointPreStdDev	L	atms1PointPreStdDev	510
2324	atms1PointPostThickness	L	atms1PointPostThickness	510
2325	atms1PointPostEncPosition	L	atms1PointPostEncPosition	510
2326	atms1PointPostValueMax	L	atms1PointPostValueMax	510
2327	atms1PointPostValueMean	L	atms1PointPostValueMean	510
2328	atms1PointPostValueMin	L	atms1PointPostValueMin	510
2329	atms1PointPostStdDev	L	atms1PointPostStdDev	510
2330	atms1SamplesPerPoint	L	atms1SamplesPerPoint	510
2401	atms2PreThickness	F4	atms2PreThickness	511
2402	atms2PostThickness	F4	atms2PostThickness	511
2403	atms2EtchedAmount	F4	atms2EtchedAmount	511
2404	atms2ChcEtchRate	F4	atms2ChcEtchRate	511
2405	atms2CalcEtchRate	F4	atms2CalcEtchRate	511
2406	atms2CalcDispTime	F4	atms2CalcDispTime	511
2407	atms2Slot	U1	atms2Slot	511
2408	atms2CarrierId	A	atms2CarrierId	511
2409	atms2MeasurementType	A	atms2MeasurementType	511
2410	atms2NumOfDatapoints	U8	atms2NumOfDatapoints	511
2411	atms2MeanThickness	F4	atms2MeanThickness	511
2412	atms2MinThickness	F4	atms2MinThickness	511
2413	atms2MaxThickness	F4	atms2MaxThickness	511
2414	atms2StandardDeviation	F4	atms2StandardDeviation	511
2415	atms2RecipeName	A	atms2RecipeName	511
2416	atms2LotId	A	atms2LotId	511
2417	atms2WaferId	A	atms2WaferId	511
2418	atms2PointPreThickness	L	atms2PointPreThickness	511
2419	atms2PointPreEncPosition	L	atms2PointPreEncPosition	511
2420	atms2PointPreValueMax	L	atms2PointPreValueMax	511
2421	atms2PointPreValueMean	L	atms2PointPreValueMean	511
2422	atms2PointPreValueMin	L	atms2PointPreValueMin	511
2423	atms2PointPreStdDev	L	atms2PointPreStdDev	511
2424	atms2PointPostThickness	L	atms2PointPostThickness	511
2425	atms2PointPostEncPosition	L	atms2PointPostEncPosition	511
2426	atms2PointPostValueMax	L	atms2PointPostValueMax	511
2427	atms2PointPostValueMean	L	atms2PointPostValueMean	511
2428	atms2PointPostValueMin	L	atms2PointPostValueMin	511
2429	atms2PointPostStdDev	L	atms2PointPostStdDev	511
2430	atms2SamplesPerPoint	L	atms2SamplesPerPoint	511
2900	PreAlignerId	U8	PreAlignerId	600
3001	Roughness1Slot	U8	Roughness 1 Slot	531
3002	Roughness1CarrierId	A	Roughness 1 Carrier Id	531
3003	Roughness1LotId	A	Roughness 1 Lot Id	531
3004	Roughness1MeasurementType	A	Roughness 1 Measurement Type	531
3005	Roughness1NumOfDatapoints	U8	Roughness 1 Number of datapoints	531
3006	Roughness1StandardDeviation	F4	Roughness 1 Standard Deviation	531

3007	Roughness1RecipeName	A	Roughness 1 Recipe Name	531
3008	Roughness1ScanPoint	L	Roughness 1 Scan Point	531
3009	Roughness1Ra	L	Roughness 1 Roughness index coefficient	531
3101	Roughness2Slot	U8	Roughness 2 Slot	532
3102	Roughness2CarrierId	A	Roughness 2 Carrier Id	532
3103	Roughness2LotId	A	Roughness 2 Lot Id	532
3104	Roughness2MeasurementType	A	Roughness 2 Measurement Type	532
3105	Roughness2NumOfDatapoints	U8	Roughness 2 Number of datapoints	532
3106	Roughness2StandardDeviation	F4	Roughness 2 Standard Deviation	532
3107	Roughness2RecipeName	A	Roughness 2 Recipe Name	532
3108	Roughness2ScanPoint	L	Roughness 2 Scan Point	532
3109	Roughness2Ra	L	Roughness 2 Roughness index coefficient	532

8.2.1.1 Legend DVVALs

a. Valid at any time/event

Status Variables (SVs)

Status variables (SV) always contain valid information

VID	Status Variable	Format	Description
8	AlarmsEnabled	L	List of enabled ALIDs
~~9~~	~~AlarmsSet~~	~~L~~	Not Supported
10	Clock	A	Equipment clock
11	ControlState	U4	Equipments current control state 1 = EquipmentOffline 2 = AttempOnline 3 = HostOffline 4 = OnlineLocal 5 = OnlineRemote
12	EventsEnabled	L	List of collection events that are enabled
13	PreviousControlState	U4	Previous equipment controls state
14	PreviousProcessState	A	Previous equipment process state
15	ProcessState	A	Equipments current process state Valid states: 0 = OFF 1 = ERROR 2 = SERVICE 3 = INIT 4 = NOT INITIALIZED 5 = IDLE 7 = PROCESS SETUP 8 = PROCESS READY 9 = PROCESSING 10 = STOP 12 = ABORT 20 = WAFER REMOVE
16	LastEventID	U4	ID of last triggered collection event
17	SpoolCountActual	A	Not supported
18	SpoolCountTotal	A	Not supported
19	SpoolStartTime	A	Not supported
20	SpoolFullTime	A	Not supported
3100	port1JobPending	BOOLEAN	A job is queued on load port 1
3110	port1MapResult	L	Map result for each slot of cassette 1. 1st list item is slot 1. 1 = FULLSLOT 2 = EMPTYSLOT 3 = CROSSSLOTTED 4 = DOUBLESLOTTED
3120	port1Status	U4	Status of load port 1. 0 = unload requested 1 = load requested 2 = unavailable
3121	port1PendingJobId	A	Job id of pending job on load port 1
3122	port1PendingLotId	A	Lot id of pending job on load port 1
3123	port1PendingPPID	A	PPID of pending job on load port 1
3124	port1PendingCid	A	Carrier id of pending job on load port 1

3130	port1JobId	A	Job id of job on load port 1
3131	port1LotId	A	Lot id of job on load port 1
3132	port1PPID	A	PPID of job on load port 1
3133	port1Cid	A	Carrier id of job on load port 1
3200	port2JobPending	BOOLEAN	A job is queued on load port 2
3210	port2MapResult	L	Map result for each slot of cassette 2. 1st list item is slot 1. 1 = FULLSLOT 2 = EMPTYSLOT 3 = CROSSSLOTTED 4 = DOUBLESLOTTED
3220	port2Status	U4	Status of load port 2. 0 = unload requested 1 = load requested 2 = unavailable
3221	port2PendingJobId	A	Job id of pending job on load port 2
3222	port2PendingLotId	A	Lot id of pending job on load port 2
3223	port2PendingPPID	A	PPID of pending job on load port 2
3224	port2PendingCid	A	Carrier id of pending job on load port 2
3230	port2JobId	A	Job id of job on load port 2
3231	port2LotId	A	Lot id of job on load port 2
3232	port2PPID	A	PPID of job on load port 2
3233	port2Cid	A	Carrier id of job on load port 2
3300	port3JobPending	BOOLEAN	A job is queued on load port 3
3310	port3MapResult	L	Map result for each slot of cassette 2. 1st list item is slot 1. 1 = FULLSLOT 2 = EMPTYSLOT 3 = CROSSSLOTTED 4 = DOUBLESLOTTED
3320	port3Status	U4	Status of load port 3. 0 = unload requested 1 = load requested 2 = unavailable
3321	port3PendingJobId	A	Job id of pending job on load port 3
3322	port3PendingLotId	A	Lot id of pending job on load port 3
3323	port3PendingPPID	A	PPID of pending job on load port 3
3324	port3PendingCid	A	Carrier id of pending job on load port 3
3330	port3JobId	A	Job id of job on load port 3
3331	port3LotId	A	Lot id of job on load port 3
3332	port3PPID	A	PPID of job on load port 3
3333	port3Cid	A	Carrier id of job on load port 3
3400	port4JobPending	BOOLEAN	A job is queued on load port 4
3410	port4MapResult	L	Map result for each slot of cassette 2. 1st list item is slot 1. 1 = FULLSLOT 2 = EMPTYSLOT 3 = CROSSSLOTTED 4 = DOUBLESLOTTED
3420	port4Status	U4	Status of load port 4. 0 = unload requested 1 = load requested

			2 = unavailable
3421	port4PendingJobId	A	Job id of pending job on load port 4
3422	port4PendingLotId	A	Lot id of pending job on load port 4
3423	port4PendingPPID	A	PPID of pending job on load port 4
3424	port4PendingCid	A	Carrier id of pending job on load port 4
3430	port4JobId	A	Job id of job on load port 4
3431	port4LotId	A	Lot id of job on load port 4
3432	port4PPID	A	PPID of job on load port 4
3433	port4Cid	A	Carrier id of job on load port 4
3500	pm1PPExec	A	Pm1 PP executing
3501	pm1N2ChuckFlow	F4	Pm1 N2 chuck flow
3502	pm1N2DryFlow	F4	Pm1 N2 dry flow
3503	pm1Med1Temp	F4	Pm1 medium 1 temperature
3504	pm1Med2Temp	F4	Pm1 medium 2 temperature
3505	pm1Med3Temp	F4	Pm1 medium 3 temperature
3506	pm1Med1Flow	F4	Pm1 medium 1 flow
3507	pm1Med2Flow	F4	Pm1 medium 2 flow
3508	pm1Med3Flow	F4	Pm1 medium 3 flow
3520	pm1DiFlow	F4	Pm1 DI flow
3521	pm1DiwO3Flow	F4	Pm1 DIW O3 flow
3530	pm1ChuckSpeed	F4	Pm1 chuck speed
3531	pm1WaferCount	F4	Pm1 wafer count
3532	pm1HpcDiwFlow	F4	Pm1 HPC Diw flow
3533	pm1HpcN2Flow	F4	Pm1 HPC N2 flow
3534	pm1Exhaust	F4	Pm1 Exhaust
3535	pm1BemMed1Flow	F4	Pm1 BEM medium 1 flow
3536	pm1BemMed2Flow	F4	Pm1 BEM medium 2 flow
3537	pm1BemMed3Flow	F4	Pm1 BEM medium 3 flow
3538	pm1BemMed4Flow	F4	Pm1 BEM medium 4 flow
3539	pm1LowFlowMedFlow	F4	Pm1 LowFlow medium x flow
3540	pm1CO2Flow	F4	Pm1 CO2 Flow
3541	pm1CO2Conductivity	F4	Pm1 CO2 Conductivity
3542	pm1CO2Temp	F4	Pm1 CO2 Temp
3543	pm1CO2Pressure	F4	Pm1 CO2 Pressure
3544	pm1Exhaust1	F4	Pm1 Exhaust 1
3545	pm1Exhaust2	F4	Pm1 Exhaust 2
3550	pm1State	U4	Pm1 status 0 = DISABLED 1 = UNKNOWN 2 = ERROR 3 = INITIALIZING 4 = READY 5 = PREFLUSH 6 = RDY_LOAD 7 = PROCESSING 8 = STOPPING 9 = STOPPED 10 = ABORTING 11 = ABORTED

			12 = RDY_UNLOAD
3700	pm2PPExec	A	Pm2 PP executing
3701	pm2N2ChuckFlow	F4	Pm2 N2 chuck flow
3702	pm2N2DryFlow	F4	Pm2 N2 dry flow
3703	pm2Med1Temp	F4	Pm2 medium 1 temperature
3704	pm2Med2Temp	F4	Pm2 medium 2 temperature
3705	pm2Med3Temp	F4	Pm2 medium 3 temperature
3706	pm2Med1Flow	F4	Pm2 medium 1 flow
3707	pm2Med2Flow	F4	Pm2 medium 2 flow
3708	pm2Med3Flow	F4	Pm2 medium 3 flow
3720	pm2DiFlow	F4	Pm2 DI flow
3721	Pm2DiwO3Flow	F4	Pm2 DIW O3 flow
3730	pm2ChuckSpeed	F4	Pm2 chuck speed
3731	pm2WaferCount	F4	Pm2 wafer count
3732	pm2HpcDiwFlow	F4	Pm2 HPC Diw flow
3733	pm2HpcN2Flow	F4	Pm2 HPC N2 flow
3734	pm2Exhaust	F4	Pm2 Exhaust
3735	pm2BemMed1Flow	F4	Pm2 BEM medium 1 flow
3736	pm2BemMed2Flow	F4	Pm2 BEM medium 2 flow
3737	pm2BemMed3Flow	F4	Pm2 BEM medium 3 flow
3738	pm2BemMed4Flow	F4	Pm2 BEM medium 4 flow
3739	pm2LowFlowMedFlow	F4	Pm2 LowFlow medium x flow
3740	pm2CO2Flow	F4	Pm2 CO2 Flow
3741	pm2CO2Conductivity	F4	Pm2 CO2 Conductivity
3742	pm2CO2Temp	F4	Pm2 CO2 Temp
3743	pm2CO2Pressure	F4	Pm2 CO2 Pressure
3744	pm2Exhaust1	F4	Pm2 Exhaust 1
3745	pm2Exhaust2	F4	Pm2 Exhaust 2
3750	pm2State	U4	Pm2 status 0 = DISABLED 1 = UNKNOWN 2 = ERROR 3 = INITIALIZING 4 = READY 5 = PREFLUSH 6 = RDY_LOAD 7 = PROCESSING 8 = STOPPING 9 = STOPPED 10 = ABORTING 11 = ABORTED 12 = RDY_UNLOAD
3900	supplyMed1Flow	F4	Supply Med1Flow
3901	supplyMed2Flow	F4	Supply Med2Flow
3902	supplyMed3Flow	F4	Supply Med3Flow
3903	supplyMed4Flow	F4	Supply Med4Flow
3904	supplyDiwFlow	F4	Supply DiwFlow
3905	supplyDiwO3Flow	F4	Supply DiwO3Flow
3906	supplyMed1Temp	F4	Supply Med1Temp

3907	supplyMed2Temp	F4	Supply Med2Temp
3908	supplyMed3Temp	F4	Supply Med3Temp
3909	supplyMed4Temp	F4	Supply Med4Temp
3910	supplyDiwTemp	F4	Supply DiwTemp
3911	supplyDiwO3Temp	F4	Supply DiwO3Temp
3912	supplyN2Temp	F4	Supply N2Temp
3913	supplyN2CoilTemp	F4	Supply N2CoilTemp
3914	supplyMed1HeatPurgeFlow	F4	Supply Med1HeatPurgeFlow
3915	supplyMed2HeatPurgeFlow	F4	Supply Med2HeatPurgeFlow
3916	supplyMed3HeatPurgeFlow	F4	Supply Med3HeatPurgeFlow
3917	supplyMed4HeatPurgeFlow	F4	Supply Med4HeatPurgeFlow
3918	supplyMed1HeatPurgePressure	F4	Supply Med1HeatPurgePressure
3919	supplyMed2HeatPurgePressure	F4	Supply Med2HeatPurgePressure
3920	supplyMed3HeatPurgePressure	F4	Supply Med3HeatPurgePressure
3921	supplyMed4HeatPurgePressure	F4	Supply Med4HeatPurgePressure
3922	supplyMed1CoilTemp	F4	Supply Med1CoilTemp
3923	supplyMed2CoilTemp	F4	Supply Med2CoilTemp
3924	supplyMed3CoilTemp	F4	Supply Med3CoilTemp
3925	supplyMed4CoilTemp	F4	Supply Med4CoilTemp
3926	supplyMed1LiquidTemp	F4	Supply Med1LiquidTemp
3927	supplyMed2LiquidTemp	F4	Supply Med2LiquidTemp
3928	supplyMed3LiquidTemp	F4	Supply Med3LiquidTemp
3929	supplyMed4LiquidTemp	F4	Supply Med4LiquidTemp
4000	facSupplyCdaPressure	F4	Facility Supply CdaPressure
4001	facSupplyN2PressureLeft	F4	Facility Supply N2PressureLeft
4002	facSupplyN2PressureRight	F4	Facility Supply N2PressureRight
4003	facSupplyVacuumPressure	F4	Facility Supply VacuumPressure
4004	facSupplyDiwPressure	F4	Facility Supply DiwPressure
4303	lightTowerStatus	U4	Light tower status
4304	buzzerStatus	U4	Buzzer status
4305	portIdLastMapped	U4	Load port id of the cassette that has been mapped most recently
4306	mapResultLastMap	L	Map result of last map sequence for each slot of cassette. 1st list item is slot 1. 1 = FULLSLOT 2 = EMPTYSLOT 3 = CROSSSLOTTED 4 = DOUBLESLOTTED
4310	med1EtchRate	F4	Medium 1 etch rate
4311	med2EtchRate	F4	Medium 2 etch rate
4312	med3EtchRate	F4	Medium 3 etch rate
4315	med1ElapsedTime	I8	Medium 1 elapsed time
4316	med2ElapsedTime	I8	Medium 2 elapsed time
4317	med3ElapsedTime	I8	Medium 3 elapsed time
4320	med1BathLifeTimeRem	I8	Medium 1 bath life time remaining
4321	med2BathLifeTimeRem	I8	Medium 2 bath life time remaining
4322	med3BathLifeTimeRem	I8	Medium 3 bath life time remaining

4325	med1BathLifeTime	I8	Medium 1 bath life time
4326	med2BathLifeTime	I8	Medium 2 bath life time
4327	med3BathLifeTime	I8	Medium 3 bath life time
4330	med1ElapsedTimeRem	I8	Medium 1 elapsed time remaining
4331	med2ElapsedTimeRem	I8	Medium 2 elapsed time remaining
4332	med3ElapsedTimeRem	I8	Medium 3 elapsed time remaining
4350	chc1State	U4	Status of chemistry cabinet 1. 0 = ERROR 1 = UNKNOWN 2 = OFF 3 = WARNING 4 = INITIALIZING 5 = EMPTYING 6 = EMPTY 7 = FILLING 8 = FULL 9 = HEATING 10 = READY 11 = PROCESSING
4351	chc2State	U4	Status of chemistry cabinet 2. 0 = ERROR 1 = UNKNOWN 2 = OFF 3 = WARNING 4 = INITIALIZING 5 = EMPTYING 6 = EMPTY 7 = FILLING 8 = FULL 9 = HEATING 10 = READY 11 = PROCESSING
4352	chc3State	U4	Status of chemistry cabinet 3. 0 = ERROR 1 = UNKNOWN 2 = OFF 3 = WARNING 4 = INITIALIZING 5 = EMPTYING 6 = EMPTY 7 = FILLING 8 = FULL 9 = HEATING 10 = READY 11 = PROCESSING
4355	waferCountChc1Actual	I8	Chemistry cabinet 1 actual wafer count
4356	waferCountChc2Actual	I8	Chemistry cabinet 2 actual wafer count
4357	waferCountChc3Actual	I8	Chemistry cabinet 3 actual wafer count
4360	chc1RefillStatus	U4	Chemistry cabinet 1 refill status. 0 = not refilling 1 = refilling
4361	chc2RefillStatus	U4	Chemistry cabinet 2 refill status.

			0 = not refilling 1 = refilling
4362	chc3RefillStatus	U4	Chemistry cabinet 3 refill status. 0 = not refilling 1 = refilling
4365	waferCountChc1Total	I8	Chemistry cabinet 1 total wafer count
4366	waferCountChc2Total	I8	Chemistry cabinet 2 total wafer count
4367	waferCountChc3Total	I8	Chemistry cabinet 3 total wafer count
4400	med1Comp1Conc	F4	Medium 1 componet 1 concentration in %
4401	med1Comp2Conc	F4	Medium 1 component 2 concentration in %
4402	med1Comp3Conc	F4	Medium 1 component 3 concentration in %
4403	med1Comp4Conc	F4	Medium 1 component 4 concentration in %
4404	med1Comp5Conc	F4	Medium 1 component 5 concentration in %
4405	med2Comp1Conc	F4	Medium 2 component 1 concentration in %
4406	med2Comp2Conc	F4	Medium 2 component 2 concentration in %
4407	med2Comp3Conc	F4	Medium 2 component 3 concentration in %
4408	med2Comp4Conc	F4	Medium 2 component 4 concentration in %
4409	med2Comp5Conc	F4	Medium 2 component 5 concentration in %
4410	med3Comp1Conc	F4	Medium 3 component 1 concentration in %
4411	med3Comp2Conc	F4	Medium 3 component 2 concentration in %
4412	med3Comp3Conc	F4	Medium 3 component 3 concentration in %
4413	med3Comp4Conc	F4	Medium 3 component 4 concentration in %
4414	med3Comp5Conc	F4	Medium 3 component 5 concentration in %
4450	diwO3Concentration	F4	DiwO3 concentration in %
4451	diwO3ConcChannel1	F4	DiwO3 concentration Channel 1
4452	diwO3ConcChannel2	F4	DiwO3 concentration Channel 2
4453	diwO3ConcChannel3	F4	DiwO3 concentration Channel 3
4454	diwO3ConcChannel4	F4	DiwO3 concentration Channel 4
4455	diwO3ConcChannel5	F4	DiwO3 concentration Channel 5
4456	diwO3ConcChannel6	F4	DiwO3 concentration Channel 6
4999	CarrierLocationMatrix	L	CarrierLocationMatrix
5000	Lp1AccessMode	U1	Lp1AccessMode
5001	Lp1CarrierID	A	Lp1CarrierID
5002	Lp1CarrierIDatFIMS	A	Lp1CarrierIDatFIMS
5003	Lp1LocationID	A	Lp1LocationID
5004	Lp1LocationIDatFIMS	A	Lp1LocationIDatFIMS
5005	Lp1PortAssociationState	U1	Lp1PortAssociationState

5006	Lp1PortID	U1	Lp1PortID
5007	Lp1PortReservationState	U1	Lp1PortReservationState
5008	Lp1PortStateInfo	L	Lp1PortStateInfo
5009	Lp1PortTransferState	U1	Lp1PortTransferState
5010	Lp2AccessMode	U1	Lp2AccessMode
5011	Lp2CarrierID	A	Lp2CarrierID
5012	Lp2CarrierIDatFIMS	A	Lp2CarrierIDatFIMS
5013	Lp2LocationID	A	Lp2LocationID
5014	Lp2LocationIDatFIMS	A	Lp2LocationIDatFIMS
5015	Lp2PortAssociationState	U1	Lp2PortAssociationState
5016	Lp2PortID	U1	Lp2PortID
5017	Lp2PortReservationState	U1	Lp2PortReservationState
5018	Lp2PortStateInfo	L	Lp2PortStateInfo
5019	Lp2PortTransferState	U1	Lp2PortTransferState
5020	Lp3AccessMode	U1	Lp3AccessMode
5021	Lp3CarrierID	A	Lp3CarrierID
5022	Lp3CarrierIDatFIMS	A	Lp3CarrierIDatFIMS
5023	Lp3LocationID	A	Lp3LocationID
5024	Lp3LocationIDatFIMS	A	Lp3LocationIDatFIMS
5025	Lp3PortAssociationState	U1	Lp3PortAssociationState
5026	Lp3PortID	U1	Lp3PortID
5027	Lp3PortReservationState	U1	Lp3PortReservationState
5028	Lp3PortStateInfo	L	Lp3PortStateInfo
5029	Lp3PortTransferState	U1	Lp3PortTransferState
5030	Lp4AccessMode	U1	Lp4AccessMode
5031	Lp4CarrierID	A	Lp4CarrierID
5032	Lp4CarrierIDatFIMS	A	Lp4CarrierIDatFIMS
5033	Lp4LocationID	A	Lp4LocationID
5034	Lp4LocationIDatFIMS	A	Lp4LocationIDatFIMS
5035	Lp4PortAssociationState	U1	Lp4PortAssociationState
5036	Lp4PortID	U1	Lp4PortID
5037	Lp4PortReservationState	U1	Lp4PortReservationState
5038	Lp4PortStateInfo	L	Lp4PortStateInfo
5039	Lp4PortTransferState	U1	Lp4PortTransferState
5100	SubstLocPM1Subst	L	SubstLocPM1Subst
5101	SubstLocPM2Subst	L	SubstLocPM2Subst
5110	SubstLocROBOTSubst	L	SubstLocROBOTSubst
5120	SubstLocLECO1Subst	L	SubstLocLECO1Subst
5121	SubstLocUECO1Subst	L	SubstLocUECO1Subst
5122	SubstLocLECO2Subst	L	SubstLocLECO2Subst
5123	SubstLocUECO2Subst	L	SubstLocUECO2Subst
5130	SubstLocPA1Subst	L	SubstLocPrealigner1Subst
5131	SubstLocPA2Subst	L	SubstLocPrealigner2Subst
5140	SubstLocATMSI1Subst	L	SubstLocATMSI1Subst
5141	SubstLocATMSI2Subst	L	SubstLocATMSI2Subst
5300	CjQueueAvailableSpace	U1	CjQueueAvailableSpace

5301	QueuedCJobs	L	QueuedCJobs

8.3 Remote Commands

All remote control commands to be processed by the equipment are sent by the host through the S2/F41-S2/F42 SECS transaction. If the CONTROL state of the equipment is not REMOTE, any remote control command that affects current processing on the equipment will be rejected.

Command names are ASCII strings up to 40 characters long. Command and parameter names are case sensitive. Special characters are not allowed.

List of remote commands

The following remote commands are available on the equipment:

Command Name	Definition
LOCAL	Equipment changes control state to LOCAL (if current state is REMOTE)
REMOTE	Equipment changes control state to REMOTE (if current state is LOCAL)
MAP	Map a cassette
PPSELECT	Select a process recipe
START	Start processing

Remote command parameters (CP)

Command parameters of type [A] (ASCII) are case sensitive.

Command LOCAL

Triggers transition of equipment control state from REMOTE to LOCAL if the equipment is in REMOTE state.

	CPNAME	Format	Description
NA	NONE	-	No parameters

Command REMOTE

Triggers transition of equipment control state from LOCAL to REMOTE if the equipment is in LOCAL state.

	CPNAME	Format	Description
NA	NONE	-	No parameters

Command PPSELECT

This command selects a recipe for the carrier of the load port defined in parameter PORTID.

	CPNAME	Format	Description
1	PPID	A	Name of the process program (recipe)
2	PORTID	U2	Id of the load port to assign the process program. 1 = Load port 1, 2 = Load port 2, 3 = Load port 3, 4 = Load port 4
3	CARRIERID	A	Carrier id
4	LOTID	A	Lot id
5	JOBID	A	Process job id

Note: Command Parameter *PPID* (recipe name is **<u>case sensitive</u>**). The parameter PPID must exactly match the PPID of the equipment.
E.g.: TESTRECIPE (PPID host) will not match Testrecipe (PPID equipment).

Command MAP

This command triggers the mappig of a load port.

CPNAME	Format	Description
PORTID	I2	Id of the load port to map. 1 = Load port 1, 2 = Load port 2, 3 = Load port 3, 4 = Load port 4

Command START

The START Command initiates processing of a lot on the equipment. Process will be initiated from the load port defined in parameter PORTID.

CPNAME	Format	Description
PORTID	I2	Id of the load port to start processing from. 1 = Load port 1, 2 = Load port 2, 3 = Load port 3, 4 = Load port 4

8.4 Equipment Constants

The following table lists all Equipment Constants (EC) that can be set by the host.

ECID	Equipment Constant	Format	Description
4	EstablishCommunicationsTimeout	U2	The length of time in seconds of the intervall between between attempts to send S1F13 when establishing communications
5	TimeFormat	A	Determines whether a 12-byte or 16-byte value (required for Year 2000 compliance) for the current date/time will be used (older Host interfaces may not be upgraded to use the new 16-byte format). The 16-byte value includes a four-digit “Year” field and a two-digit centisecond (one hundredth of a second) field. SECS messages which use the STIME or TIME data item will then use the appropriate 12-byte or 16- byte value specified by GemTimeFormat. 0 = 12-byte format 1 = 16-byte format default=1

8.5 Alarms

Alarm Overview

Alarm ID	Group ID	Group Name	Description
12	1	General	Watchdog error
13	1	General	Handling error
14	1	General	Stop error
15	1	General	Abort error
16	1	General	Process Module 1 error
17	1	General	Process Module 2 error
18	1	General	Tool error
19	1	General	Safety error
501	50	System	System initialization timeout error
502	50	System	System initialization attempt with invalid parameter set.
1001	100	Handling	Initialization timeout error
1002	100	Handling	Process module 1 wafer unload error
1003	100	Handling	Wafer mapping time out error
1004	100	Handling	Process module 1 wafer on chuck error
1005	100	Handling	Cassette 1 removed during process
1006	100	Handling	PM1 wafer handling sequence timeout error
1007	100	Handling	PM2 wafer handling sequence timeout error
1008	100	Handling	Wafer handling timeout error
1009	100	Handling	Robot could not get wafer from UECO1, please check robot's vacuum.
10010	100	Handling	Wafer detection on LECO1 error, please check robot's vacuum.
10011	100	Handling	Wafer detection on UECO1 error, please check ECO's pins closing action and/or chuck's unload position and/or ECO's on chuck position.
10012	100	Handling	Robot could not get wafer from UECO2, please check robot's vacuum.
10013	100	Handling	Wafer detection on LECO2 error, please check robot's vacuum.
10014	100	Handling	Wafer detection on UECO2 error, please check ECO's pins closing action and/or chuck's unload position and/or ECO's on chuck position.
10015	100	Handling	Robot could not get wafer from Loadport 1, please check robot's vacuum and/or loadport station teaching.
10016	100	Handling	Robot could not get wafer from Loadport 2, please check robot's vacuum and/or loadport station teaching.
10017	100	Handling	Robot could not get wafer from Loadport 3, please check robot's vacuum and/or loadport station teaching.
10018	100	Handling	Robot could not get wafer from Loadport 4, please check robot's vacuum and/or loadport station teaching.
10019	100	Handling	Robot could not get wafer from LECO1, please check robot's vacuum.
10020	100	Handling	Robot could not get wafer from LECO2, please check robot's vacuum.
10021	100	Handling	Robot Bernoulli flow ramp down timeout.
10022	100	Handling	Robot Bernoulli flow ramp up timeout.
10023	100	Handling	LECO1 Bernoulli flow ramp down timeout.
10024	100	Handling	LECO1 Bernoulli flow ramp up timeout.
10025	100	Handling	UECO1 Bernoulli flow ramp down timeout.
10026	100	Handling	UECO1 Bernoulli flow ramp up timeout.

10027	100	Handling	**LECO2 Bernoulli flow ramp down timeout.**
10028	100	Handling	**LECO2 Bernoulli flow ramp up timeout.**
10029	100	Handling	**UECO2 Bernoulli flow ramp down timeout.**
10030	100	Handling	**UECO2 Bernoulli flow ramp up timeout.**
10031	100	Handling	**Cassette 2 removed during process**
10032	100	Handling	**Cassette 2 removed during process**
10033	100	Handling	**Cassette 3 removed during process**
10034	100	Handling	**Cassette 4 removed during process**
10035	100	Handling	**Handling Configuration Mismatch**
10036	100	Handling	**Robot Bernoulli flow ramp down timeout.**
10037	100	Handling	**Robot Bernoulli flow ramp up timeout.**
10038	100	Handling	**Load Port not available**
10039	100	Handling	**Unload Port not available**
2001	200	PM1	**Process module 1 medium dispenser not mounted**
2002	200	PM1	**Process module 1 DIW/N2 dispenser module not mounted**
2003	200	PM1	**Process module 1 wafer on chuck detection failure**
2004	200	PM1	**Process module 1 no wafer on chuck detection failure**
2005	200	PM1	**Process module 1 medium dispenser not over chuck**
2006	200	PM1	**Process module 1 chuck N2 flow not reached**
2007	200	PM1	**Process module 1 chuck speed not reached**
2008	200	PM1	**Process module 1 Level 1 sequence timeout**
2009	200	PM1	**Process module 1 Level 2 sequence timeout**
20010	200	PM1	**Process module 1 Level 3 sequence timeout**
20011	200	PM1	**Process module 1 Level 4 sequence timeout**
20012	200	PM1	**Process module 1 Level Diw/N2 sequence timeout**
20013	200	PM1	**Process module 1 Medium dispenser error**
20014	200	PM1	**Process module 1 N2 dispenser error**
20015	200	PM1	**Process module 1 Diw dispenser error**
20016	200	PM1	**Process module 1 Bem Axis X error**
20017	200	PM1	**Process module 1 Bem Axis Z error**
20018	200	PM1	**Process module 1 Elevating Unit error**
20019	200	PM1	**Process module 1 Cdr error**
2101	210	PM1 MedDisp	**Process module 1 medium dispenser motion control error**
2102	210	PM1 MedDisp	**Process module 1 medium dispenser motor module power supply error**
2103	210	PM1 MedDisp	**Process module 1 medium dispenser motor stall error**
2104	210	PM1 MedDisp	**Process module 1 medium dispenser overtemperature error**
2105	210	PM1 MedDisp	**Process module 1 medium dispenser motor current error**
2106	210	PM1 MedDisp	**Process module 1 medium dispenser motor overcurrent error**
2107	210	PM1 MedDisp	**Process module 1 medium dispenser positioning error**
2108	210	PM1 MedDisp	**Process module 1 medium dispenser belt broken**
2201	220	PM1 N2Disp	**Process module 1 N2 dispenser motion control error**
2202	220	PM1 N2Disp	**Process module 1 N2 dispenser motor module power supply error**
2203	220	PM1 N2Disp	**Process module 1 N2 dispenser motor stall error**
2204	220	PM1 N2Disp	**Process module 1 N2 dispenser overtemperature error**
2205	220	PM1 N2Disp	**Process module 1 N2 dispenser motor current error**
2206	220	PM1 N2Disp	**Process module 1 N2 dispenser motor overcurrent error**

2207	220	PM1 N2Disp	Process module 1 N2 dispenser positioning error
2208	220	PM1 N2Disp	Process module 1 N2 dispenser belt broken
2301	230	PM1 DiwDisp	Process module 1 Diw dispenser motion control error
2302	230	PM1 DiwDisp	Process module 1 Diw dispenser motor module power supply error
2303	230	PM1 DiwDisp	Process module 1 Diw dispenser motor stall error
2304	230	PM1 DiwDisp	Process module 1 Diw dispenser overtemperature error
2305	230	PM1 DiwDisp	Process module 1 Diw dispenser motor current error
2306	230	PM1 DiwDisp	Process module 1 Diw dispenser motor overcurrent error
2307	230	PM1 DiwDisp	Process module 1 Diw dispenser positioning error
2308	230	PM1 DiwDisp	Process module 1 Diw dispenser belt broken
2401	240	PM1 Elevating Unit	Process module 1 elevating unit amplifier error
2402	240	PM1 Elevating Unit	Process module 1 elevating unit load/unload level positioning error
2403	240	PM1 Elevating Unit	Process module 1 elevating unit level 4 positioning error
2404	240	PM1 Elevating Unit	Process module 1 elevating unit level 3 positioning error
2405	240	PM1 Elevating Unit	Process module 1 elevating unit level 2 positioning error
2406	240	PM1 Elevating Unit	Process module 1 elevating unit level 1 positioning error
2407	240	PM1 Elevating Unit	Process module 1 elevating unit initialization timeout error
2408	240	PM1 Elevating Unit	Process module 1 elevating unit positioning error at pre-flush level
2501	250	PM1 Chuck Drive	Process module 1 chuck drive amplifier error
2502	250	PM1 Chuck Drive	Process module 1 chuck drive belt broken error
2503	250	PM1 Chuck Drive	Process module 1 chuck drive release brake error
2504	250	PM1 Chuck Drive	Process module 1 chuck drive apply brake error
2505	250	PM1 Chuck Drive	Process module 1 chuck drive initialization timeout error
2506	250	PM1 Chuck Drive	Process module 1 chuck drive cloase timeout error
2507	250	PM1 Chuck Drive	Process module 1 chuck drive open unload partial timeout error
2508	250	PM1 Chuck Drive	Process module 1 chuck drive open unload full timeout error
2509	250	PM1 Chuck Drive	Process module 1 chuck drive open load timeout error
25010	250	PM1 Chuck Drive	Process module 1 brake feedback error
2601	260	PM1 Facility	Process module 1 exhaust 1 low limit error
2602	260	PM1 Facility	Process module 1 exhaust 1 high limit error
2603	260	PM1 Facility	Process module 1 exhaust 2 low limit error
2604	260	PM1 Facility	Process module 1 exhaust 2 high limit error
2701	270	PM1 Bevel Etch	Process module 1 Bevel Etch module disabled
2702	270	PM1 Bevel Etch	Process module 1 Bevel Etch module axis error
2703	270	PM1 Bevel Etch	Process module 1 Bevel Etch module Level 1 position switch not reached
2704	270	PM1 Bevel Etch	Process module 1 Bevel Etch module Level 2 position switch not reached
2705	270	PM1 Bevel Etch	Process module 1 Bevel Etch module Level 3 position switch not reached
2706	270	PM1 Bevel Etch	Process module 1 Bevel Etch module Level 4 position switch not reached
2707	270	PM1 Bevel Etch	Process module 1 Bevel Etch module Level 1 not configured
2708	270	PM1 Bevel Etch	Process module 1 Bevel Etch module Level 2 not configured
2709	270	PM1 Bevel Etch	Process module 1 Bevel Etch module Level 3 not configured
27010	270	PM1 Bevel Etch	Process module 1 Bevel Etch module Level 4 not configured
27011	270	PM1 Bevel Etch	Process module 1 Bevel Etch module intialization timeout error

2801	280	PM1 Epd	**Process module 1 one or more recipes not found on EPD system**
2802	280	PM1 Epd	**Process module 1 invalid EPD recipe selected**
2803	280	PM1 Epd	**Process module 1 EPD system error**
2804	280	PM1 Epd	**Process module 1 endpoint not detected during process**
2851	285	PM1 Bem X Axis	**Process module 1 Bem X Axis dispenser motion control error**
2852	285	PM1 Bem X Axis	**Process module 1 Bem X Axis dispenser motor module power supply error**
2853	285	PM1 Bem X Axis	**Process module 1 Bem X Axis dispenser motor stall error**
2854	285	PM1 Bem X Axis	**Process module 1 Bem X Axis dispenser overtemperature error**
2855	285	PM1 Bem X Axis	**Process module 1 Bem X Axis dispenser motor current error**
2856	285	PM1 Bem X Axis	**Process module 1 Bem X Axis dispenser motor overcurrent error**
2857	285	PM1 Bem X Axis	**Process module 1 Bem X Axis dispenser positioning error**
2858	285	PM1 Bem X Axis	**Process module 1 Bem X Axis dispenser belt broken**
2901	290	PM1 Bem Z Axis	**Process module 1 Bem Z Axis dispenser motion control error**
2902	290	PM1 Bem Z Axis	**Process module 1 Bem Z Axis dispenser motor module power supply error**
2903	290	PM1 Bem Z Axis	**Process module 1 Bem Z Axis dispenser motor stall error**
2904	290	PM1 Bem Z Axis	**Process module 1 Bem Z Axis dispenser overtemperature error**
2905	290	PM1 Bem Z Axis	**Process module 1 Bem Z Axis dispenser motor current error**
2906	290	PM1 Bem Z Axis	**Process module 1 Bem Z Axis dispenser motor overcurrent error**
2907	290	PM1 Bem Z Axis	**Process module 1 Bem Z Axis dispenser positioning error**
2908	290	PM1 Bem Z Axis	**Process module 1 Bem Z Axis dispenser belt broken**
3001	300	Pm2	**Process module 2 medium dispenser not mounted**
3002	300	Pm2	**Process module 2 DIW/N2 dispenser module not mounted**
3003	300	Pm2	**Process module 2 wafer on chuck detection failure**
3004	300	Pm2	**Process module 2 no wafer on chuck detection failure**
3005	300	Pm2	**Process module 2 medium dispenser not over chuck**
3006	300	Pm2	**Process module 2 chuck N2 flow not reached**
3007	300	Pm2	**Process module 2 chuck speed not reached**
3008	300	Pm2	**Process module 2 Level 1 sequence timeout**
3009	300	Pm2	**Process module 2 Level 2 sequence timeout**
30010	300	Pm2	**Process module 2 Level 3 sequence timeout**
30011	300	Pm2	**Process module 2 Level 4 sequence timeout**
30012	300	Pm2	**Process module 2 Level Diw/N2 sequence timeout**
30013	300	Pm2	**Process module 2 Medium dispenser error**
30014	300	Pm2	**Process module 2 N2 dispenser error**
30015	300	Pm2	**Process module 2 Diw dispenser error**
30016	300	Pm2	**Process module 2 Bem Axis X error**
30017	300	Pm2	**Process module 2 Bem Axis Z error**
30018	300	Pm2	**Process module 2 Elevating Unit error**
30019	300	Pm2	**Process module 2 Cdr error**
3101	310	Pm2 MedDisp	**Process module 2 medium dispenser motion control error**
3102	310	Pm2 MedDisp	**Process module 2 medium dispenser motor module power supply error**
3103	310	Pm2 MedDisp	**Process module 2 medium dispenser motor stall error**
3104	310	Pm2 MedDisp	**Process module 2 medium dispenser overtemperature error**
3105	310	Pm2 MedDisp	**Process module 2 medium dispenser motor current error**

3106	310	Pm2 MedDisp	Process module 2 medium dispenser motor overcurrent error
3107	310	Pm2 MedDisp	Process module 2 medium dispenser positioning error
3108	310	Pm2 MedDisp	Process module 2 medium dispenser belt broken
3201	320	Pm2 N2Disp	Process module 2 N2 dispenser motion control error
3202	320	Pm2 N2Disp	Process module 2 N2 dispenser motor module power supply error
3203	320	Pm2 N2Disp	Process module 2 N2 dispenser motor stall error
3204	320	Pm2 N2Disp	Process module 2 N2 dispenser overtemperature error
3205	320	Pm2 N2Disp	Process module 2 N2 dispenser motor current error
3206	320	Pm2 N2Disp	Process module 2 N2 dispenser motor overcurrent error
3207	320	Pm2 N2Disp	Process module 2 N2 dispenser positioning error
3208	320	Pm2 N2Disp	Process module 2 N2 dispenser belt broken
3301	330	Pm2 DiwDisp	Process module 2 Diw dispenser motion control error
3302	330	Pm2 DiwDisp	Process module 2 Diw dispenser motor module power supply error
3303	330	Pm2 DiwDisp	Process module 2 Diw dispenser motor stall error
3304	330	Pm2 DiwDisp	Process module 2 Diw dispenser overtemperature error
3305	330	Pm2 DiwDisp	Process module 2 Diw dispenser motor current error
3306	330	Pm2 DiwDisp	Process module 2 Diw dispenser motor overcurrent error
3307	330	Pm2 DiwDisp	Process module 2 Diw dispenser positioning error
3308	330	Pm2 DiwDisp	Process module 2 Diw dispenser belt broken
3401	340	Pm2 Elevating Unit	Process module 2 elevating unit amplifier error
3402	340	Pm2 Elevating Unit	Process module 2 elevating unit load/unload level positioning error
3403	340	Pm2 Elevating Unit	Process module 2 elevating unit level 4 positioning error
3404	340	Pm2 Elevating Unit	Process module 2 elevating unit level 3 positioning error
3405	340	Pm2 Elevating Unit	Process module 2 elevating unit level 2 positioning error
3406	340	Pm2 Elevating Unit	Process module 2 elevating unit level 1 positioning error
3407	340	Pm2 Elevating Unit	Process module 2 elevating unit initialization timeout error
3408	340	Pm2 Elevating Unit	Process module 2 elevating unit positioning error at pre-flush level
3501	350	Pm2 Chuck Drive	Process module 2 chuck drive amplifier error
3502	350	Pm2 Chuck Drive	Process module 2 chuck drive belt broken error
3503	350	Pm2 Chuck Drive	Process module 2 chuck drive release brake error
3504	350	Pm2 Chuck Drive	Process module 2 chuck drive apply brake error
3505	350	Pm2 Chuck Drive	Process module 2 chuck drive initialization timeout error
3506	350	Pm2 Chuck Drive	Process module 2 chuck drive clease timeout error
3507	350	Pm2 Chuck Drive	Process module 2 chuck drive open unload partial timeout error
3508	350	Pm2 Chuck Drive	Process module 2 chuck drive open unload full timeout error
3509	350	Pm2 Chuck Drive	Process module 2 chuck drive open load timeout error
35010	350	Pm2 Chuck Drive	Process module 2 brake feedback error
3601	360	Pm2 Facility	Process module 2 exhaust 1 low limit error
3602	360	Pm2 Facility	Process module 2 exhaust 1 high limit error
3603	360	Pm2 Facility	Process module 2 exhaust 2 low limit error
3604	360	Pm2 Facility	Process module 2 exhaust 2 high limit error
3701	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module disabled
3702	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module axis error
3703	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module Level 1 position switch not reached
3704	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module Level 2 position

			switch not reached
3705	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module Level 3 position switch not reached
3706	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module Level 4 position switch not reached
3707	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module Level 1 not configured
3708	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module Level 2 not configured
3709	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module Level 3 not configured
37010	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module Level 4 not configured
37011	370	Pm2 Bevel Etch	Process module 2 Bevel Etch module intialization timeout error
3801	380	Pm2 Epd	Process module 2 one or more recipes not found on EPD system
3802	380	Pm2 Epd	Process module 2 invalid EPD recipe selected
3803	380	Pm2 Epd	Process module 2 EPD system error
3804	380	Pm2 Epd	Process module 2 endpoint not detected during process
3851	385	Pm2 Bem X Axis	Process module 2 Bem X Axis dispenser motion control error
3852	385	Pm2 Bem X Axis	Process module 2 Bem X Axis dispenser motor module power supply error
3853	385	Pm2 Bem X Axis	Process module 2 Bem X Axis dispenser motor stall error
3854	385	Pm2 Bem X Axis	Process module 2 Bem X Axis dispenser overtemperature error
3855	385	Pm2 Bem X Axis	Process module 2 Bem X Axis dispenser motor current error
3856	385	Pm2 Bem X Axis	Process module 2 Bem X Axis dispenser motor overcurrent error
3857	385	Pm2 Bem X Axis	Process module 2 Bem X Axis dispenser positioning error
3858	385	Pm2 Bem X Axis	Process module 2 Bem X Axis dispenser belt broken
3901	390	Pm2 Bem Z Axis	Process module 2 Bem Z Axis dispenser motion control error
3902	390	Pm2 Bem Z Axis	Process module 2 Bem Z Axis dispenser motor module power supply error
3903	390	Pm2 Bem Z Axis	Process module 2 Bem Z Axis dispenser motor stall error
3904	390	Pm2 Bem Z Axis	Process module 2 Bem Z Axis dispenser overtemperature error
3905	390	Pm2 Bem Z Axis	Process module 2 Bem Z Axis dispenser motor current error
3906	390	Pm2 Bem Z Axis	Process module 2 Bem Z Axis dispenser motor overcurrent error
3907	390	Pm2 Bem Z Axis	Process module 2 Bem Z Axis dispenser positioning error
3908	390	Pm2 Bem Z Axis	Process module 2 Bem Z Axis dispenser belt broken
4001	400	Shutter	Load port 1 shutter error
4002	400	Shutter	Load port 2 shutter error
4003	400	Shutter	Load port 3 shutter error
4004	400	Shutter	Load port 4 shutter error
5101	510	LoadPort1	Load port 1 mapping error
5102	510	LoadPort1	Load port 1 double slotted wafer detected
5103	510	LoadPort1	Load port 1 cross slotted wafer detected
5104	510	LoadPort1	Load port 1 protrusion detected Front Sensor
5105	510	LoadPort1	Load port 1 CDA supply not ok
5106	510	LoadPort1	Load port 1 Vacuum supply not ok
5107	510	LoadPort1	Load port 1 protrusion detected
5108	510	LoadPort1	Load port 1 communication error
5109	510	LoadPort1	Load port 1 access denied
51010	510	LoadPort1	Load port 1 ABS Error
51011	510	LoadPort1	Load port 1 Interlock error

51012	510	LoadPort1	**Load port 1 failed to initialize**
51013	510	LoadPort1	**Load port 1 protrusion detected Back Sensor**
51014	510	LoadPort1	**Load Port 1 Abort Calibration**
51015	510	LoadPort1	**Load Port 1 Abort Lock**
51016	510	LoadPort1	**Load Port 1 Abort Unlock**
51017	510	LoadPort1	**Load Port 1 Abort Stage**
51018	510	LoadPort1	**Load Port 1 Abort Map**
51019	510	LoadPort1	**Load Port 1 Abort Shuttle Advance**
51020	510	LoadPort1	**Load Port 1 Abort Shuttle Retract**
51021	510	LoadPort1	**Load Port 1 Abort Home**
5201	520	LoadPort2	**Load port 2 mapping error**
5202	520	LoadPort2	**Load port 2 double slotted wafer detected**
5203	520	LoadPort2	**Load port 2 cross slotted wafer detected**
5204	520	LoadPort2	**Load port 2 protrusion detected Front Sensor**
5205	520	LoadPort2	**Load port 2 CDA supply not ok**
5206	520	LoadPort2	**Load port 2 Vacuum supply not ok**
5207	520	LoadPort2	**Load port 2 protrusion detected**
5208	520	LoadPort2	**Load port 2 communication error**
5209	520	LoadPort2	**Load port 2 access denied**
52010	520	LoadPort2	**Load port 2 ABS Error**
52011	520	LoadPort2	**Load port 2 Interlock error**
52012	520	LoadPort2	**Load port 2 failed to initialize**
52013	520	LoadPort2	**Load port 2 protrusion detected Back Sensor**
52014	520	LoadPort2	**Load Port 2 Abort Calibration**
52015	520	LoadPort2	**Load Port 2 Abort Lock**
52016	520	LoadPort2	**Load Port 2 Abort Unlock**
52017	520	LoadPort2	**Load Port 2 Abort Stage**
52018	520	LoadPort2	**Load Port 2 Abort Map**
52019	520	LoadPort2	**Load Port 2 Abort Shuttle Advance**
52020	520	LoadPort2	**Load Port 2 Abort Shuttle Retract**
52021	520	LoadPort2	**Load Port 2 Abort Home**
5301	530	LoadPort3	**Load port 3 mapping error**
5302	530	LoadPort3	**Load port 3 double slotted wafer detected**
5303	530	LoadPort3	**Load port 3 cross slotted wafer detected**
5304	530	LoadPort3	**Load port 3 protrusion detected Front Sensor**
5305	530	LoadPort3	**Load port 3 CDA supply not ok**
5306	530	LoadPort3	**Load port 3 Vacuum supply not ok**
5307	530	LoadPort3	**Load port 3 protrusion detected**
5308	530	LoadPort3	**Load port 3 communication error**
5309	530	LoadPort3	**Load port 3 access denied**
53010	530	LoadPort3	**Load port 3 ABS Error**
53011	530	LoadPort3	**Load port 3 Interlock error**
53012	530	LoadPort3	**Load port 3 failed to initialize**
53013	530	LoadPort3	**Load port 3 protrusion detected Back Sensor**
53014	530	LoadPort3	**Load Port 3 Abort Calibration**
53015	530	LoadPort3	**Load Port 3 Abort Lock**

53016	530	LoadPort3	**Load Port 3 Abort Unlock**
53017	530	LoadPort3	**Load Port 3 Abort Stage**
53018	530	LoadPort3	**Load Port 3 Abort Map**
53019	530	LoadPort3	**Load Port 3 Abort Shuttle Advance**
53020	530	LoadPort3	**Load Port 3 Abort Shuttle Retract**
53021	530	LoadPort3	**Load Port 3 Abort Home**
5401	540	LoadPort4	**Load port 4 mapping error**
5402	540	LoadPort4	**Load port 4 double slotted wafer detected**
5403	540	LoadPort4	**Load port 4 cross slotted wafer detected**
5404	540	LoadPort4	**Load port 4 protrusion detected Front Sensor**
5405	540	LoadPort4	**Load port 4 CDA supply not ok**
5406	540	LoadPort4	**Load port 4 Vacuum supply not ok**
5407	540	LoadPort4	**Load port 4 protrusion detected**
5408	540	LoadPort4	**Load port 4 communication error**
5409	540	LoadPort4	**Load port 4 access denied**
54010	540	LoadPort4	**Load port 4 ABS Error**
54011	540	LoadPort4	**Load port 4 Interlock error**
54012	540	LoadPort4	**Load port 4 failed to initialize**
54013	540	LoadPort4	**Load port 4 protrusion detected Back Sensor**
54014	540	LoadPort4	**Load Port 4 Abort Calibration**
54015	540	LoadPort4	**Load Port 4 Abort Lock**
54016	540	LoadPort4	**Load Port 4 Abort Unlock**
54017	540	LoadPort4	**Load Port 4 Abort Stage**
54018	540	LoadPort4	**Load Port 4 Abort Map**
54019	540	LoadPort4	**Load Port 4 Abort Shuttle Advance**
54020	540	LoadPort4	**Load Port 4 Abort Shuttle Retract**
54021	540	LoadPort4	**Load Port 4 Abort Home**
6001	600	Robot	**Robot interface initialization error**
6002	600	Robot	**Robot interface write error**
6003	600	Robot	**Robot interface read error**
6004	600	Robot	**Robot status error**
6005	600	Robot	**Robot safety error**
6006	600	Robot	**Robot communication error**
6007	600	Robot	**A39006: EnDat encoder: Alarm bit - Position value contains an error**
6008	600	Robot	**A39010: EnDat encoder: Alarm-bit - Battery change required**
6009	600	Robot	**A39014: EnDat encoder: Warning bit - Battery charge too low**
60010	600	Robot	**A39022: EnDat encoder: Warning bit is set**
60011	600	Robot	**A39024: EnDat encoder: Operating status error sources: M ALL Power down**
60012	600	Robot	**A50000: Macro ... was not able to create directory**
60013	600	Robot	**A50001: Macro ... was not able to link device**
60014	600	Robot	**A50002: Macro ... was not able to unlink device**
60015	600	Robot	**A50003: Macro ... was not able to create file**
60016	600	Robot	**A50004: Macro ... was not able to write file**
60017	600	Robot	**A50005: Macro ... was not able to create network trace**
60018	600	Robot	**A50006: Macro ... was not able to create system dump**
60019	600	Robot	**A50007: Macro ... was used with invalid motion parameter**

60020	600	Robot	**A50008: Macro ... was not able to find edge**
60021	600	Robot	**A50009: Macro ... was not able to read file**
60022	600	Robot	**A50010: Macro ...: Axis not ready to home**
60023	600	Robot	**A50011: Macro ...: Axis not ready to move relative**
60024	600	Robot	**A50012: Macro ...: Axis not ready to move absolute**
60025	600	Robot	**A50013: Macro ...: Axis not in standstill**
60026	600	Robot	**A50014: Macro ... was used with invalid parameter set**
60027	600	Robot	**A50015: Macro ... was used with invalid input number**
60028	600	Robot	**A50016: Macro ... was aborted by cnc**
60029	600	Robot	**A50017: Macro ... was aborted by internal function**
60030	600	Robot	**A50018: Macro ...: invalid password**
60031	600	Robot	**A50019: Macro ... is only use in password level 1**
60032	600	Robot	**A50020: Macro ... is only use in password level 2**
60033	600	Robot	**A50021: Macro ... was used with invalid pusher type**
60034	600	Robot	**A50022: Invalid station name length**
60035	600	Robot	**A50023: Invalid fault - contact Support**
60036	600	Robot	**A50024: no USB device available**
60037	600	Robot	**A50025: Station not completely configured for Align on Fly**
60038	600	Robot	**A50026: Macro ...: System is turned off**
60039	600	Robot	**A50027: Macro ...: Transformation Polar to Cartesian not valid**
60040	600	Robot	**A50028: Macro ...: Transformation degree to polar not valid**
60041	600	Robot	**A50029: Macro ...: Transformation axis coupling not valid**
60042	600	Robot	**A50030: Macro ...: Homing on autoposition**
60043	600	Robot	**A50031: Macro ...: End effector change not possible**
60044	600	Robot	**A50032: Macro ...: Invalid hardware configuration**
60045	600	Robot	**A50033: Macro ...: Found end effector on clutch**
60046	600	Robot	**A50034: Macro ...: EE-Changer Station configured: Only 0 or 3 allowed**
60047	600	Robot	**A50035: Macro ...: Wrong Beginposition**
60048	600	Robot	**A50036: Macro ... is only use in password level 3**
60049	600	Robot	**A50037: Macro ...: Invalid INF mode**
60050	600	Robot	**A50038: Macro ... was used with invalid limit parameter**
60051	600	Robot	**A50039: Macro ...: Flipmodul in wrong orientation**
60052	600	Robot	**A50041: Active macro was aborted by error**
60053	600	Robot	**A50049: Macro ...: Error Technology Guard**
60054	600	Robot	**A50050: Invalid macro name**
60055	600	Robot	**A50051: Invalid number of parameter**
60056	600	Robot	**A50052: Invalid station name**
60057	600	Robot	**A50053: Reference run required**
60058	600	Robot	**A50054: Invalid axis name**
60059	600	Robot	**A50055: Invalid slot number**
60060	600	Robot	**A50056: Maximum number of station reached**
60061	600	Robot	**A50057: This station already exist**
60062	600	Robot	**A50058: Invalid parameter option**
60063	600	Robot	**A50059: Homing found a wafer on end effector**
60064	600	Robot	**A50060: Found no wafer on end effector**
60065	600	Robot	**A50061: wrong parameter value**

60066	600	Robot	**A50062: Found wafer on end effector**
60067	600	Robot	**A50063: Lost wafer during transport**
60068	600	Robot	**A50064: Wafer not deposit**
60069	600	Robot	**A50065: Station was not teached with primary arm**
60070	600	Robot	**A50066: Wrong Station Type**
60071	600	Robot	**A50067: Invalid Station Pick-up parameters in Station**
60072	600	Robot	**A50068: Invalid Station Cassette parameters in Station**
60073	600	Robot	**A50069: Invalid Station Pushing parameters in Station**
60074	600	Robot	**A50070: Invalid Station Inline parameters in Station**
60075	600	Robot	**A50071: Timeout exceeded**
60076	600	Robot	**A50072: Move over approach to station ... not possible**
60077	600	Robot	**A50073: Actual position of yaw axis is not allowed**
60078	600	Robot	**A50074: Macro name ... not allowed while invalid software key**
60079	600	Robot	**A50075: Macro name ... not allowed while robot in error state**
60080	600	Robot	**A50076: Macro not allowed while Handwheelmode is activated**
60081	600	Robot	**A50077: Syntax Error**
60082	600	Robot	**A50078: Viapoint already exist**
60083	600	Robot	**A50079: Maximum number of viapointsets reached**
60084	600	Robot	**A50080: Invalid viapoint name**
60085	600	Robot	**A50081: Macro not allowed while Teachpendant is activated (switch to manual mode)**
60086	600	Robot	**A50082: Paddlechange not possible when Station is coupled**
60087	600	Robot	**A50083: Station is coupled with**
60088	600	Robot	**A50084: Actual Yaw-Angle not allowed .**
60089	600	Robot	**A50085: There is no Yaw-Axis Configured**
60090	600	Robot	**A50086: Macro ... will reach Softwarelimit**
60091	600	Robot	**A50087: Macro ...: Station is not coupled.**
60092	600	Robot	**A50088: Macro ...: Invalid Station Wafer search parameters in Station.**
60093	600	Robot	**A50100: Homing procedure for single arm robot.**
60094	600	Robot	**A50101: Homing procedure for dual arm robot**
60095	600	Robot	**A50102: Direct homing procedure for single arm robot**
60096	600	Robot	**A50103: Direct homing procedure for dual arm robot**
60097	600	Robot	**A50104: Actual position of yaw axis is not allowed to home**
60098	600	Robot	**A50300: Invalid address for variable**
60099	600	Robot	**A50301: Timeout while opens the edge gripper**
600100	600	Robot	**A50302: Timeout while closes the edge gripper**
600101	600	Robot	**A50303: Invalid Fault**
600102	600	Robot	**A50400: Error while reading the sensors: Trigger(s) missing**
600103	600	Robot	**A50401: No circle-calculation possible**
600104	600	Robot	**A50450: Max. X-Offset exceeded**
600105	600	Robot	**A50451: Max. Y-Offset exceeded**
600106	600	Robot	**A50452: Max. X- and Y-Offset exceeded**
600107	600	Robot	**A50500: Read XML system parameter**
600108	600	Robot	**A50501: Read XML robot configuration**
600109	600	Robot	**A50502: Read XML global parameter**
600110	600	Robot	**A50503: Read XML axis parameter**

600111	600	Robot	**A50504: Read XML motion parameter**
600112	600	Robot	**A50505: Read XML motion parameter sets**
600113	600	Robot	**A50506: Read XML station parameter**
600114	600	Robot	**A50507: Read XML host communication parameter**
600115	600	Robot	**A50508: Read XML via point sets**
600116	600	Robot	**A50509: Read XML (see logger file)**
600117	600	Robot	**A50511: Invalid robot type**
600118	600	Robot	**A50512: Unable to write axis parameter table**
600119	600	Robot	**A50513: Unable to initialize system**
600120	600	Robot	**A50514: Unable to link file device**
600121	600	Robot	**A50515: Unable to delete file rcp_load_all.log**
600122	600	Robot	**A50516: Unable to start safety**
600123	600	Robot	**A50517: Unable to set ripple compensation**
600124	600	Robot	**A50518: Unable to set eth configuration**
600125	600	Robot	**A50519: Read XML logging parameter**
600126	600	Robot	**A50520: Read XML gantry parameter**
600127	600	Robot	**A50521: Invalid ACOPOSmicro hardware**
600128	600	Robot	**A50522: Read XML end effector-Changer parameter**
600129	600	Robot	**A50523: Hardware module not found**
600130	600	Robot	**A50550: Read XML - for more information see read-log-file**
600131	600	Robot	**A50600: Invalid address for variable**
600132	600	Robot	**A50601: Timeout while opens the vacuum end effector**
600133	600	Robot	**A50602: Timeout while closes the vacuum end effector**
600134	600	Robot	**A50603: Invalid Fault**
600135	600	Robot	**A50700: Log error**
600136	600	Robot	**A50701: Logging1**
600137	600	Robot	**A50702: Logging2**
600138	600	Robot	**A50703: Logging3**
600139	600	Robot	**A50800: Invalid address for variable**
600140	600	Robot	**A50801: Is disabled**
600141	600	Robot	**A50802: Timeout while leaving the switch**
600142	600	Robot	**A50803: Timeout while reaching the switch**
600143	600	Robot	**A50804: Invalid Fault**
600144	600	Robot	**A50805: Is not homed**
600145	600	Robot	**A50900: Invalid signal combination at mode selector detected**
600146	600	Robot	**A50901: A static reset-signal was detected by safety controller**
600147	600	Robot	**A50902: Emergency stop on controller or manual control unit**
600148	600	Robot	**A50903: A static reset-signal was detected by safety controller**
600149	600	Robot	**A50904: Emergency stop by parent safety circuit**
600150	600	Robot	**A50905: A static reset-signal was detected by safety controller**
600151	600	Robot	**A50906: Emergency stop by acknowledge switch**
600152	600	Robot	**A50907: A static reset-signal was detected by safety controller**
600153	600	Robot	**A50908: Protective circuit opened while automatic mode**
600154	600	Robot	**A50909: Protective circuit closed while manual mode**
600155	600	Robot	**A50910: Safety not in Run (Safe)**

600156	600	Robot	**A51000: Gantry axis: position difference to large**
600157	600	Robot	**A51001: Gantry axis: torque difference to large**
600158	600	Robot	**A51002: Gantry axis: the first gantry axis stopped**
600159	600	Robot	**A51003: Gantry axis: the second gantry axis stopped**
600160	600	Robot	**A51100: EE changer: Invalid end effector Id**
600161	600	Robot	**A51101: EE changer: Found wafer on end effector or end effector is closed**
600162	600	Robot	**A51102: EE changer: Invalid end effector Input-ID**
600163	600	Robot	**A51103: EE changer: Readout EE-Id not found. Please check your end effector!**
600164	600	Robot	**A51104: EE changer: Readout EE-Id does not match to actual end effector!**
600165	600	Robot	**A51105: EE changer: System is off**
600166	600	Robot	**A51200: Telnet error: (...)**
600167	600	Robot	**A51250: RS232 error: (...)**
600168	600	Robot	**A51300: 1-Axis LPA error: (...)**
600169	600	Robot	**A51304: 1-Axis LPA error: To many BAL macro repetitions**
600170	600	Robot	**A51305: 1-Axis LPA error: Previous command error reason (See PER command in Logosol manual)**
600171	600	Robot	**A51312: 1-Axis LPA error: Single-Axis prealigner is busy**
600172	600	Robot	**A51313: 1-Axis LPA error: Name of Single-Axis prealigner is unknown**
600173	600	Robot	**A51314: 1-Axis LPA error: Problem during (...). Question mark detected**
600174	600	Robot	**A51315: 1-Axis LPA error: Status error. Maybe wrong INF Mode set.**
600175	600	Robot	**A51316: 1-Axis LPA error: Invalid IP address entered**
600176	600	Robot	**A51317: 1-Axis LPA error: BAL did not reached the required accuracy**
600177	600	Robot	**A51318: 1-Axis LPA error: Prealigner name already exist: (...). Double declaration not allowed.**
600178	600	Robot	**A51319: 1-Axis LPA error: Parameter set already exist: (...). Double declaration not allowed.**
600179	600	Robot	**A51320: 1-Axis LPA error: Single-Axis prealigner support not enabled.**
600180	600	Robot	**A59996: Quickstop Error: Quick Stop input active**
600181	600	Robot	**A59997: Error buffer full**
600182	600	Robot	**A59998: Warning buffer full**
600183	600	Robot	**A59999: Invalid software key**
600184	600	Robot	**Robot detected Wafer on EE**
7001	700	Recipe	**Failed to load recipe**
7002	700	Recipe	**Failed to save recipe**
7003	700	Recipe	**Zero N2 dry time in recipe**
8001	800	ECO1	**ECO1 unload error**
8002	800	ECO1	**ECO1 load error**
8101	810	ECO1 Load Swivel	**ECO1 load swivel axis motion control error**
8102	810	ECO1 Load Swivel	**ECO1 load swivel axis motor module undervoltage error**
8103	810	ECO1 Load Swivel	**ECO1 load swivel axis motor module overvoltage error**
8104	810	ECO1 Load Swivel	**ECO1 load swivel axis overtemperature error**
8105	810	ECO1 Load Swivel	**ECO1 load swivel axis motor current error**
8106	810	ECO1 Load Swivel	**ECO1 load swivel axis motor overcurrent error**
8107	810	ECO1 Load Swivel	**ECO1 load swivel axis encoder wire break channel A**
8108	810	ECO1 Load Swivel	**ECO1 load swivel axis encoder wire break channel B**
8109	810	ECO1 Load Swivel	**ECO1 load swivel axis encoder wire break channel R**
8201	820	ECO1 Load Turn	**ECO1 load turn axis motion control error**

8202	820	ECO1 Load Turn	**ECO1 load turn axis motor module undervoltage error**
8203	820	ECO1 Load Turn	**ECO1 load turn axis motor module overvoltage error**
8204	820	ECO1 Load Turn	**ECO1 load turn axis overtemperature error**
8205	820	ECO1 Load Turn	**ECO1 load turn axis motor current error**
8206	820	ECO1 Load Turn	**ECO1 load turn axis motor overcurrent error**
8207	820	ECO1 Load Turn	**ECO1 load turn axis encoder wire break channel A**
8208	820	ECO1 Load Turn	**ECO1 load turn axis encoder wire break channel B**
8209	820	ECO1 Load Turn	**ECO1 load turn axis encoder wire break channel R**
8301	830	ECO1 Unload Swivel	**ECO1 unload swivel axis motion control error**
8302	830	ECO1 Unload Swivel	**ECO1 unload swivel axis motor module undervoltage error**
8303	830	ECO1 Unload Swivel	**ECO1 unload swivel axis motor module overvoltage error**
8304	830	ECO1 Unload Swivel	**ECO1 unload swivel axis overtemperature error**
8305	830	ECO1 Unload Swivel	**ECO1 unload swivel axis motor current error**
8306	830	ECO1 Unload Swivel	**ECO1 unload swivel axis motor overcurrent error**
8307	830	ECO1 Unload Swivel	**ECO1 unload swivel axis encoder wire break channel A**
8308	830	ECO1 Unload Swivel	**ECO1 unload swivel axis encoder wire break channel B**
8309	830	ECO1 Unload Swivel	**ECO1 unload swivel axis encoder wire break channel R**
8401	840	ECO1 Unload Turn	**ECO1 unload turn axis motion control error**
8402	840	ECO1 Unload Turn	**ECO1 unload turn axis motor module undervoltage error**
8403	840	ECO1 Unload Turn	**ECO1 unload turn axis motor module overvoltage error**
8404	840	ECO1 Unload Turn	**ECO1 unload turn axis overtemperature error**
8405	840	ECO1 Unload Turn	**ECO1 unload turn axis motor current error**
8406	840	ECO1 Unload Turn	**ECO1 unload turn axis motor overcurrent error**
8407	840	ECO1 Unload Turn	**ECO1 unload turn axis encoder wire break channel A**
8408	840	ECO1 Unload Turn	**ECO1 unload turn axis encoder wire break channel B**
8409	840	ECO1 Unload Turn	**ECO1 unload turn axis encoder wire break channel R**
9001	900	ECO2	**ECO2 unload error**
9002	900	ECO2	**ECO2 load error**
9101	910	ECO2 Load Swivel	**ECO2 load swivel axis motion control error**
9102	910	ECO2 Load Swivel	**ECO2 load swivel axis motor module undervoltage error**
9103	910	ECO2 Load Swivel	**ECO2 load swivel axis motor module overvoltage error**
9104	910	ECO2 Load Swivel	**ECO2 load swivel axis overtemperature error**
9105	910	ECO2 Load Swivel	**ECO2 load swivel axis motor current error**
9106	910	ECO2 Load Swivel	**ECO2 load swivel axis motor overcurrent error**
9107	910	ECO2 Load Swivel	**ECO2 load swivel axis encoder wire break channel A**
9108	910	ECO2 Load Swivel	**ECO2 load swivel axis encoder wire break channel B**
9109	910	ECO2 Load Swivel	**ECO2 load swivel axis encoder wire break channel R**
9201	920	ECO2 Load Turn	**ECO2 load turn axis motion control error**
9202	920	ECO2 Load Turn	**ECO2 load turn axis motor module undervoltage error**
9203	920	ECO2 Load Turn	**ECO2 load turn axis motor module overvoltage error**
9204	920	ECO2 Load Turn	**ECO2 load turn axis overtemperature error**
9205	920	ECO2 Load Turn	**ECO2 load turn axis motor current error**
9206	920	ECO2 Load Turn	**ECO2 load turn axis motor overcurrent error**
9207	920	ECO2 Load Turn	**ECO2 load turn axis encoder wire break channel A**
9208	920	ECO2 Load Turn	**ECO2 load turn axis encoder wire break channel B**
9209	920	ECO2 Load Turn	**ECO2 load turn axis encoder wire break channel R**

9301	930	ECO2 Unload Swivel	**ECO2 unload swivel axis motion control error**
9302	930	ECO2 Unload Swivel	**ECO2 unload swivel axis motor module undervoltage error**
9303	930	ECO2 Unload Swivel	**ECO2 unload swivel axis motor module overvoltage error**
9304	930	ECO2 Unload Swivel	**ECO2 unload swivel axis overtemperature error**
9305	930	ECO2 Unload Swivel	**ECO2 unload swivel axis motor current error**
9306	930	ECO2 Unload Swivel	**ECO2 unload swivel axis motor overcurrent error**
9307	930	ECO2 Unload Swivel	**ECO2 unload swivel axis encoder wire break channel A**
9308	930	ECO2 Unload Swivel	**ECO2 unload swivel axis encoder wire break channel B**
9309	930	ECO2 Unload Swivel	**ECO2 unload swivel axis encoder wire break channel R**
9401	940	ECO2 Unload Turn	**ECO2 unload turn axis motion control error**
9402	940	ECO2 Unload Turn	**ECO2 unload turn axis motor module undervoltage error**
9403	940	ECO2 Unload Turn	**ECO2 unload turn axis motor module overvoltage error**
9404	940	ECO2 Unload Turn	**ECO2 unload turn axis overtemperature error**
9405	940	ECO2 Unload Turn	**ECO2 unload turn axis motor current error**
9406	940	ECO2 Unload Turn	**ECO2 unload turn axis motor overcurrent error**
9407	940	ECO2 Unload Turn	**ECO2 unload turn axis encoder wire break channel A**
9408	940	ECO2 Unload Turn	**ECO2 unload turn axis encoder wire break channel B**
9409	940	ECO2 Unload Turn	**ECO2 unload turn axis encoder wire break channel R**
10101	1010	Monitoring PM1	**Process module 1 chuck speed out of range**
10102	1010	Monitoring PM1	**Process module 1 medium 1 flow out of range**
10103	1010	Monitoring PM1	**Process module 1 medium 2 flow out of range**
10104	1010	Monitoring PM1	**Process module 1 medium 3 flow out of range**
10105	1010	Monitoring PM1	**Process module 1 medium 4 flow out of range**
10106	1010	Monitoring PM1	**Process module 1 DI water flow out of range**
10107	1010	Monitoring PM1	**Process module 1 DIW+O3 flow out of range**
10108	1010	Monitoring PM1	**Process module 1 chuck N2 flow out of range**
10109	1010	Monitoring PM1	**Process module 1 N2 dry flow out of range**
101010	1010	Monitoring PM1	**Process module 1 medium 1 minimum flow error**
101011	1010	Monitoring PM1	**Process module 1 medium 2 minimum flow error**
101012	1010	Monitoring PM1	**Process module 1 medium 3 minimum flow error**
101013	1010	Monitoring PM1	**Process module 1 medium 4 minimum flow error**
101014	1010	Monitoring PM1	**Process module 1 DI water minimum flow error**
101015	1010	Monitoring PM1	**Process module 1 DIW+O3 minimum flow error**
101016	1010	Monitoring PM1	**Process module 1 medium 1 temperature out of range**
101017	1010	Monitoring PM1	**Process module 1 medium 2 temperature out of range**
101018	1010	Monitoring PM1	**Process module 1 medium 3 temperature out of range**
101019	1010	Monitoring PM1	**Process module 1 medium 4 temperature out of range**
101020	1010	Monitoring PM1	**Process module 1 HPC N2 flow out of range**
101021	1010	Monitoring PM1	**Process module 1 HPC minimum N2 flow error**
101022	1010	Monitoring PM1	**Process module 1 HPC DI water flow out of range**
101023	1010	Monitoring PM1	**Process module 1 HPC minimum DI water flow error**
101024	1010	Monitoring PM1	**Process module 1 BEM medium 1 flow out of range**
101025	1010	Monitoring PM1	**Process module 1 BEM medium 2 flow out of range**
101026	1010	Monitoring PM1	**Process module 1 BEM medium 3 flow out of range**
101027	1010	Monitoring PM1	**Process module 1 BEM medium 4 flow out of range**
101028	1010	Monitoring PM1	**Process module 1 BEM medium 1 minimum flow error**

101029	1010	Monitoring PM1	**Process module 1 BEM medium 2 minimum flow error**
101030	1010	Monitoring PM1	**Process module 1 BEM medium 3 minimum flow error**
101031	1010	Monitoring PM1	**Process module 1 BEM medium 4 minimum flow error**
101032	1010	Monitoring PM1	**Process module 1 N2 temperature out of range**
101033	1010	Monitoring PM1	**Process module 1 Flowmeter communication error**
101034	1010	Monitoring PM1	**Process module 1 low flow flow out of range**
101035	1010	Monitoring PM1	**Process module 1 low flow minimum flow error**
10201	1020	Monitoring PM2	**Process module 2 chuck speed out of range**
10202	1020	Monitoring PM2	**Process module 2 medium 1 flow out of range**
10203	1020	Monitoring PM2	**Process module 2 medium 2 flow out of range**
10204	1020	Monitoring PM2	**Process module 2 medium 3 flow out of range**
10205	1020	Monitoring PM2	**Process module 2 medium 4 flow out of range**
10206	1020	Monitoring PM2	**Process module 2 DI water flow out of range**
10207	1020	Monitoring PM2	**Process module 2 DIW+O3 flow out of range**
10208	1020	Monitoring PM2	**Process module 2 chuck N2 flow out of range**
10209	1020	Monitoring PM2	**Process module 2 N2 dry flow out of range**
102010	1020	Monitoring PM2	**Process module 2 medium 1 minimum flow error**
102011	1020	Monitoring PM2	**Process module 2 medium 2 minimum flow error**
102012	1020	Monitoring PM2	**Process module 2 medium 3 minimum flow error**
102013	1020	Monitoring PM2	**Process module 2 medium 4 minimum flow error**
102014	1020	Monitoring PM2	**Process module 2 DI water minimum flow error**
102015	1020	Monitoring PM2	**Process module 2 DIW+O3 minimum flow error**
102016	1020	Monitoring PM2	**Process module 2 medium 1 temperature out of range**
102017	1020	Monitoring PM2	**Process module 2 medium 2 temperature out of range**
102018	1020	Monitoring PM2	**Process module 2 medium 3 temperature out of range**
102019	1020	Monitoring PM2	**Process module 2 medium 4 temperature out of range**
102020	1020	Monitoring PM2	**Process module 2 HPC N2 flow out of range**
102021	1020	Monitoring PM2	**Process module 2 HPC minimum N2 flow error**
102022	1020	Monitoring PM2	**Process module 2 HPC DI water flow out of range**
102023	1020	Monitoring PM2	**Process module 2 HPC minimum DI water flow error**
102024	1020	Monitoring PM2	**Process module 2 BEM medium 1 flow out of range**
102025	1020	Monitoring PM2	**Process module 2 BEM medium 2 flow out of range**
102026	1020	Monitoring PM2	**Process module 2 BEM medium 3 flow out of range**
102027	1020	Monitoring PM2	**Process module 2 BEM medium 4 flow out of range**
102028	1020	Monitoring PM2	**Process module 2 BEM medium 1 minimum flow error**
102029	1020	Monitoring PM2	**Process module 2 BEM medium 2 minimum flow error**
102030	1020	Monitoring PM2	**Process module 2 BEM medium 3 minimum flow error**
102031	1020	Monitoring PM2	**Process module 2 BEM medium 4 minimum flow error**
102032	1020	Monitoring PM2	**Process module 2 N2 temperature out of range**
102033	1020	Monitoring PM2	**Process module 2 Flowmeter communication error**
102034	1020	Monitoring PM2	**Process module 2 low flow flow out of range**
102035	1020	Monitoring PM2	**Process module 2 low flow minimum flow error**
11001	1100	Monitoring Supply	**medium 1 supply temperature out of range**
11002	1100	Monitoring Supply	**medium 2 supply temperature out of range**
11003	1100	Monitoring Supply	**medium 3 supply temperature out of range**
11004	1100	Monitoring Supply	**medium 4 supply temperature out of range**

11005	1100	Monitoring Supply	N2 Heater error
11006	1100	Monitoring Supply	DIW+CO2 conductivity error
11007	1100	Monitoring Supply	DIW+O3 concentration out of range
11008	1100	Monitoring Supply	DIW+CO2 flow error
11009	1100	Monitoring Supply	DIW+CO2 pressure error
110010	1100	Monitoring Supply	DIW temperature out of range error
12101	1210	MED1 Analyzer	Medium 1 component 1 concentration out of range
12102	1210	MED1 Analyzer	Medium 1 component 2 concentration out of range
12103	1210	MED1 Analyzer	Medium 1 component 3 concentration out of range
12104	1210	MED1 Analyzer	Medium 1 component 4 concentration out of range
12201	1220	MED2 Analyzer	Medium 2 component 1 concentration out of range
12202	1220	MED2 Analyzer	Medium 2 component 2 concentration out of range
12203	1220	MED2 Analyzer	Medium 2 component 3 concentration out of range
12204	1220	MED2 Analyzer	Medium 2 component 4 concentration out of range
12301	1230	MED3 Analyzer	Medium 3 component 1 concentration out of range
12302	1230	MED3 Analyzer	Medium 3 component 2 concentration out of range
12303	1230	MED3 Analyzer	Medium 3 component 3 concentration out of range
12304	1230	MED3 Analyzer	Medium 3 component 4 concentration out of range
12401	1240	MED4 Analyzer	Medium 4 component 1 concentration out of range
12402	1240	MED4 Analyzer	Medium 4 component 2 concentration out of range
12403	1240	MED4 Analyzer	Medium 4 component 3 concentration out of range
12404	1240	MED4 Analyzer	Medium 4 component 4 concentration out of range
13001	1300	Safety	EMS button on MainUnit pressed
13002	1300	Safety	EMS button on rear panel pressed
13003	1300	Safety	EMS button on front panel pressed
13004	1300	Safety	EMS button on Robot Controller pressed
13005	1300	Safety	EMS button on jog terminal pressed
13006	1300	Safety	EMS button in left chemistry area pressed
13007	1300	Safety	EMS button in right chemistry area pressed
13008	1300	Safety	EMS button on process module 1 pressed
13009	1300	Safety	EMS button on process module 2 pressed
130010	1300	Safety	External EMS pressed
130011	1300	Safety	Main Unit safety system not running
130012	1300	Safety	EMS button on CDS pressed
130013	1300	Safety	EMS button on CHC1 pressed
130014	1300	Safety	EMS button on CHC2 pressed
130015	1300	Safety	EMS button on CHC3 pressed
130016	1300	Safety	EMS button on CHC4 pressed
130017	1300	Safety	CDA safety system valve not opened
130018	1300	Safety	Main Unit doors unlocked/opened
130019	1300	Safety	Heater supply contactor error medium 1
130020	1300	Safety	Heater supply contactor error medium 2
130021	1300	Safety	Heater supply contactor error medium 3
130022	1300	Safety	Heater supply contactor error medium 4
130023	1300	Safety	EMS button left robot area pressed
130024	1300	Safety	EMS button right robot area pressed

130025	1300	Safety	**EMS N2 Heater error**
130026	1300	Safety	**Heat supply contactor N2 error**
130027	1300	Safety	**Fire Suppression System Disabled / Fault Signal**
130028	1300	Safety	**Fire Suppression System fire detected**
130029	1300	Safety	**Fire Suppression System triggered**
14101	1410	Heater MED1	**Heater medium 1 purge error**
14102	1410	Heater MED1	**Heater medium 1 internal leak error**
14103	1410	Heater MED1	**Heater medium 1 overtemperature error**
14104	1410	Heater MED1	**Heater medium 1 thermal cutoff sensor error**
14105	1410	Heater MED1	**Heater medium 1 coil 1 overtemperature error**
14106	1410	Heater MED1	**Heater medium 1 coil 2 overtemperature error**
14107	1410	Heater MED1	**Heater medium 1 coil 3 overtemperature error**
14108	1410	Heater MED1	**Heater medium 1 gas N2 overtemperature**
14109	1410	Heater MED1	**Heater medium 1 gas overtemperature sensor 1 error**
141010	1410	Heater MED1	**Heater medium 1 gas overtemperature sensor 2 error**
141011	1410	Heater MED1	**Heater medium 1 solid state relay error**
141012	1410	Heater MED1	**Heater medium 1 contactor error**
141013	1410	Heater MED1	**Heater medium 1 overtemperature sensor 1 error**
141014	1410	Heater MED1	**Heater medium 1 overtemperature sensor 2 error**
141015	1410	Heater MED1	**Heater medium 1 temperature out of range 0-150CÂ°**
14201	1420	Heater MED2	**Heater medium 2 purge error**
14202	1420	Heater MED2	**Heater medium 2 internal leak error**
14203	1420	Heater MED2	**Heater medium 2 overtemperature error**
14204	1420	Heater MED2	**Heater medium 2 thermal cutoff sensor error**
14205	1420	Heater MED2	**Heater medium 2 coil 1 overtemperature error**
14206	1420	Heater MED2	**Heater medium 2 coil 2 overtemperature error**
14207	1420	Heater MED2	**Heater medium 2 coil 3 overtemperature error**
14208	1420	Heater MED2	**Heater medium 2 gas N2 overtemperature**
14209	1420	Heater MED2	**Heater medium 2 gas overtemperature sensor 1 error**
142010	1420	Heater MED2	**Heater medium 2 gas overtemperature sensor 2 error**
142011	1420	Heater MED2	**Heater medium 2 solid state relay error**
142012	1420	Heater MED2	**Heater medium 2 contactor error**
142013	1420	Heater MED2	**Heater medium 2 overtemperature sensor 1 error**
142014	1420	Heater MED2	**Heater medium 2 overtemperature sensor 2 error**
142015	1420	Heater MED2	**Heater medium 2 temperature out of range 0-150CÂ°**
14301	1430	Heater MED3	**Heater medium 3 purge error**
14302	1430	Heater MED3	**Heater medium 3 internal leak error**
14303	1430	Heater MED3	**Heater medium 3 overtemperature error**
14304	1430	Heater MED3	**Heater medium 3 thermal cutoff sensor error**
14305	1430	Heater MED3	**Heater medium 3 coil 1 overtemperature error**
14306	1430	Heater MED3	**Heater medium 3 coil 2 overtemperature error**
14307	1430	Heater MED3	**Heater medium 3 coil 3 overtemperature error**
14308	1430	Heater MED3	**Heater medium 3 gas N2 overtemperature**
14309	1430	Heater MED3	**Heater medium 3 gas overtemperature sensor 1 error**
143010	1430	Heater MED3	**Heater medium 3 gas overtemperature sensor 2 error**
143011	1430	Heater MED3	**Heater medium 3 solid state relay error**

143012	1430	Heater MED3	**Heater medium 3 contactor error**
143013	1430	Heater MED3	**Heater medium 3 overtemperature sensor 1 error**
143014	1430	Heater MED3	**Heater medium 3 overtemperature sensor 2 error**
143015	1430	Heater MED3	**Heater medium 3 temperature out of range 0-150CÂ°**
14401	1440	Heater MED4	**Heater medium 4 purge error**
14402	1440	Heater MED4	**Heater medium 4 internal leak error**
14403	1440	Heater MED4	**Heater medium 4 overtemperature error**
14404	1440	Heater MED4	**Heater medium 4 thermal cutoff sensor error**
14405	1440	Heater MED4	**Heater medium 4 coil 1 overtemperature error**
14406	1440	Heater MED4	**Heater medium 4 coil 2 overtemperature error**
14407	1440	Heater MED4	**Heater medium 4 coil 3 overtemperature error**
14408	1440	Heater MED4	**Heater medium 4 gas N2 overtemperature**
14409	1440	Heater MED4	**Heater medium 4 gas overtemperature sensor 1 error**
144010	1440	Heater MED4	**Heater medium 4 gas overtemperature sensor 2 error**
144011	1440	Heater MED4	**Heater medium 4 solid state relay error**
144012	1440	Heater MED4	**Heater medium 4 contactor error**
144013	1440	Heater MED4	**Heater medium 4 overtemperature sensor 1 error**
144014	1440	Heater MED4	**Heater medium 4 overtemperature sensor 2 error**
144015	1440	Heater MED4	**Heater medium 4 temperature out of range 0-150CÂ°**
15101	1510	Cooler MED1	**Medium 1 cooler pH-measurment error**
15201	1520	Cooler MED2	**Medium 2 cooler pH-measurment error**
15301	1530	Cooler MED3	**Medium 3 cooler pH-measurment error**
15401	1540	Cooler MED4	**Medium 4 cooler pH-measurment error**
16001	1600	Thickness Measurement	**AMTSi system error**
16002	1600	Thickness Measurement	**ATMSi measurment error**
16003	1600	Thickness Measurement	**ATMSi minimum process time violation error**
16004	1600	Thickness Measurement	**ATMSi maximum process time violation error**
17001	1700	Job Manager	**Load port 1 no wafer selected for processing**
17002	1700	Job Manager	**Load port 2 no wafer selected for processing**
17003	1700	Job Manager	**No recipe selected for processing**
17004	1700	Job Manager	**Wafer still in system, needs to be removed before processing**
17005	1700	Job Manager	**Equipment not ready for processing**
18001	1800	CDS	**CDS communication error**
18002	1800	CDS	**CDS module error**
18003	1800	CDS	**CHC1 not ready for processing**
18004	1800	CDS	**CHC2 not ready for processing**
18005	1800	CDS	**CHC3 not ready for processing**
18006	1800	CDS	**CHC4 not ready for processing**
18007	1800	CDS	**DIWO3 module not ready for processing**
18008	1800	CDS	**CHC1 preparation tank filling time out**
18009	1800	CDS	**CHC1 Preparation Tank Supplying Time Out**
180010	1800	CDS	**CHC1 Preparation Tank Pump Error**
180011	1800	CDS	**CHC1 Preparation Tank Alarm Level Sensor Active**
180012	1800	CDS	**CHC1 Canister Fill Supplying Time Out**

180013	1800	CDS	**CHC1 Mix System Communication Error**
180014	1800	CDS	**CHC1 Mix System Not Ready**
180015	1800	CDS	**CHC1 Mix System Volume Set Error Comp.1**
180016	1800	CDS	**CHC1 Mix System Volume Set Error Comp.2**
180017	1800	CDS	**CHC1 Mix System Volume Set Error Comp.3**
180018	1800	CDS	**CHC1 Mix System Volume Set Error Comp.4**
180019	1800	CDS	**CHC1 Mix System Volume Set Error Comp.5**
180020	1800	CDS	**CHC1 Leak Error Drain Box Sensor 1**
180021	1800	CDS	**CHC1 Leak Error Drain Box Sensor 2**
180022	1800	CDS	**CHC1 Leak Error Drain Box Sensor 3**
180023	1800	CDS	**CHC1 Leak Error Drain Box Sensor 4**
180024	1800	CDS	**CHC1 Leak Error Drain Box Sensor 5**
180025	1800	CDS	**CHC1 Leak Error Drain Box Sensor 6**
180026	1800	CDS	**CHC1 Leak Error Drain Box Sensor 7**
180027	1800	CDS	**CHC1 Leak Error Drain Box Sensor 8**
180028	1800	CDS	**CHC1 Leak Error Drain Box Sensor 9**
180029	1800	CDS	**CHC1 Leak Error Drain Box Sensor 10**
180030	1800	CDS	**CHC1 Leak Gravity Drain Sensor 1**
180031	1800	CDS	**CHC1 Leak Gravity Drain Sensor 2**
180032	1800	CDS	**CHC1 Leak Gravity Drain Sensor 3**
180033	1800	CDS	**CHC1 Leak Gravity Drain Sensor 4**
180034	1800	CDS	**CHC1 Leak Gravity Drain Sensor 5**
180035	1800	CDS	**CHC1 Leak Gravity Drain Sensor 6**
180036	1800	CDS	**CHC1 Leak Gravity Drain Sensor 7**
180037	1800	CDS	**CHC1 Leak Gravity Drain Sensor 8**
180038	1800	CDS	**CHC1 Leak Gravity Drain Sensor 9**
180039	1800	CDS	**CHC1 Leak Gravity Drain Sensor 10**
180040	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 1**
180041	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 2**
180042	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 3**
180043	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 4**
180044	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 5**
180045	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 6**
180046	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 7**
180047	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 8**
180048	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 9**
180049	1800	CDS	**CHC1 Leak Error Canister Fill Sensor 10**
180050	1800	CDS	**CHC1 Leak Emergency Drain Sensor 1**
180051	1800	CDS	**CHC1 Leak Emergency Drain Sensor 2**
180052	1800	CDS	**CHC1 Leak Emergency Drain Sensor 3**
180053	1800	CDS	**CHC1 Leak Emergency Drain Sensor 4**
180054	1800	CDS	**CHC1 Leak Emergency Drain Sensor 5**
180055	1800	CDS	**CHC1 Leak Emergency Drain Sensor 6**
180056	1800	CDS	**CHC1 Leak Emergency Drain Sensor 7**
180057	1800	CDS	**CHC1 Leak Emergency Drain Sensor 8**
180058	1800	CDS	**CHC1 Leak Emergency Drain Sensor 9**

180059	1800	CDS	**CHC1 Leak Emergency Drain Sensor 10**
180060	1800	CDS	**CHC1 Alarm Manual Drain Valve Sensor 1**
180061	1800	CDS	**CHC1 Alarm Manual Drain Valve Sensor 2**
180062	1800	CDS	**CHC1 Alarm Manual Drain Valve Sensor 3**
180063	1800	CDS	**CHC1 Alarm Manual Drain Valve Sensor 4**
180064	1800	CDS	**CHC1 Alarm Manual Drain Valve Sensor 5**
180065	1800	CDS	**CHC1 Exhaust High Limit Error**
180066	1800	CDS	**CHC1 Exhaust Low Limit Error**
180067	1800	CDS	**CHC1 Pressurized Drain 1 Facility not OK**
180068	1800	CDS	**CHC1 Pressurized Drain 2 Facility not OK**
180069	1800	CDS	**CHC1 Component Pressure 1 not OK**
180070	1800	CDS	**CHC1 Component Pressure 2 not OK**
180071	1800	CDS	**CHC1 Component Pressure 3 not OK**
180072	1800	CDS	**CHC1 Component Pressure 4 not OK**
180073	1800	CDS	**CHC1 Component Pressure 5 not OK**
180074	1800	CDS	**CHC1 Component Pressure 6 not OK**
180075	1800	CDS	**CHC1 Component Pressure 7 not OK**
180076	1800	CDS	**CHC1 Component Pressure 8 not OK**
180077	1800	CDS	**CHC1 Component Pressure 9 not OK**
180078	1800	CDS	**CHC1 Component Pressure 10 not OK**
180079	1800	CDS	**CHC1 Component Supply 1 not OK**
180080	1800	CDS	**CHC1 Component Supply 2 not OK**
180081	1800	CDS	**CHC1 Component Supply 3 not OK**
180082	1800	CDS	**CHC1 Component Supply 4 not OK**
180083	1800	CDS	**CHC1 Component Supply 5 not OK**
180084	1800	CDS	**CHC1 Component Supply 6 not OK**
180085	1800	CDS	**CHC1 Component Supply 7 not OK**
180086	1800	CDS	**CHC1 Component Supply 8 not OK**
180087	1800	CDS	**CHC1 Component Supply 9 not OK**
180088	1800	CDS	**CHC1 Component Supply 10 not OK**
180089	1800	CDS	**CHC1 Blending System Supply Error**
180090	1800	CDS	**CHC1 Blending System Pump 1 Error**
180091	1800	CDS	**CHC1 Blending System Pump 2 Error**
180092	1800	CDS	**CHC1 Blending System Pump 3 Error**
180093	1800	CDS	**CHC1 Blending System Pump 4 Error**
180094	1800	CDS	**CHC1 Blending System Pump 5 Error**
180095	1800	CDS	**CHC1 XML Transfer Recipe Error**
180096	1800	CDS	**CHC1 XML Transfer Parameter Error**
180097	1800	CDS	**CHC1 XML Transfer Config Error**
180098	1800	CDS	**CHC1 XML Transfer Buffer Table Error**
180099	1800	CDS	**CHC1 XML Transfer User Error**
1800100	1800	CDS	**CHC1 Cooling Water Flow Out Of Range**
1800101	1800	CDS	**CHC1 Cooling Differential Water Flow Out Of Range**
1800102	1800	CDS	**CHC1 Alarm Drain Dilution Inlet Pressure Low Limit**
1800103	1800	CDS	**CHC1 Alarm Drain Dilution Outlet Flow Low Limit**
1800104	1800	CDS	**CHC1 Alarm Drain Dilution Outlet Temperature High Limit**

1800105	1800	CDS	**CHC1 Alarm Drain Dilution Outlet Idle Flow High Limit**
1800106	1800	CDS	**CHC1 Alarm Drain Dilution Booster Pump Error**
1800107	1800	CDS	**CHC2 preparation tank filling time out**
1800108	1800	CDS	**CHC2 Preparation Tank Supplying Time Out**
1800109	1800	CDS	**CHC2 Preparation Tank Pump Error**
1800110	1800	CDS	**CHC2 Preparation Tank Alarm Level Sensor Active**
1800111	1800	CDS	**CHC2 Canister Fill Supplying Time Out**
1800112	1800	CDS	**CHC2 Mix System Communication Error**
1800113	1800	CDS	**CHC2 Mix System Not Ready**
1800114	1800	CDS	**CHC2 Mix System Volume Set Error Comp.1**
1800115	1800	CDS	**CHC2 Mix System Volume Set Error Comp.2**
1800116	1800	CDS	**CHC2 Mix System Volume Set Error Comp.3**
1800117	1800	CDS	**CHC2 Mix System Volume Set Error Comp.4**
1800118	1800	CDS	**CHC2 Mix System Volume Set Error Comp.5**
1800119	1800	CDS	**CHC2 Leak Error Drain Box Sensor 1**
1800120	1800	CDS	**CHC2 Leak Error Drain Box Sensor 2**
1800121	1800	CDS	**CHC2 Leak Error Drain Box Sensor 3**
1800122	1800	CDS	**CHC2 Leak Error Drain Box Sensor 4**
1800123	1800	CDS	**CHC2 Leak Error Drain Box Sensor 5**
1800124	1800	CDS	**CHC2 Leak Error Drain Box Sensor 6**
1800125	1800	CDS	**CHC2 Leak Error Drain Box Sensor 7**
1800126	1800	CDS	**CHC2 Leak Error Drain Box Sensor 8**
1800127	1800	CDS	**CHC2 Leak Error Drain Box Sensor 9**
1800128	1800	CDS	**CHC2 Leak Error Drain Box Sensor 10**
1800129	1800	CDS	**CHC2 Leak Gravity Drain Sensor 1**
1800130	1800	CDS	**CHC2 Leak Gravity Drain Sensor 2**
1800131	1800	CDS	**CHC2 Leak Gravity Drain Sensor 3**
1800132	1800	CDS	**CHC2 Leak Gravity Drain Sensor 4**
1800133	1800	CDS	**CHC2 Leak Gravity Drain Sensor 5**
1800134	1800	CDS	**CHC2 Leak Gravity Drain Sensor 6**
1800135	1800	CDS	**CHC2 Leak Gravity Drain Sensor 7**
1800136	1800	CDS	**CHC2 Leak Gravity Drain Sensor 8**
1800137	1800	CDS	**CHC2 Leak Gravity Drain Sensor 9**
1800138	1800	CDS	**CHC2 Leak Gravity Drain Sensor 10**
1800139	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 1**
1800140	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 2**
1800141	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 3**
1800142	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 4**
1800143	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 5**
1800144	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 6**
1800145	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 7**
1800146	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 8**
1800147	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 9**
1800148	1800	CDS	**CHC2 Leak Error Canister Fill Sensor 10**
1800149	1800	CDS	**CHC2 Leak Emergency Drain Sensor 1**
1800150	1800	CDS	**CHC2 Leak Emergency Drain Sensor 2**

1800151	1800	CDS	**CHC2 Leak Emergency Drain Sensor 3**
1800152	1800	CDS	**CHC2 Leak Emergency Drain Sensor 4**
1800153	1800	CDS	**CHC2 Leak Emergency Drain Sensor 5**
1800154	1800	CDS	**CHC2 Leak Emergency Drain Sensor 6**
1800155	1800	CDS	**CHC2 Leak Emergency Drain Sensor 7**
1800156	1800	CDS	**CHC2 Leak Emergency Drain Sensor 8**
1800157	1800	CDS	**CHC2 Leak Emergency Drain Sensor 9**
1800158	1800	CDS	**CHC2 Leak Emergency Drain Sensor 10**
1800159	1800	CDS	**CHC2 Alarm Manual Drain Valve Sensor 1**
1800160	1800	CDS	**CHC2 Alarm Manual Drain Valve Sensor 2**
1800161	1800	CDS	**CHC2 Alarm Manual Drain Valve Sensor 3**
1800162	1800	CDS	**CHC2 Alarm Manual Drain Valve Sensor 4**
1800163	1800	CDS	**CHC2 Alarm Manual Drain Valve Sensor 5**
1800164	1800	CDS	**CHC2 Exhaust High Limit Error**
1800165	1800	CDS	**CHC2 Exhaust Low Limit Error**
1800166	1800	CDS	**CHC2 Pressurized Drain 1 Facility not OK**
1800167	1800	CDS	**CHC2 Pressurized Drain 2 Facility not OK**
1800168	1800	CDS	**CHC2 Component Pressure 1 not OK**
1800169	1800	CDS	**CHC2 Component Pressure 2 not OK**
1800170	1800	CDS	**CHC2 Component Pressure 3 not OK**
1800171	1800	CDS	**CHC2 Component Pressure 4 not OK**
1800172	1800	CDS	**CHC2 Component Pressure 5 not OK**
1800173	1800	CDS	**CHC2 Component Pressure 6 not OK**
1800174	1800	CDS	**CHC2 Component Pressure 7 not OK**
1800175	1800	CDS	**CHC2 Component Pressure 8 not OK**
1800176	1800	CDS	**CHC2 Component Pressure 9 not OK**
1800177	1800	CDS	**CHC2 Component Pressure 10 not OK**
1800178	1800	CDS	**CHC2 Component Supply 1 not OK**
1800179	1800	CDS	**CHC2 Component Supply 2 not OK**
1800180	1800	CDS	**CHC2 Component Supply 3 not OK**
1800181	1800	CDS	**CHC2 Component Supply 4 not OK**
1800182	1800	CDS	**CHC2 Component Supply 5 not OK**
1800183	1800	CDS	**CHC2 Component Supply 6 not OK**
1800184	1800	CDS	**CHC2 Component Supply 7 not OK**
1800185	1800	CDS	**CHC2 Component Supply 8 not OK**
1800186	1800	CDS	**CHC2 Component Supply 9 not OK**
1800187	1800	CDS	**CHC2 Component Supply 10 not OK**
1800188	1800	CDS	**CHC2 Blending System Supply Error**
1800189	1800	CDS	**CHC2 Blending System Pump 1 Error**
1800190	1800	CDS	**CHC2 Blending System Pump 2 Error**
1800191	1800	CDS	**CHC2 Blending System Pump 3 Error**
1800192	1800	CDS	**CHC2 Blending System Pump 4 Error**
1800193	1800	CDS	**CHC2 Blending System Pump 5 Error**
1800194	1800	CDS	**CHC2 XML Transfer Recipe Error**
1800195	1800	CDS	**CHC2 XML Transfer Parameter Error**
1800196	1800	CDS	**CHC2 XML Transfer Config Error**

1800197	1800	CDS	CHC2 XML Transfer Buffer Table Error
1800198	1800	CDS	CHC2 XML Transfer User Error
1800199	1800	CDS	CHC2 Cooling Water Flow Out Of Range
1800200	1800	CDS	CHC2 Cooling Differential Water Flow Out Of Range
1800201	1800	CDS	CHC2 Alarm Drain Dilution Inlet Pressure Low Limit
1800202	1800	CDS	CHC2 Alarm Drain Dilution Outlet Flow Low Limit
1800203	1800	CDS	CHC2 Alarm Drain Dilution Outlet Temperature High Limit
1800204	1800	CDS	CHC2 Alarm Drain Dilution Outlet Idle Flow High Limit
1800205	1800	CDS	CHC2 Alarm Drain Dilution Booster Pump Error
1800206	1800	CDS	CHC3 preparation tank filling time out
1800207	1800	CDS	CHC3 Preparation Tank Supplying Time Out
1800208	1800	CDS	CHC3 Preparation Tank Pump Error
1800209	1800	CDS	CHC3 Preparation Tank Alarm Level Sensor Active
1800210	1800	CDS	CHC3 Canister Fill Supplying Time Out
1800211	1800	CDS	CHC3 Mix System Communication Error
1800212	1800	CDS	CHC3 Mix System Not Ready
1800213	1800	CDS	CHC3 Mix System Volume Set Error Comp.1
1800214	1800	CDS	CHC3 Mix System Volume Set Error Comp.2
1800215	1800	CDS	CHC3 Mix System Volume Set Error Comp.3
1800216	1800	CDS	CHC3 Mix System Volume Set Error Comp.4
1800217	1800	CDS	CHC3 Mix System Volume Set Error Comp.5
1800218	1800	CDS	CHC3 Leak Error Drain Box Sensor 1
1800219	1800	CDS	CHC3 Leak Error Drain Box Sensor 2
1800220	1800	CDS	CHC3 Leak Error Drain Box Sensor 3
1800221	1800	CDS	CHC3 Leak Error Drain Box Sensor 4
1800222	1800	CDS	CHC3 Leak Error Drain Box Sensor 5
1800223	1800	CDS	CHC3 Leak Error Drain Box Sensor 6
1800224	1800	CDS	CHC3 Leak Error Drain Box Sensor 7
1800225	1800	CDS	CHC3 Leak Error Drain Box Sensor 8
1800226	1800	CDS	CHC3 Leak Error Drain Box Sensor 9
1800227	1800	CDS	CHC3 Leak Error Drain Box Sensor 10
1800228	1800	CDS	CHC3 Leak Gravity Drain Sensor 1
1800229	1800	CDS	CHC3 Leak Gravity Drain Sensor 2
1800230	1800	CDS	CHC3 Leak Gravity Drain Sensor 3
1800231	1800	CDS	CHC3 Leak Gravity Drain Sensor 4
1800232	1800	CDS	CHC3 Leak Gravity Drain Sensor 5
1800233	1800	CDS	CHC3 Leak Gravity Drain Sensor 6
1800234	1800	CDS	CHC3 Leak Gravity Drain Sensor 7
1800235	1800	CDS	CHC3 Leak Gravity Drain Sensor 8
1800236	1800	CDS	CHC3 Leak Gravity Drain Sensor 9
1800237	1800	CDS	CHC3 Leak Gravity Drain Sensor 10
1800238	1800	CDS	CHC3 Leak Error Canister Fill Sensor 1
1800239	1800	CDS	CHC3 Leak Error Canister Fill Sensor 2
1800240	1800	CDS	CHC3 Leak Error Canister Fill Sensor 3
1800241	1800	CDS	CHC3 Leak Error Canister Fill Sensor 4
1800242	1800	CDS	CHC3 Leak Error Canister Fill Sensor 5

1800243	1800	CDS	**CHC3 Leak Error Canister Fill Sensor 6**
1800244	1800	CDS	**CHC3 Leak Error Canister Fill Sensor 7**
1800245	1800	CDS	**CHC3 Leak Error Canister Fill Sensor 8**
1800246	1800	CDS	**CHC3 Leak Error Canister Fill Sensor 9**
1800247	1800	CDS	**CHC3 Leak Error Canister Fill Sensor 10**
1800248	1800	CDS	**CHC3 Leak Emergency Drain Sensor 1**
1800249	1800	CDS	**CHC3 Leak Emergency Drain Sensor 2**
1800250	1800	CDS	**CHC3 Leak Emergency Drain Sensor 3**
1800251	1800	CDS	**CHC3 Leak Emergency Drain Sensor 4**
1800252	1800	CDS	**CHC3 Leak Emergency Drain Sensor 5**
1800253	1800	CDS	**CHC3 Leak Emergency Drain Sensor 6**
1800254	1800	CDS	**CHC3 Leak Emergency Drain Sensor 7**
1800255	1800	CDS	**CHC3 Leak Emergency Drain Sensor 8**
1800256	1800	CDS	**CHC3 Leak Emergency Drain Sensor 9**
1800257	1800	CDS	**CHC3 Leak Emergency Drain Sensor 10**
1800258	1800	CDS	**CHC3 Alarm Manual Drain Valve Sensor 1**
1800259	1800	CDS	**CHC3 Alarm Manual Drain Valve Sensor 2**
1800260	1800	CDS	**CHC3 Alarm Manual Drain Valve Sensor 3**
1800261	1800	CDS	**CHC3 Alarm Manual Drain Valve Sensor 4**
1800262	1800	CDS	**CHC3 Alarm Manual Drain Valve Sensor 5**
1800263	1800	CDS	**CHC3 Exhaust High Limit Error**
1800264	1800	CDS	**CHC3 Exhaust Low Limit Error**
1800265	1800	CDS	**CHC3 Pressurized Drain 1 Facility not OK**
1800266	1800	CDS	**CHC3 Pressurized Drain 2 Facility not OK**
1800267	1800	CDS	**CHC3 Component Pressure 1 not OK**
1800268	1800	CDS	**CHC3 Component Pressure 2 not OK**
1800269	1800	CDS	**CHC3 Component Pressure 3 not OK**
1800270	1800	CDS	**CHC3 Component Pressure 4 not OK**
1800271	1800	CDS	**CHC3 Component Pressure 5 not OK**
1800272	1800	CDS	**CHC3 Component Pressure 6 not OK**
1800273	1800	CDS	**CHC3 Component Pressure 7 not OK**
1800274	1800	CDS	**CHC3 Component Pressure 8 not OK**
1800275	1800	CDS	**CHC3 Component Pressure 9 not OK**
1800276	1800	CDS	**CHC3 Component Pressure 10 not OK**
1800277	1800	CDS	**CHC3 Component Supply 1 not OK**
1800278	1800	CDS	**CHC3 Component Supply 2 not OK**
1800279	1800	CDS	**CHC3 Component Supply 3 not OK**
1800280	1800	CDS	**CHC3 Component Supply 4 not OK**
1800281	1800	CDS	**CHC3 Component Supply 5 not OK**
1800282	1800	CDS	**CHC3 Component Supply 6 not OK**
1800283	1800	CDS	**CHC3 Component Supply 7 not OK**
1800284	1800	CDS	**CHC3 Component Supply 8 not OK**
1800285	1800	CDS	**CHC3 Component Supply 9 not OK**
1800286	1800	CDS	**CHC3 Component Supply 10 not OK**
1800287	1800	CDS	**CHC3 Blending System Supply Error**
1800288	1800	CDS	**CHC3 Blending System Pump 1 Error**

1800289	1800	CDS	CHC3 Blending System Pump 2 Error
1800290	1800	CDS	CHC3 Blending System Pump 3 Error
1800291	1800	CDS	CHC3 Blending System Pump 4 Error
1800292	1800	CDS	CHC3 Blending System Pump 5 Error
1800293	1800	CDS	CHC3 XML Transfer Recipe Error
1800294	1800	CDS	CHC3 XML Transfer Parameter Error
1800295	1800	CDS	CHC3 XML Transfer Config Error
1800296	1800	CDS	CHC3 XML Transfer Buffer Table Error
1800297	1800	CDS	CHC3 XML Transfer User Error
1800298	1800	CDS	CHC3 Cooling Water Flow Out Of Range
1800299	1800	CDS	CHC3 Cooling Differential Water Flow Out Of Range
1800300	1800	CDS	CHC3 Alarm Drain Dilution Inlet Pressure Low Limit
1800301	1800	CDS	CHC3 Alarm Drain Dilution Outlet Flow Low Limit
1800302	1800	CDS	CHC3 Alarm Drain Dilution Outlet Temperature High Limit
1800303	1800	CDS	CHC3 Alarm Drain Dilution Outlet Idle Flow High Limit
1800304	1800	CDS	CHC3 Alarm Drain Dilution Booster Pump Error
1800305	1800	CDS	CHC4 preparation tank filling time out
1800306	1800	CDS	CHC4 Preparation Tank Supplying Time Out
1800307	1800	CDS	CHC4 Preparation Tank Pump Error
1800308	1800	CDS	CHC4 Preparation Tank Alarm Level Sensor Active
1800309	1800	CDS	CHC4 Canister Fill Supplying Time Out
1800310	1800	CDS	CHC4 Mix System Communication Error
1800311	1800	CDS	CHC4 Mix System Not Ready
1800312	1800	CDS	CHC4 Mix System Volume Set Error Comp.1
1800313	1800	CDS	CHC4 Mix System Volume Set Error Comp.2
1800314	1800	CDS	CHC4 Mix System Volume Set Error Comp.3
1800315	1800	CDS	CHC4 Mix System Volume Set Error Comp.4
1800316	1800	CDS	CHC4 Mix System Volume Set Error Comp.5
1800317	1800	CDS	CHC4 Leak Error Drain Box Sensor 1
1800318	1800	CDS	CHC4 Leak Error Drain Box Sensor 2
1800319	1800	CDS	CHC4 Leak Error Drain Box Sensor 3
1800320	1800	CDS	CHC4 Leak Error Drain Box Sensor 4
1800321	1800	CDS	CHC4 Leak Error Drain Box Sensor 5
1800322	1800	CDS	CHC4 Leak Error Drain Box Sensor 6
1800323	1800	CDS	CHC4 Leak Error Drain Box Sensor 7
1800324	1800	CDS	CHC4 Leak Error Drain Box Sensor 8
1800325	1800	CDS	CHC4 Leak Error Drain Box Sensor 9
1800326	1800	CDS	CHC4 Leak Error Drain Box Sensor 10
1800327	1800	CDS	CHC4 Leak Gravity Drain Sensor 1
1800328	1800	CDS	CHC4 Leak Gravity Drain Sensor 2
1800329	1800	CDS	CHC4 Leak Gravity Drain Sensor 3
1800330	1800	CDS	CHC4 Leak Gravity Drain Sensor 4
1800331	1800	CDS	CHC4 Leak Gravity Drain Sensor 5
1800332	1800	CDS	CHC4 Leak Gravity Drain Sensor 6
1800333	1800	CDS	CHC4 Leak Gravity Drain Sensor 7
1800334	1800	CDS	CHC4 Leak Gravity Drain Sensor 8

1800335	1800	CDS	**CHC4 Leak Gravity Drain Sensor 9**
1800336	1800	CDS	**CHC4 Leak Gravity Drain Sensor 10**
1800337	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 1**
1800338	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 2**
1800339	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 3**
1800340	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 4**
1800341	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 5**
1800342	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 6**
1800343	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 7**
1800344	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 8**
1800345	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 9**
1800346	1800	CDS	**CHC4 Leak Error Canister Fill Sensor 10**
1800347	1800	CDS	**CHC4 Leak Emergency Drain Sensor 1**
1800348	1800	CDS	**CHC4 Leak Emergency Drain Sensor 2**
1800349	1800	CDS	**CHC4 Leak Emergency Drain Sensor 3**
1800350	1800	CDS	**CHC4 Leak Emergency Drain Sensor 4**
1800351	1800	CDS	**CHC4 Leak Emergency Drain Sensor 5**
1800352	1800	CDS	**CHC4 Leak Emergency Drain Sensor 6**
1800353	1800	CDS	**CHC4 Leak Emergency Drain Sensor 7**
1800354	1800	CDS	**CHC4 Leak Emergency Drain Sensor 8**
1800355	1800	CDS	**CHC4 Leak Emergency Drain Sensor 9**
1800356	1800	CDS	**CHC4 Leak Emergency Drain Sensor 10**
1800357	1800	CDS	**CHC4 Alarm Manual Drain Valve Sensor 1**
1800358	1800	CDS	**CHC4 Alarm Manual Drain Valve Sensor 2**
1800359	1800	CDS	**CHC4 Alarm Manual Drain Valve Sensor 3**
1800360	1800	CDS	**CHC4 Alarm Manual Drain Valve Sensor 4**
1800361	1800	CDS	**CHC4 Alarm Manual Drain Valve Sensor 5**
1800362	1800	CDS	**CHC4 Exhaust High Limit Error**
1800363	1800	CDS	**CHC4 Exhaust Low Limit Error**
1800364	1800	CDS	**CHC4 Pressurized Drain 1 Facility not OK**
1800365	1800	CDS	**CHC4 Pressurized Drain 2 Facility not OK**
1800366	1800	CDS	**CHC4 Component Pressure 1 not OK**
1800367	1800	CDS	**CHC4 Component Pressure 2 not OK**
1800368	1800	CDS	**CHC4 Component Pressure 3 not OK**
1800369	1800	CDS	**CHC4 Component Pressure 4 not OK**
1800370	1800	CDS	**CHC4 Component Pressure 5 not OK**
1800371	1800	CDS	**CHC4 Component Pressure 6 not OK**
1800372	1800	CDS	**CHC4 Component Pressure 7 not OK**
1800373	1800	CDS	**CHC4 Component Pressure 8 not OK**
1800374	1800	CDS	**CHC4 Component Pressure 9 not OK**
1800375	1800	CDS	**CHC4 Component Pressure 10 not OK**
1800376	1800	CDS	**CHC4 Component Supply 1 not OK**
1800377	1800	CDS	**CHC4 Component Supply 2 not OK**
1800378	1800	CDS	**CHC4 Component Supply 3 not OK**
1800379	1800	CDS	**CHC4 Component Supply 4 not OK**
1800380	1800	CDS	**CHC4 Component Supply 5 not OK**

1800381	1800	CDS	CHC4 Component Supply 6 not OK
1800382	1800	CDS	CHC4 Component Supply 7 not OK
1800383	1800	CDS	CHC4 Component Supply 8 not OK
1800384	1800	CDS	CHC4 Component Supply 9 not OK
1800385	1800	CDS	CHC4 Component Supply 10 not OK
1800386	1800	CDS	CHC4 Blending System Supply Error
1800387	1800	CDS	CHC4 Blending System Pump 1 Error
1800388	1800	CDS	CHC4 Blending System Pump 2 Error
1800389	1800	CDS	CHC4 Blending System Pump 3 Error
1800390	1800	CDS	CHC4 Blending System Pump 4 Error
1800391	1800	CDS	CHC4 Blending System Pump 5 Error
1800392	1800	CDS	CHC4 XML Transfer Recipe Error
1800393	1800	CDS	CHC4 XML Transfer Parameter Error
1800394	1800	CDS	CHC4 XML Transfer Config Error
1800395	1800	CDS	CHC4 XML Transfer Buffer Table Error
1800396	1800	CDS	CHC4 XML Transfer User Error
1800397	1800	CDS	CHC4 Cooling Water Flow Out Of Range
1800398	1800	CDS	CHC4 Cooling Differential Water Flow Out Of Range
1800399	1800	CDS	CHC4 Alarm Drain Dilution Inlet Pressure Low Limit
1800400	1800	CDS	CHC4 Alarm Drain Dilution Inlet Pressure Low Limit
1800401	1800	CDS	CHC4 Alarm Drain Dilution Outlet Temperature High Limit
1800402	1800	CDS	CHC4 Alarm Drain Dilution Outlet Idle Flow High Limit
1800403	1800	CDS	CHC4 Alarm Drain Dilution Booster Pump Error
1800404	1800	CDS	Drip Pan Leak Sensor 1 triggered
1800405	1800	CDS	Drip Pan Leak Sensor 2 triggered
1800406	1800	CDS	Drip Pan Leak Sensor 3 triggered
1800407	1800	CDS	Drip Pan Leak Sensor 4 triggered
1800408	1800	CDS	Drip Pan Leak Sensor 5 triggered
1800409	1800	CDS	Drip Pan Leak Sensor 6 triggered
1800410	1800	CDS	Drip Pan Leak Sensor 7 triggered
1800411	1800	CDS	Drip Pan Leak Sensor 8 triggered
1800412	1800	CDS	Drip Pan Leak Sensor 9 triggered
1800413	1800	CDS	Drip Pan Leak Sensor 10 triggered
1800414	1800	CDS	Chemistry Interconnection Box Leak Sensor 1 triggered
1800415	1800	CDS	Chemistry Interconnection Box Leak Sensor 2 triggered
1800416	1800	CDS	Chemistry Interconnection Box Leak Sensor 3 triggered
1800417	1800	CDS	Chemistry Interconnection Box Leak Sensor 4 triggered
1800418	1800	CDS	Chemistry Interconnection Box Leak Sensor 5 triggered
1800419	1800	CDS	Chemistry Interconnection Box Leak Sensor 6 triggered
1800420	1800	CDS	Chemistry Interconnection Box Leak Sensor 7 triggered
1800421	1800	CDS	Chemistry Interconnection Box Leak Sensor 8 triggered
1800422	1800	CDS	Chemistry Interconnection Box Leak Sensor 9 triggered
1800423	1800	CDS	Chemistry Interconnection Box Leak Sensor 10 triggered
1800424	1800	CDS	CDS Facility CDA Pressure Error
1800425	1800	CDS	CDS Facility DIW Pressure Error
1800426	1800	CDS	CDS Facility Component 1 Pressure Error

1800427	1800	CDS	**CDS Facility Component 2 Pressure Error**
1800428	1800	CDS	**CDS Facility Component 3 Pressure Error**
1800429	1800	CDS	**CDS Facility Component 4 Pressure Error**
1800430	1800	CDS	**CDS Facility Component 5 Pressure Error**
1800431	1800	CDS	**CDS Facility Component 6 Pressure Error**
1800432	1800	CDS	**CDS Facility Component 7 Pressure Error**
1800433	1800	CDS	**CDS Facility Component 8 Pressure Error**
1800434	1800	CDS	**CDS Facility Component 9 Pressure Error**
1800435	1800	CDS	**CDS Facility DIW Supply Error**
1800436	1800	CDS	**CDS Facility Component 1 Supply Error**
1800437	1800	CDS	**CDS Facility Component 2 Supply Error**
1800438	1800	CDS	**CDS Facility Component 3 Supply Error**
1800439	1800	CDS	**CDS Facility Component 4 Supply Error**
1800440	1800	CDS	**CDS Facility Component 5 Supply Error**
1800441	1800	CDS	**CDS Facility Component 6 Supply Error**
1800442	1800	CDS	**CDS Facility Component 7 Supply Error**
1800443	1800	CDS	**CDS Facility Component 8 Supply Error**
1800444	1800	CDS	**CDS Facility Component 9 Supply Error**
1800445	1800	CDS	**CDS Main Power CB tripped**
1800446	1800	CDS	**CHC1 Process Pump CB tripped**
1800447	1800	CDS	**CHC2 Process Pump CB tripped**
1800448	1800	CDS	**CHC3 Process Pump CB tripped**
1800449	1800	CDS	**CHC4 Process Pump CB tripped**
1800450	1800	CDS	**CDS Cooling Line pH-Value Out of Range**
1800451	1800	CDS	**XML Transfer Recipe Error**
1800452	1800	CDS	**XML Transfer Parameter Error**
1800453	1800	CDS	**XML Transfer Config Error**
1800454	1800	CDS	**XML Transfer Buffer Table Error**
1800455	1800	CDS	**XML Transfer Correction Table Error**
1800456	1800	CDS	**XML Transfer User Error**
1800457	1800	CDS	**CHC1 XML Transfer Correction Table Error**
1800458	1800	CDS	**CHC2 XML Transfer Correction Table Error**
1800459	1800	CDS	**CHC3 XML Transfer Correction Table Error**
1800460	1800	CDS	**CHC4 XML Transfer Correction Table Error**
1800461	1800	CDS	**CHC1 Canister Fill Door Open Time Out**
1800462	1800	CDS	**CHC2 Canister Fill Door Open Time Out**
1800463	1800	CDS	**CHC3 Canister Fill Door Open Time Out**
1800464	1800	CDS	**CHC4 Canister Fill Door Open Time Out**
18101	1810	CDS Setup	**CHC1 not ready for processing - CHC is OFF or in ERROR**
18102	1810	CDS Setup	**CHC2 not ready for processing - CHC is OFF or in ERROR**
18103	1810	CDS Setup	**CHC3 not ready for processing - CHC is OFF or in ERROR**
18104	1810	CDS Setup	**CHC4 not ready for processing - CHC is OFF or in ERROR**
18105	1810	CDS Setup	**CHC1 refill not ready for processing - refill started without request**
18106	1810	CDS Setup	**CHC2 refill not ready for processing - refill started without request**
18107	1810	CDS Setup	**CHC3 refill not ready for processing - refill started without request**
18108	1810	CDS Setup	**CHC4 refill not ready for processing - refill started without request**

18109	1810	CDS Setup	**CDS Communication Error**
181010	1810	CDS Setup	**CDS Module Error**
181011	1810	CDS Setup	**CDS DiwO3 Not Ready**
19001	1900	N2 Heater	**N2 Heater purge error**
19002	1900	N2 Heater	**N2 Heater internal leak error**
19003	1900	N2 Heater	**N2 Heater overtemperature error**
19004	1900	N2 Heater	**N2 Heater thermal cutoff sensor error**
19005	1900	N2 Heater	**N2 Heater coil 1 overtemperature error**
19006	1900	N2 Heater	**N2 Heater coil 2 overtemperature error**
19007	1900	N2 Heater	**N2 Heater coil 3 overtemperature error**
19008	1900	N2 Heater	**N2 Heater gas N2 overtemperature**
19009	1900	N2 Heater	**N2 Heater gas overtemperature sensor 1 error**
190010	1900	N2 Heater	**N2 Heater 1 gas overtemperature sensor 2 error**
190011	1900	N2 Heater	**N2 Heater solid state relay error**
190012	1900	N2 Heater	**N2 Heater contactor error**
190013	1900	N2 Heater	**N2 Heater overtemperature sensor 1 error**
190014	1900	N2 Heater	**N2 Heater overtemperature sensor 2 error**
190015	1900	N2 Heater	**N2 Heater gas overtemperature sensor 1 error**
20001	2000	Leak	**Connection box leak sensor 1 triggered**
20002	2000	Leak	**Connection box leak sensor 2 triggered**
20003	2000	Leak	**Connection box leak sensor 3 triggered**
20004	2000	Leak	**Connection box leak sensor 4 triggered**
20005	2000	Leak	**Connection box leak sensor 5 triggered**
20006	2000	Leak	**Connection box leak sensor 6 triggered**
20007	2000	Leak	**Connection box leak sensor 7 triggered**
20008	2000	Leak	**Connection box leak sensor 8 triggered**
20009	2000	Leak	**Connection box leak sensor 9 triggered**
200010	2000	Leak	**Connection box leak sensor 10 triggered**
200011	2000	Leak	**Main Unit drip pan leak sensor 1 triggered**
200012	2000	Leak	**Main Unit drip pan leak sensor 2 triggered**
200013	2000	Leak	**Main Unit drip pan leak sensor 3 triggered**
200014	2000	Leak	**Main Unit drip pan leak sensor 4 triggered**
200015	2000	Leak	**Main Unit drip pan leak sensor 5 triggered**
200016	2000	Leak	**Main Unit drip pan leak sensor 6 triggered**
200017	2000	Leak	**Main Unit drip pan leak sensor 7 triggered**
200018	2000	Leak	**Main Unit drip pan leak sensor 8 triggered**
200019	2000	Leak	**Main Unit drip pan leak sensor 9 triggered**
200020	2000	Leak	**Main Unit drip pan leak sensor 10 triggered**
200021	2000	Leak	**Heater area leak sensor 1 triggered**
200022	2000	Leak	**Heater area leak sensor 2 triggered**
200023	2000	Leak	**Heater area leak sensor 3 triggered**
200024	2000	Leak	**Heater area leak sensor 4 triggered**
200025	2000	Leak	**Heater area leak sensor 5 triggered**
200026	2000	Leak	**Heater area leak sensor 6 triggered**
200027	2000	Leak	**Heater area leak sensor 7 triggered**
200028	2000	Leak	**Heater area leak sensor 8 triggered**

200029	2000	Leak	**Heater area leak sensor 9 triggered**
200030	2000	Leak	**Heater area leak sensor 10 triggered**
2000101	2000	Leak	**PM1 Suck back box leak sensor 1 triggered**
2000102	2000	Leak	**PM1 Suck back box leak sensor 2 triggered**
2000103	2000	Leak	**PM1 Suck back box leak sensor 3 triggered**
2000104	2000	Leak	**PM1 Suck back box leak sensor 4 triggered**
2000105	2000	Leak	**PM1 Suck back box leak sensor 5 triggered**
2000106	2000	Leak	**PM1 Suck back box leak sensor 6 triggered**
2000107	2000	Leak	**PM1 Suck back box leak sensor 7 triggered**
2000108	2000	Leak	**PM1 Suck back box leak sensor 8 triggered**
2000109	2000	Leak	**PM1 Suck back box leak sensor 9 triggered**
2000110	2000	Leak	**PM1 Suck back box leak sensor 10 triggered**
2000111	2000	Leak	**PM1 Process Chamber leak sensor 1 triggered**
2000112	2000	Leak	**PM1 Process Chamber leak sensor 2 triggered**
2000113	2000	Leak	**PM1 Process Chamber leak sensor 3 triggered**
2000114	2000	Leak	**PM1 Process Chamber leak sensor 4 triggered**
2000115	2000	Leak	**PM1 Process Chamber leak sensor 5 triggered**
2000116	2000	Leak	**PM1 Process Chamber leak sensor 6 triggered**
2000117	2000	Leak	**PM1 Process Chamber leak sensor 7 triggered**
2000118	2000	Leak	**PM1 Process Chamber leak sensor 8 triggered**
2000119	2000	Leak	**PM1 Process Chamber leak sensor 9 triggered**
2000120	2000	Leak	**PM1 Process Chamber leak sensor 10 triggered**
2000121	2000	Leak	**PM1 Elevating Unit leak sensor 1 triggered**
2000122	2000	Leak	**PM1 Elevating Unit leak sensor 2 triggered**
2000123	2000	Leak	**PM1 Elevating Unit leak sensor 3 triggered**
2000124	2000	Leak	**PM1 Elevating Unit leak sensor 4 triggered**
2000125	2000	Leak	**PM1 Elevating Unit leak sensor 5 triggered**
2000126	2000	Leak	**PM1 Elevating Unit leak sensor 6 triggered**
2000127	2000	Leak	**PM1 Elevating Unit leak sensor 7 triggered**
2000128	2000	Leak	**PM1 Elevating Unit leak sensor 8 triggered**
2000129	2000	Leak	**PM1 Elevating Unit leak sensor 9 triggered**
2000130	2000	Leak	**PM1 Elevating Unit leak sensor 10 triggered**
2000131	2000	Leak	**PM1 Drip Pan leak sensor 1 triggered**
2000132	2000	Leak	**PM1 Drip Pan leak sensor 2 triggered**
2000133	2000	Leak	**PM1 Drip Pan leak sensor 3 triggered**
2000134	2000	Leak	**PM1 Drip Pan leak sensor 4 triggered**
2000135	2000	Leak	**PM1 Drip Pan leak sensor 5 triggered**
2000136	2000	Leak	**PM1 Drip Pan leak sensor 6 triggered**
2000137	2000	Leak	**PM1 Drip Pan leak sensor 7 triggered**
2000138	2000	Leak	**PM1 Drip Pan leak sensor 8 triggered**
2000139	2000	Leak	**PM1 Drip Pan leak sensor 9 triggered**
2000140	2000	Leak	**PM1 Drip Pan leak sensor 10 triggered**
2000201	2000	Leak	**PM2 Suck back box leak sensor 1 triggered**
2000202	2000	Leak	**PM2 Suck back box leak sensor 2 triggered**
2000203	2000	Leak	**PM2 Suck back box leak sensor 3 triggered**
2000204	2000	Leak	**PM2 Suck back box leak sensor 4 triggered**

2000205	2000	Leak	**PM2 Suck back box leak sensor 5 triggered**
2000206	2000	Leak	**PM2 Suck back box leak sensor 6 triggered**
2000207	2000	Leak	**PM2 Suck back box leak sensor 7 triggered**
2000208	2000	Leak	**PM2 Suck back box leak sensor 8 triggered**
2000209	2000	Leak	**PM2 Suck back box leak sensor 9 triggered**
2000210	2000	Leak	**PM2 Suck back box leak sensor 10 triggered**
2000211	2000	Leak	**PM2 Process Chamber leak sensor 1 triggered**
2000212	2000	Leak	**PM2 Process Chamber leak sensor 2 triggered**
2000213	2000	Leak	**PM2 Process Chamber leak sensor 3 triggered**
2000214	2000	Leak	**PM2 Process Chamber leak sensor 4 triggered**
2000215	2000	Leak	**PM2 Process Chamber leak sensor 5 triggered**
2000216	2000	Leak	**PM2 Process Chamber leak sensor 6 triggered**
2000217	2000	Leak	**PM2 Process Chamber leak sensor 7 triggered**
2000218	2000	Leak	**PM2 Process Chamber leak sensor 8 triggered**
2000219	2000	Leak	**PM2 Process Chamber leak sensor 9 triggered**
2000220	2000	Leak	**PM2 Process Chamber leak sensor 10 triggered**
2000221	2000	Leak	**PM2 Elevating Unit leak sensor 1 triggered**
2000222	2000	Leak	**PM2 Elevating Unit leak sensor 2 triggered**
2000223	2000	Leak	**PM2 Elevating Unit leak sensor 3 triggered**
2000224	2000	Leak	**PM2 Elevating Unit leak sensor 4 triggered**
2000225	2000	Leak	**PM2 Elevating Unit leak sensor 5 triggered**
2000226	2000	Leak	**PM2 Elevating Unit leak sensor 6 triggered**
2000227	2000	Leak	**PM2 Elevating Unit leak sensor 7 triggered**
2000228	2000	Leak	**PM2 Elevating Unit leak sensor 8 triggered**
2000229	2000	Leak	**PM2 Elevating Unit leak sensor 9 triggered**
2000230	2000	Leak	**PM2 Elevating Unit leak sensor 10 triggered**
2000231	2000	Leak	**PM2 Drip Pan leak sensor 1 triggered**
2000232	2000	Leak	**PM2 Drip Pan leak sensor 2 triggered**
2000233	2000	Leak	**PM2 Drip Pan leak sensor 3 triggered**
2000234	2000	Leak	**PM2 Drip Pan leak sensor 4 triggered**
2000235	2000	Leak	**PM2 Drip Pan leak sensor 5 triggered**
2000236	2000	Leak	**PM2 Drip Pan leak sensor 6 triggered**
2000237	2000	Leak	**PM2 Drip Pan leak sensor 7 triggered**
2000238	2000	Leak	**PM2 Drip Pan leak sensor 8 triggered**
2000239	2000	Leak	**PM2 Drip Pan leak sensor 9 triggered**
2000240	2000	Leak	**PM2 Drip Pan leak sensor 10 triggered**
2000241	2000	Leak	**Return Pump Drip Pan leak sensor 1 triggered**
2000242	2000	Leak	**Return Pump Drip Pan leak sensor 2 triggered**
2000243	2000	Leak	**Return Pump Drip Pan leak sensor 3 triggered**
2000244	2000	Leak	**Return Pump Drip Pan leak sensor 4 triggered**
2000245	2000	Leak	**Return Pump Drip Pan leak sensor 5 triggered**
2000246	2000	Leak	**Return Pump Drip Pan leak sensor 6 triggered**
2000247	2000	Leak	**Return Pump Drip Pan leak sensor 7 triggered**
2000248	2000	Leak	**Return Pump Drip Pan leak sensor 8 triggered**
2000249	2000	Leak	**Return Pump Drip Pan leak sensor 9 triggered**
2000250	2000	Leak	**Return Pump Drip Pan leak sensor 10 triggered**

2000251	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 1 triggered
2000252	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 2 triggered
2000253	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 3 triggered
2000254	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 4 triggered
2000255	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 5 triggered
2000256	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 6 triggered
2000257	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 7 triggered
2000258	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 8 triggered
2000259	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 9 triggered
2000260	2000	Leak	Chc Suck Back Box Process Pump Drip Pan leak sensor 10 triggered
2000261	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 1 triggered
2000262	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 2 triggered
2000263	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 3 triggered
2000264	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 4 triggered
2000265	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 5 triggered
2000266	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 6 triggered
2000267	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 7 triggered
2000268	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 8 triggered
2000269	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 9 triggered
2000270	2000	Leak	Chc Process Chamber Process Pump Drip Pan leak sensor 10 triggered
2000271	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 1 triggered
2000272	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 2 triggered
2000273	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 3 triggered
2000274	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 4 triggered
2000275	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 5 triggered
2000276	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 6 triggered
2000277	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 7 triggered
2000278	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 8 triggered
2000279	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 9 triggered
2000280	2000	Leak	Chc EU Process Pump Drip Pan leak sensor 10 triggered
2000281	2000	Leak	Chc Drip Pan leak sensor 1 triggered
2000282	2000	Leak	Chc Drip Pan leak sensor 2 triggered
2000283	2000	Leak	Chc Drip Pan leak sensor 3 triggered
2000284	2000	Leak	Chc Drip Pan leak sensor 4 triggered
2000285	2000	Leak	Chc Drip Pan leak sensor 5 triggered
2000286	2000	Leak	Chc Drip Pan leak sensor 6 triggered
2000287	2000	Leak	Chc Drip Pan leak sensor 7 triggered
2000288	2000	Leak	Chc Drip Pan leak sensor 8 triggered
2000289	2000	Leak	Chc Drip Pan leak sensor 9 triggered
2000290	2000	Leak	Chc Drip Pan leak sensor 10 triggered
21001	2100	Facility	Facility CDA supply low error
21002	2100	Facility	Facility CDA supply high error

21003	2100	Facility	**Facility N2 supply low error**
21004	2100	Facility	**Facility N2 supply high error**
21005	2100	Facility	**Facility N2 supply low error**
21006	2100	Facility	**Facility N2 supply high error**
21007	2100	Facility	**Facility DI water supply low error**
21008	2100	Facility	**Facility DI water supply high error**
21009	2100	Facility	**Facility vacuum supply low error**
210010	2100	Facility	**Facility vacuum supply high error**
210011	2100	Facility	**FFU1 error**
210012	2100	Facility	**FFU2 error**
210013	2100	Facility	**Ionization system error**
210014	2100	Facility	**Facility drain1 error**
210015	2100	Facility	**Facility drain2 error**
210016	2100	Facility	**Facility external process release error**
22001	2200	cbMon	**Chuck drive 1 axis amplifier circuit breaker tripped**
22002	2200	cbMon	**Elevating unit 1 axis amplifier circuit breaker tripped**
22003	2200	cbMon	**Chuck drive 2 axis amplifier circuit breaker tripped**
22004	2200	cbMon	**Elevating unit 2 axis amplifier circuit breaker tripped**
22005	2200	cbMon	**Robot power circuit breaker tripped**
22006	2200	cbMon	**FFU1 power circuit breaker tripped**
22007	2200	cbMon	**FFU2 power circuit breaker tripped**
22008	2200	cbMon	**Ionization system power circuit breaker tripped**
23001	2300	DIW O3 Supply	**DIWO3 module power circuit breaker tripped**
23002	2300	DIW O3 Supply	**DIWO3 module EMO pressed**
23003	2300	DIW O3 Supply	**DIWO3 module leak sensor 1 triggered**
23004	2300	DIW O3 Supply	**DIWO3 module leak sensor 2 triggered**
23005	2300	DIW O3 Supply	**DIWO3 module supply not ready**
23006	2300	DIW O3 Supply	**DIWO3 module critical alarm**
23007	2300	DIW O3 Supply	**DIWO3 module communication error**
23008	2300	DIW O3 Supply	**DIWO3 module pump error**
23009	2300	DIW O3 Supply	**DIWO3 module supply pressure error**
230010	2300	DIW O3 Supply	**DIWO3 Modbus communication error**
24001	2400	O3 Monitoring	**Ozone monitoring system error**
24002	2400	O3 Monitoring	**Ozone monitoring UV-lamp error**
24003	2400	O3 Monitoring	**Ozone monitoring EMO group 1 triggered**
24004	2400	O3 Monitoring	**Ozone monitoring EMO group 2 triggered**
24005	2400	O3 Monitoring	**Ozone monitoring EMO group 3 triggered**
24006	2400	O3 Monitoring	**Ozone monitoring concentration low error**
24007	2400	O3 Monitoring	**Ozone monitoring concentration high error**
25001	2500	Handling Monitoring PM1	**Robot wafer position plausibility error**
25002	2500	Handling Monitoring PM1	**PM1 load ECO wafer position plausibility error**
25003	2500	Handling Monitoring PM1	**PM1 unload ECO wafer position plausibility error**
26001	2600	Handling Monitoring PM2	**Robot wafer position plausibility error**
26002	2600	Handling Monitoring	**PM2 load ECO wafer position plausibility error**

		PM2	
26003	2600	Handling Monitoring PM2	**PM2 unload ECO wafer position plausibility error**
27101	2710	PreAligner1	**PreAligner 1 not available**
27102	2710	PreAligner1	**PreAligner 1 init error**
27103	2710	PreAligner1	**PreAligner 1 module disabled**
27104	2710	PreAligner1	**PreAligner 1 system not ready**
27105	2710	PreAligner1	**PreAligner 1 pre process load error**
27106	2710	PreAligner1	**PreAligner 1 pre process align error**
27107	2710	PreAligner1	**PreAligner 1 pre process unload error**
27108	2710	PreAligner1	**PreAligner 1 post process load error**
27109	2710	PreAligner1	**PreAligner 1 post process align error**
271010	2710	PreAligner1	**PreAligner 1 post process unload error**
271011	2710	PreAligner1	**PreAligner 1 system error**
271012	2710	PreAligner1	**PreAligner 1 alignment error**
271013	2710	PreAligner1	**PreAligner 1 no wafer error**
271014	2710	PreAligner1	**PreAligner 1 recipe mismatch**
271015	2710	PreAligner1	**PreAligner 1 pre process devices load error**
271016	2710	PreAligner1	**PreAligner 1 post process devices load error**
271017	2710	PreAligner1	**PreAligner 1 PreProcess Wid1 Alignment error**
271018	2710	PreAligner1	**PreAligner 1 PreProcess Wid2 Alignment error**
271019	2710	PreAligner1	**PreAligner 1 PreProcess Dmc Alignment error**
271020	2710	PreAligner1	**PreAligner 1 PreProcess Irm Alignment error**
271021	2710	PreAligner1	**PreAligner 1 PostProcess Irm Alignment error**
271022	2710	PreAligner1	**PreAligner 1 External Sensor is disconected**
271023	2710	PreAligner1	**PreAligner 1 External Sensor is disabled**
271024	2710	PreAligner1	**PreAligner 1 Fatal Info Error**
271025	2710	PreAligner1	**PreAligner 1 External Sensor ATMSi not in home position**
271026	2710	PreAligner1	**PreAligner 1 Milara SensorHead Movement TimeOut**
271027	2710	PreAligner1	**PreAligner 1 Milara SensorHead Error**
27201	2720	PreAligner2	**PreAligner 2 not available**
27202	2720	PreAligner2	**PreAligner 2 init error**
27203	2720	PreAligner2	**PreAligner 2 module disabled**
27204	2720	PreAligner2	**PreAligner 2 system not ready**
27205	2720	PreAligner2	**PreAligner 2 pre process load error**
27206	2720	PreAligner2	**PreAligner 2 pre process align error**
27207	2720	PreAligner2	**PreAligner 2 pre process unload error**
27208	2720	PreAligner2	**PreAligner 2 post process load error**
27209	2720	PreAligner2	**PreAligner 2 post process align error**
272010	2720	PreAligner2	**PreAligner 2 post process unload error**
272011	2720	PreAligner2	**PreAligner 2 system error**
272012	2720	PreAligner2	**PreAligner 2 alignment error**
272013	2720	PreAligner2	**PreAligner 2 no wafer error**
272014	2720	PreAligner2	**PreAligner 2 recipe mismatch**
272015	2720	PreAligner2	**PreAligner 2 pre process devices load error**
272016	2720	PreAligner2	**PreAligner 2 post process devices load error**
272017	2720	PreAligner2	**PreAligner 2 PreProcess Wid1 Alignment error**

272018	2720	PreAligner2	**PreAligner 2 PreProcess Wid2 Alignment error**
272019	2720	PreAligner2	**PreAligner 2 PreProcess Dmc Alignment error**
272020	2720	PreAligner2	**PreAligner 2 PreProcess Irm Alignment error**
272021	2720	PreAligner2	**PreAligner 2 PostProcess Irm Alignment error**
272022	2720	PreAligner2	**PreAligner 2 External Sensor is disconected**
272023	2720	PreAligner2	**PreAligner 2 External Sensor is disabled**
272024	2720	PreAligner2	**PreAligner 2 Fatal Info Error**
272025	2720	PreAligner2	**PreAligner 2 External Sensor ATMSi not in home position**
272026	2720	PreAligner2	**PreAligner 2 Milara SensorHead Movement TimeOut**
272027	2720	PreAligner2	**PreAligner 2 Milara SensorHead Error**
28101	2810	Atmsi1	**Atmsi 1 not available**
28102	2810	Atmsi1	**Atmsi 1 init error**
28103	2810	Atmsi1	**Atmsi 1 module disabled**
28104	2810	Atmsi1	**Atmsi 1 system not ready**
28105	2810	Atmsi1	**Atmsi 1 recipe mismatch**
28106	2810	Atmsi1	**Atmsi 1 pre process load error**
28107	2810	Atmsi1	**Atmsi 1 pre process align error**
28108	2810	Atmsi1	**Atmsi 1 pre process unload error**
28109	2810	Atmsi1	**Atmsi 1 post process load error**
281010	2810	Atmsi1	**Atmsi 1 post process align error**
281011	2810	Atmsi1	**Atmsi 1 post process unload error**
281012	2810	Atmsi1	**Atmsi1 system error**
281013	2810	Atmsi1	**Atmsi 1 alignment error**
281014	2810	Atmsi1	**Atmsi 1 no wafer error**
281015	2810	Atmsi1	**Atmsi 1 X Axis motion control error**
281016	2810	Atmsi1	**Atmsi 1 X Axis motor module power supply error**
281017	2810	Atmsi1	**Atmsi 1 X Axis motor stall error**
281018	2810	Atmsi1	**Atmsi 1 X Axis overtemperature error**
281019	2810	Atmsi1	**Atmsi 1 X Axis motor current error**
281020	2810	Atmsi1	**Atmsi 1 X Axis motor overcurrent error**
281021	2810	Atmsi1	**Atmsi 1 X Axis positioning error**
281022	2810	Atmsi1	**Atmsi 1 X Axis belt broken**
281023	2810	Atmsi1	**Atmsi 1 Z Axis motion control error**
281024	2810	Atmsi1	**Atmsi 1 Z Axis motor module power supply error**
281025	2810	Atmsi1	**Atmsi 1 Z Axis motor stall error**
281026	2810	Atmsi1	**Atmsi 1 Z Axis overtemperature error**
281027	2810	Atmsi1	**Atmsi 1 Z Axis motor current error**
281028	2810	Atmsi1	**Atmsi 1 Z Axis motor overcurrent error**
281029	2810	Atmsi1	**Atmsi 1 Z Axis positioning error**
281030	2810	Atmsi1	**Atmsi 1 Z Axis belt broken**
28201	2820	Atmsi2	**Atmsi 2 not available**
28202	2820	Atmsi2	**Atmsi 2 init error**
28203	2820	Atmsi2	**Atmsi 2 module disabled**
28204	2820	Atmsi2	**Atmsi 2 system not ready**
28205	2820	Atmsi2	**Atmsi 2 recipe mismatch**
28206	2820	Atmsi2	**Atmsi 2 pre process load error**

28207	2820	Atmsi2	Atmsi 2 pre process align error
28208	2820	Atmsi2	Atmsi 2 pre process unload error
28209	2820	Atmsi2	Atmsi 2 post process load error
282010	2820	Atmsi2	Atmsi 2 post process align error
282011	2820	Atmsi2	Atmsi 2 post process unload error
282012	2820	Atmsi2	Atmsi2 system error
282013	2820	Atmsi2	Atmsi 2 alignment error
282014	2820	Atmsi2	Atmsi 2 no wafer error
282015	2820	Atmsi2	Atmsi 2 X Axis motion control error
282016	2820	Atmsi2	Atmsi 2 X Axis motor module power supply error
282017	2820	Atmsi2	Atmsi 2 X Axis motor stall error
282018	2820	Atmsi2	Atmsi 2 X Axis overtemperature error
282019	2820	Atmsi2	Atmsi 2 X Axis motor current error
282020	2820	Atmsi2	Atmsi 2 X Axis motor overcurrent error
282021	2820	Atmsi2	Atmsi 2 X Axis positioning error
282022	2820	Atmsi2	Atmsi 2 X Axis belt broken
282023	2820	Atmsi2	Atmsi 2 Z Axis motion control error
282024	2820	Atmsi2	Atmsi 2 Z Axis motor module power supply error
282025	2820	Atmsi2	Atmsi 2 Z Axis motor stall error
282026	2820	Atmsi2	Atmsi 2 Z Axis overtemperature error
282027	2820	Atmsi2	Atmsi 2 Z Axis motor current error
282028	2820	Atmsi2	Atmsi 2 Z Axis motor overcurrent error
282029	2820	Atmsi2	Atmsi 2 Z Axis positioning error
282030	2820	Atmsi2	Atmsi 2 Z Axis belt broken
29101	2910	FireSuppresionSys	Fire Suppression System Disabled / Fault Signal
29102	2910	FireSuppresionSys	Fire Suppression System fire detected
29103	2910	FireSuppresionSys	Fire Suppression System triggered
29201	2920	Wid11	Wid11 System Error
29202	2920	Wid11	Wid11 ID Reading Failed
29203	2920	Wid11	Wid11 System Not Ready
29204	2920	Wid11	Wid11 ID Module Disabled
29205	2920	Wid11	Wid11 No Wafer Id Available
29301	2930	Wid21	Wid21 System Error
29302	2930	Wid21	Wid21 ID Reading Failed
29303	2930	Wid21	Wid21 System Not Ready
29304	2930	Wid21	Wid21 ID Module Disabled
29305	2930	Wid21	Wid21 No Wafer Id Available
29401	2940	Wid12	Wid12 System Error
29402	2940	Wid12	Wid12 ID Reading Failed
29403	2940	Wid12	Wid12 System Not Ready
29404	2940	Wid12	Wid12 ID Module Disabled
29405	2940	Wid12	Wid12 No Wafer Id Available
29501	2950	Wid22	Wid22 System Error
29502	2950	Wid22	Wid22 ID Reading Failed
29503	2950	Wid22	Wid22 System Not Ready
29504	2950	Wid22	Wid22 ID Module Disabled

29505	2950	Wid22	**Wid22 No Wafer Id Available**
29601	2960	Dmc1	**Dmc1 System Error**
29602	2960	Dmc1	**Dmc1 ID Reading Failed**
29603	2960	Dmc1	**Dmc1 System Not Ready**
29604	2960	Dmc1	**Dmc1 ID Module Disabled**
29605	2960	Dmc1	**Dmc1 No Wafer Id Available**
29701	2970	Dmc2	**Dmc2 System Error**
29702	2970	Dmc2	**Dmc2 ID Reading Failed**
29703	2970	Dmc2	**Dmc2 System Not Ready**
29704	2970	Dmc2	**Dmc2 ID Module Disabled**
29705	2970	Dmc2	**Dmc2 No Wafer Id Available**
29801	2980	Irm1	**Irm1 System Error**
29802	2980	Irm1	**Irm1 ID Reading Failed**
29803	2980	Irm1	**Irm1 System Not Ready**
29804	2980	Irm1	**Irm1 System Not Ready**
29805	2980	Irm1	**Irm1 Module Disabled**
29806	2980	Irm1	**Irm1 No Irm Available**
29807	2980	Irm1	**Irm1 Recipe Mismatch**
29808	2980	Irm1	**Irm1 Pre Process Read Error**
29809	2980	Irm1	**Irm1 Post Process Read Error**
298010	2980	Irm1	**Irm1 Pre Process Load Error**
298011	2980	Irm1	**Irm1 Post Process Load Error**
29901	2990	Irm2	**Irm1 System Error**
29902	2990	Irm2	**Irm1 ID Reading Failed**
29903	2990	Irm2	**Irm1 System Not Ready**
29904	2990	Irm2	**Irm1 System Not Ready**
29905	2990	Irm2	**Irm1 Module Disabled**
29906	2990	Irm2	**Irm1 No Irm Available**
29907	2990	Irm2	**Irm1 Recipe Mismatch**
29908	2990	Irm2	**Irm1 Pre Process Read Error**
29909	2990	Irm2	**Irm1 Post Process Read Error**
299010	2990	Irm2	**Irm1 Pre Process Load Error**
299011	2990	Irm2	**Irm1 Post Process Load Error**
30001	3000	Abort	**Abort Cause: Cds Error**
30002	3000	Abort	**Abort Cause: Heater Error**
30003	3000	Abort	**Abort Cause: Cooler Error**
30004	3000	Abort	**Abort Cause: Leak Error**
30005	3000	Abort	**Abort Cause: O3 Error**
30006	3000	Abort	**Abort Cause: Safety Error**
30007	3000	Abort	**Abort Cause: Facility Error**
30008	3000	Abort	**Abort Cause: Atmsi Error**
30009	3000	Abort	**Abort Cause: Aligner Error**
300010	3000	Abort	**Abort Cause: Wid error**
300011	3000	Abort	**Abort Cause: Irm error**
300012	3000	Abort	**Abort Cause: ChemAnalzyer Error**
30101	3010	Stop	**Stop Cause: Cds Error**

30102	3010	Stop	**Stop Cause: Facility Error**
30103	3010	Stop	**Stop Cause: Atmsi Error**
30104	3010	Stop	**Stop Cause: Aligner Error**
30105	3010	Stop	**Stop Cause: Wid error**
30106	3010	Stop	**Stop Cause: ChemAnalzyer Error**
30107	3010	Stop	**Stop Cause: Irm error**

9 Examples

This chapter provides examples for common scenarios and sequences provided by the MG-Series Wet Chemical Processor.

9.1 Lot Start – Load Cassette to Unload Cassette processing

The following example shows how to start a lot beginning from the initial connection from Host to Equipment.

- Sequence
  1. Host establishes communication
  2. Host deletes all existing report definitions
  3. Host deletes all existing report links
  4. Host disables all event reports
  5. Host defines report definitions
  6. Host defines report links
  7. Host enables events
  8. Host checks if all necessary events are enabled
  9. Host requests transition to Online Remote control state
  10. Host checks the current control state
  11. Host requests to map the first cassette
  12. Host selects a recipe and assigns it to a LoadPort
  13. Host requests to start processing

Sequence Details

9.1.1.1 S1F13 Establish Communication Request H → E

Host initiates the communication to the equipment.

```
13:57:33        [SENDING MESSAGE]
13:57:33        S1F13 [00000012] W
13:57:33        <L[0]
13:57:33        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
13:57:34        [RECEIVED MESSAGE]
13:57:34        S1F14 [00000012]
13:57:34        <L[2]
13:57:34          <B[1] 00>
13:57:34          <L[2]
13:57:34            <A[4] 'MG22'>
13:57:34            <A[7] '3.7.0.0'>
13:57:34          >
13:57:34        >
```

9.1.1.2 S2F33 Define Report H → E

Host sends a zero-length list to delete all report definitions and associated links.

```
14:20:51        [SENDING MESSAGE]
14:20:51        S2F33 [00000030] W
14:20:51        <L[2]
14:20:51          <U4[1] 0>
14:20:51          <L[0]
14:20:51          >
14:20:51        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:20:52        [RECEIVED MESSAGE]
14:20:52        S2F34 [00000030]
14:20:52        <B[1] 00>
```

9.1.1.3 S2F35 Link Event Report H → E

Host sends a zero-length list to unlink all reports.

```
14:20:51	[SENDING MESSAGE]
14:20:51	S2F35 [00000021] W
14:20:51	<L[2]
14:20:51	  <U4[1] 0>
14:20:51	  <L[0]
14:20:51	  >
14:20:51	>
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:20:52	[RECEIVED MESSAGE]
14:20:52	S2F36 [00000021]
14:20:52	<B[1] 00>
```

9.1.1.4 S2F37 Enable/Disable Event Report H → E

Host sends a zero-length list following CEED (false) to deactivate all CEIDs.

```
14:23:44	[SENDING MESSAGE]
14:23:44	S2F37 [00000034] W
14:23:44	<L[2]
14:23:44	  <Bool[5] false>
14:23:44	  <L[0]
14:23:44	  >
14:23:44	>
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:23:44	[RECEIVED MESSAGE]
14:23:44	S2F38 [00000034]
14:23:44	<B[1] 00>
```

9.1.1.5 S2F33 Define Report H → E

Host defines a group of (three) event reports.

- Report 1 (RPTID 1)
    - DVAR 1835, jobIdLastStartedLot
    - DVAR 1836, lotIdLastStartedLot
- Report 2 (RPTID 2)
    - DVAR 1830, jobIdLastFinishedLot
    - DVAR 1831, lotIdLastFinishedLot
- Report 3 (RPTID 3)
    - SVAR 4306, mapResultLastMap

Note: SVAR 4306 is a list representing the result of the last map operation that has been performed on equipment. By using this variable to get the map result one only needs to map one SVAR to get the map result for each cassette slot.

```
14:32:26    [SENDING MESSAGE]
14:32:26    S2F33 [00000044] W
14:32:26    <L[2]
14:32:26      <U4[3] 100>
14:32:26      <L[3]
14:32:26        <L[2]
14:32:26          <U4[1] 1>
14:32:26          <L[2]
14:32:26            <U4[4] 1835>
14:32:26            <U4[4] 1836>
14:32:26          >
14:32:26        >
14:32:26        <L[2]
14:32:26          <U4[1] 2>
14:32:26          <L[2]
14:32:26            <U4[4] 1830>
14:32:26            <U4[4] 1831>
14:32:26          >
14:32:26        >
14:32:26        <L[2]
14:32:26          <U4[1] 3>
14:32:26          <L[1]
14:32:26            <U4[3] 4306>
14:32:26          >
14:32:26        >
14:32:26      >
14:32:26    >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:32:26    [RECEIVED MESSAGE]
14:32:26    S2F34 [00000044]
14:32:26    <B[1] 00>
```

9.1.1.6 S2F35 Link Event Report H → E

Host links the event reports defined in the previous scenario to CEIDs.

- RPTID 1 → CEID 4, ProcessingStarted
- RPTID 2 → CEID 5, ProcessingCompleted
- RPTID 3 → CEID 140, port1CasMapped
- RPTID 3 → CEID 141, port2CasMapped

```
14:41:16      [SENDING MESSAGE]
14:41:16      S2F35 [00000054] W
14:41:16      <L[5]
14:41:16        <U4[1] 0>
14:41:16        <L[1]
14:41:16          <L[2]
14:41:16            <U4[1] 4>
14:41:16            <L[1]
14:41:16              <U4[1] 1>
14:41:16            >
14:41:16          >
14:41:16        >
14:41:16        <L[1]
14:41:16          <L[2]
14:41:16            <U4[1] 5>
14:41:16            <L[1]
14:41:16              <U4[1] 2>
14:41:16            >
14:41:16          >
14:41:16        >
14:41:16        <L[1]
14:41:16          <L[2]
14:41:16            <U4[3] 140>
14:41:16            <L[1]
14:41:16              <U4[1] 3>
14:41:16            >
14:41:16          >
14:41:16        >
14:41:16        <L[1]
14:41:16          <L[2]
14:41:16            <U4[3] 141>
14:41:16            <L[1]
14:41:16              <U4[1] 3>
14:41:16            >
14:41:16          >
14:41:16        >
14:41:16      >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:41:16      [RECEIVED MESSAGE]
14:41:16      S2F36 [00000054]
14:41:16      <B[1] 00>
```

9.1.1.7 S2F37 Enable/Disable Event Report H → E

Host enables relevant CEIDs on equipment. Note that events 4,5,140,141 that have been linked to reports in the previous step, while 13(processRecipeSelected), 130(port1CasPlaced) and 131(port2CasPlaced) are enabled without linked report. Not all events must have an event report linked to it.

```
14:47:40	[SENDING MESSAGE]
14:47:40	S2F37 [00000062] W
14:47:40	<L[2]
14:47:40	  <Bool[4] true>
14:47:40	  <L[7]
14:47:40	    <U4[1] 4>
14:47:40	    <U4[1] 5>
14:47:40	    <U4[2] 13>
14:47:40	    <U4[3] 130>
14:47:40	    <U4[3] 131>
14:47:40	    <U4[3] 140>
14:47:40	    <U4[3] 141>
14:47:40	  >
14:47:40	>
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:47:41	[RECEIVED MESSAGE]
14:47:41	S2F38 [00000062]
14:47:41	<B[1] 00>
```

9.1.1.8 S1F3 Query Status Variable 12(EventsEnabled) H → E

Host requests the list of enabled CEIDs to ensure that all relevant events are activated.

```
14:52:36	[SENDING MESSAGE]
14:52:36	S1F3 [00000068] W
14:52:36	<L[1]
14:52:36	  <U4[2] 12>
14:52:36	>
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:52:37	[RECEIVED MESSAGE]
14:52:37	S1F4 [00000068]
14:52:37	<L[1]
14:52:37	  <L[7]
14:52:37	    <U4[1] 4>
14:52:37	    <U4[1] 5>
14:52:37	    <U4[2] 13>
14:52:37	    <U4[3] 143>
14:52:37	    <U4[3] 144>
14:52:37	    <U4[3] 140>
14:52:37	    <U4[3] 141>
14:52:37	  >
14:52:37	>
```

9.1.1.9 S2F41 Host command Send – REMOTE H → E

Host requests a transition to control state remote

```
14:54:34        [SENDING MESSAGE]
14:54:34        S2F41 [00000071] W
14:54:34        <L[1]
14:54:34          <A[6] 'REMOTE'>
14:54:34        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:54:34        [RECEIVED MESSAGE]
14:54:34        S2F42 [00000071]
14:54:34        <L[2]
14:54:34          <B[1] 00>
14:54:34          <L[0]
14:54:34          >
14:54:34        >
```

9.1.1.10 S1F3 Query Status Variable 11(ControlState) H → E

Host queries the current control state before continuing to ensure that the equipment is in Online Remote control state (reply: 5).

```
14:56:02        [SENDING MESSAGE]
14:56:02        S1F3 [00000073] W
14:56:02        <L[1]
14:56:02          <U4[2] 11>
14:56:02        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:56:03        [RECEIVED MESSAGE]
14:56:03        S1F4 [00000073]
14:56:03        <L[1]
14:56:03          <U4[1] 5>
14:56:03        >
```

9.1.1.11 S2F41 Host Command Send – PPSELECT H → E

Host selects a recipe on the equipment for a specific port/cassette (load port 1) and assigns CarrierId, LotId and JobId.

```
14:59:02        [SENDING MESSAGE]
14:59:02        S2F41 [00000079] W
14:59:02        <L[2]
14:59:02          <A[8] 'PPSELECT'>
14:59:02          <L[5]
14:59:02            <L[2]
14:59:02              <A[4] 'PPID'>
14:59:02              <A[4] 'DIN2'>
14:59:02            >
14:59:02            <L[2]
14:59:02              <A[6] 'PORTID'>
14:59:02              <U2[1] 1>
14:59:02            >
14:59:02            <L[2]
14:59:02              <A[9] 'CARRIERID'>
14:59:02              <A[7] 'THECID1'>
14:59:02            >
14:59:02            <L[2]
14:59:02              <A[5] 'LOTID'>
14:59:02              <A[8] 'THELOTID'>
14:59:02            >
14:59:02            <L[2]
14:59:02              <A[5] 'JOBID'>
14:59:02              <A[6] 081588>
14:59:02            >
14:59:02          >
14:59:02        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:59:03        [RECEIVED MESSAGE]
14:59:03        S2F42 [00000079]
14:59:03        <L[2]
14:59:03          <B[1] 04>
14:59:03          <L[0]
14:59:03          >
14:59:03        >
```

Host receives event reporting for CEID 13 (processRecipeSelected).

```
14:59:03        [RECEIVED MESSAGE]
14:59:03        S6F11 [00000005] W
14:59:03        <L[3]
14:59:03          <U4[1] 0>
14:59:03          <U4[2] 13>
14:59:03          <L[0]
14:59:03          >
14:59:03        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:59:03        [SENDING MESSAGE]
14:59:03        S6F12 [00000005]
```

```
14:59:03          <B[1] 00>
```

9.1.1.12 S2F41 Host Command Send – MAP(Port1) H → E

Host requests the equipment to map the cassette on port 1.

```
15:03:46          [SENDING MESSAGE]
15:03:46          S2F41 [00000085]
15:03:46          <L[2]
15:03:46            <A[3] 'MAP'>
15:03:46            <L[1]
15:03:46              <L[2]
15:03:46                <A[6] 'PORTID'>
15:03:46                <U4[1] 1>
15:03:46              >
15:03:46            >
15:03:46          >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
15:03:46          [RECEIVED MESSAGE]
15:03:46          S2F42 [00000085]
15:03:46          <L[2]
15:03:46            <B[1] 04>
15:03:46            <L[0]
15:03:46            >
15:03:46          >
```

Once the cassette on port 1 has been mapped (CEID 140) the host receives the map result via the linked report (RPTID 3).

```
15:04:46          [RECEIVED MESSAGE]
15:04:46          S6F11 [00000007] W
15:04:46          <L[3]
15:04:46            <U4[1] 0>
15:04:46            <U4[3] 140>
15:04:46            <L[1]
15:04:46              <L[2]
15:04:46                <U4[2] 3
15:04:46                <L[1]
15:04:46                  <L[25]
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
15:04:46                    <U1[1] 1>
```

```
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46          <U1[1] 1>
15:04:46        >
15:04:46      >
15:04:46    >
15:04:46  >
15:04:46 >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
15:04:46 [SENDING MESSAGE]
15:04:46 S6F12 [00000007]
15:04:46 <B[1] 00>
```

9.1.1.13 S2F41 Host Command Send – MAP(Port2) H → E

Host requests the equipment to map the cassette on port 2.

```
15:09:52        [SENDING MESSAGE]
15:09:52        S2F41 [00000092]
15:09:52        <L[2]
15:09:52          <A[3] 'MAP'>
15:09:52          <L[1]
15:09:52            <L[2]
15:09:52              <A[6] 'PORTID'>
15:09:52              <U4[1] 2>
15:09:52            >
15:09:52          >
15:09:52        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
15:09:52        [RECEIVED MESSAGE]
15:09:52        S2F42 [00000092]
15:09:52        <L[2]
15:09:52          <B[1] 04>
15:09:52          <L[0]
15:09:52          >
15:09:52        >
```

Once the cassette on port 2 has been mapped (CEID 141) the host receives the map result via the linked report (RPTID 3).

```
15:10:52        [RECEIVED MESSAGE]
15:10:52        S6F11 [00000009] W
15:10:52        <L[3]
15:10:52          <U4[1] 0>
15:10:52          <U4[3] 141>
15:10:52          <L[1]
15:10:52            <L[2]
15:10:52              <U4[2] 3
15:10:52              <L[1]
15:10:52                <L[25]
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
15:10:52                  <U1[1] 2>
```

```
15:10:52                    <U1[1] 2>
15:10:52                    <U1[1] 2>
15:10:52                    <U1[1] 2>
15:10:52                    <U1[1] 2>
15:10:52                    <U1[1] 2>
15:10:52                    <U1[1] 2>
15:10:52                    <U1[1] 2>
15:10:52                    <U1[1] 2>
15:10:52                  >
15:10:52                >
15:10:52              >
15:10:52            >
15:10:52          >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
15:10:52          [SENDING MESSAGE]
15:10:52          S6F12 [00000009]
15:10:52          <B[1] 00>
```

9.1.1.14 S2F41 Host Command Send – START H → E

Host requests the equipment to start processing from load port 1.

```
15:13:29        [SENDING MESSAGE]
15:13:29        S2F41 [00000096] W
15:13:29        <L[2]
15:13:29          <A[5] 'START'>
15:13:29          <L[1]
15:13:29            <L[2]
15:13:29              <A[6] 'PORTID'>
15:13:29              <I2[1] 1>
15:13:29            >
15:13:29          >
15:13:29        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
15:13:29        [RECEIVED MESSAGE]
15:13:29        S2F42 [00000096]
15:13:29        <L[2]
15:13:29          <B[1] 04>
15:13:29          <L[0]
15:13:29          >
15:13:29        >
```

Host received event reporting of linked report RPTID 1 for CEID 4 (ProcessingStarted) once the equipment started processing from load port 1.

```
15:13:29        [RECEIVED MESSAGE]
15:13:29        S6F11 [00000011] W
15:13:29        <L[3]
15:13:29          <U4[1] 0>
15:13:29          <U4[1] 4>
15:13:29          <L[1]
15:13:29            <L[2]
15:13:29              <U4[1] 1>
15:13:29              <L[2]
15:13:29                <U1[1] 1>
15:13:29                <A[8] 'THELOTID'>
15:13:29              >
15:13:29            >
15:13:29          >
15:13:29        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
15:13:29        [SENDING MESSAGE]
15:13:29        S6F12 [00000011]
15:13:29        <B[1] 00>
```

9.2 GEM300 Lot Start

The following example shows how to start a lot beginning from the initial connection from Host to Equipment. Note that GEM300 sequence is only supported on equipment with FOUP ports.

- Sequence
  1. Host establishes communication
  2. Host deletes all existing report definitions
  3. Host deletes all existing report links
  4. Host disables all event reports
  5. Host defines report definitions
  6. Host defines report links
  7. Host enables events
  8. Host requests transition to Online Remote control state
  9. Host checks the current control state
  10. Host sends Carrier Bind request
  11. FOUP loaded and mapped
  12. Host creates Process Job
  13. Host creates Control Job to start processing

Sequence Details

9.2.1.1 S1F13 Establish Communication Request H → E

Host initiates the communication to the equipment.

```
13:57:33        [SENDING MESSAGE]
13:57:33        S1F13 [00000012] W
13:57:33        <L[0]
13:57:33        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
13:57:34        [RECEIVED MESSAGE]
13:57:34        S1F14 [00000012]
13:57:34        <L[2]
13:57:34          <B[1] 00>
13:57:34          <L[2]
13:57:34            <A[4] 'MG22'>
13:57:34            <A[7] '3.7.0.0'>
13:57:34          >
13:57:34        >
```

9.2.1.2 S2F33 Define Report H → E

Host sends a zero-length list to delete all report definitions and associated links.

```
14:20:51        [SENDING MESSAGE]
14:20:51        S2F33 [00000030] W
14:20:51        <L[2]
14:20:51          <U4[1] 0>
14:20:51          <L[0]
14:20:51          >
14:20:51        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:20:52        [RECEIVED MESSAGE]
14:20:52        S2F34 [00000030]
14:20:52        <B[1] 00>
```

9.2.1.3 S2F35 Link Event Report H → E

Host sends a zero-length list to unlink all reports.

```
14:20:51    [SENDING MESSAGE]
14:20:51    S2F35 [00000021] W
14:20:51    <L[2]
14:20:51      <U4[1] 0>
14:20:51      <L[0]
14:20:51      >
14:20:51    >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:20:52    [RECEIVED MESSAGE]
14:20:52    S2F36 [00000021]
14:20:52    <B[1] 00>
```

9.2.1.4 S2F37 Enable/Disable Event Report H → E

Host sends a zero-length list following CEED (false) to deactivate all CEIDs.

```
14:23:44    [SENDING MESSAGE]
14:23:44    S2F37 [00000034] W
14:23:44    <L[2]
14:23:44      <Bool[5] false>
14:23:44      <L[0]
14:23:44      >
14:23:44    >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:23:44    [RECEIVED MESSAGE]
14:23:44    S2F38 [00000034]
14:23:44    <B[1] 00>
```

9.2.1.5 S2F33 Define Report H → E

Host defines a group of (three) event reports.

- Report 1 (RPTID 1)
    - DVAR 1835, jobIdLastStartedLot
    - DVAR 1836, lotIdLastStartedLot
- Report 2 (RPTID 2)
    - DVAR 1830, jobIdLastFinishedLot
    - DVAR 1831, lotIdLastFinishedLot
- Report 3 (RPTID 3)
    - SVAR 4306, mapResultLastMap

Note: SVAR 4306 is a list representing the result of the last map operation that has been performed on equipment. By using this variable to get the map result one only needs to map one SVAR to get the map result for each cassette slot.

```
14:32:26    [SENDING MESSAGE]
14:32:26    S2F33 [00000044] W
14:32:26    <L[2]
14:32:26      <U4[3] 100>
14:32:26      <L[3]
14:32:26        <L[2]
14:32:26          <U4[1] 1>
14:32:26          <L[2]
14:32:26            <U4[4] 1835>
14:32:26            <U4[4] 1836>
14:32:26          >
14:32:26        >
14:32:26        <L[2]
14:32:26          <U4[1] 2>
14:32:26          <L[2]
14:32:26            <U4[4] 1830>
14:32:26            <U4[4] 1831>
14:32:26          >
14:32:26        >
14:32:26        <L[2]
14:32:26          <U4[1] 3>
14:32:26          <L[1]
14:32:26            <U4[3] 4306>
14:32:26          >
14:32:26        >
14:32:26      >
14:32:26    >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:32:26    [RECEIVED MESSAGE]
14:32:26    S2F34 [00000044]
14:32:26    <B[1] 00>
```

9.2.1.6 S2F35 Link Event Report H → E

Host links the event reports defined in the previous scenario to CEIDs.

- RPTID 1 → CEID 4, ProcessingStarted
- RPTID 2 → CEID 5, ProcessingCompleted
- RPTID 3 → CEID 140, port1CasMapped
- RPTID 3 → CEID 141, port2CasMapped

```
14:41:16        [SENDING MESSAGE]
14:41:16        S2F35 [00000054] W
14:41:16        <L[5]
14:41:16          <U4[1] 0>
14:41:16          <L[1]
14:41:16            <L[2]
14:41:16              <U4[1] 4>
14:41:16              <L[1]
14:41:16                <U4[1] 1>
14:41:16              >
14:41:16            >
14:41:16          >
14:41:16          <L[1]
14:41:16            <L[2]
14:41:16              <U4[1] 5>
14:41:16              <L[1]
14:41:16                <U4[1] 2>
14:41:16              >
14:41:16            >
14:41:16          >
14:41:16          <L[1]
14:41:16            <L[2]
14:41:16              <U4[3] 140>
14:41:16              <L[1]
14:41:16                <U4[1] 3>
14:41:16              >
14:41:16            >
14:41:16          >
14:41:16          <L[1]
14:41:16            <L[2]
14:41:16              <U4[3] 141>
14:41:16              <L[1]
14:41:16                <U4[1] 3>
14:41:16              >
14:41:16            >
14:41:16          >
14:41:16        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:41:16        [RECEIVED MESSAGE]
14:41:16        S2F36 [00000054]
14:41:16        <B[1] 00>
```

9.2.1.7 S2F37 Enable/Disable Event Report H → E

Host enables relevant CEIDs on equipment. Note that events 4,5,140,141 that have been linked to reports in the previous step, while 13(processRecipeSelected), 130(port1CasPlaced) and 131(port2CasPlaced) are enabled without linked report. Not all events must have an event report linked to it.

```
14:47:40    [SENDING MESSAGE]
14:47:40    S2F37 [00000062] W
14:47:40    <L[2]
14:47:40      <Bool[4] true>
14:47:40      <L[7]
14:47:40        <U4[1] 4>
14:47:40        <U4[1] 5>
14:47:40        <U4[2] 13>
14:47:40        <U4[3] 130>
14:47:40        <U4[3] 131>
14:47:40        <U4[3] 140>
14:47:40        <U4[3] 141>
14:47:40      >
14:47:40    >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:47:41    [RECEIVED MESSAGE]
14:47:41    S2F38 [00000062]
14:47:41    <B[1] 00>
```

9.2.1.8 S2F41 Host command Send – REMOTE H → E

Host requests a transition to control state remote

```
14:54:34        [SENDING MESSAGE]
14:54:34        S2F41 [00000071] W
14:54:34        <L[1]
14:54:34          <A[6] 'REMOTE'>
14:54:34        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:54:34        [RECEIVED MESSAGE]
14:54:34        S2F42 [00000071]
14:54:34        <L[2]
14:54:34          <B[1] 00>
14:54:34          <L[0]
14:54:34          >
14:54:34        >
```

9.2.1.9 S1F3 Query Status Variable 11(ControlState) H → E

Host queries the current control state before continuing to ensure that the equipment is in Online Remote control state (reply: 5).

```
14:56:02        [SENDING MESSAGE]
14:56:02        S1F3 [00000073] W
14:56:02        <L[1]
14:56:02          <U4[2] 11>
14:56:02        >
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
14:56:03        [RECEIVED MESSAGE]
14:56:03        S1F4 [00000073]
14:56:03        <L[1]
14:56:03          <U4[1] 5>
14:56:03        >
```

9.2.1.10 S3F17 Carrier Action Request – Bind H → E

Host sends Carrier Bind request with ContentMap definition and expected CarrierID and SlotMap

```
17:49:05    [SENDING MESSAGE]
17:49:05    S3F17 [00000021] W
17:49:05    <L[5]
17:49:05      <U4[1] 1>
17:49:05      <A[4] 'Bind'>
17:49:05      <A[16] '0039640900000000'>
17:49:05      <U1[1] 1>
17:49:05      <L[5]
17:49:05        <L[2]
17:49:05          <A[8] 'Capacity'>
17:49:05          <U1[2] 25>
17:49:05        >
17:49:05        <L[2]
17:49:05          <A[5] 'Usage'>
17:49:05          <A[7] 'Product'>
17:49:05        >
17:49:05        <L[2]
17:49:05          <A[14] 'SubstrateCount'>
17:49:05          <U1[2] 12>
17:49:05        >
17:49:05        <L[2]
17:49:05          <A[7] 'SlotMap'>
17:49:05          <L[25]
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 1>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 3>
17:49:05            <U1[1] 3>
17:49:05          >
```

```
17:49:05        >
17:49:05        <L[2]
17:49:05          <A[10] 'ContentMap'>
17:49:05          <L[25]
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer01'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer02'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer03'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer04'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer05'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer06'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer07'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer08'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer09'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer10'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer11'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer12'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
```

```
17:49:05              <A[7] 'Wafer13'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer14'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer15'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer16'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer17'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer18'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer19'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer20'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer21'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer22'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer23'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer24'>
17:49:05            >
17:49:05            <L[2]
17:49:05              <A[5] 'Lot01'>
17:49:05              <A[7] 'Wafer25'>
17:49:05            >
17:49:05          >
17:49:05        >
17:49:05      >
17:49:05    >
```

```
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
17:49:06        [RECEIVED MESSAGE]
17:49:06        S3F18 [00000021]
17:49:06        <L[2]
17:49:06          <U1[1] 0>
17:49:06          <L[0]
17:49:06          >
17:49:06        >
```

9.2.1.11 FOUP loaded and mapped

A carrier is placed on FOUP and auto mapped.

9.2.1.12 S16F15 PRJobMultiCreate H → E

Host creates Process Job with auto start and specifies slots to process and Recipe to run with

```
17:49:08	[SENDING MESSAGE]
17:49:08	S16F15 [00000022] W
17:49:08	<L[2]
17:49:08	  <U4[1] 1>
17:49:08	  <L[1]
17:49:08	    <L[6]
17:49:08	      <A[3] 'PJ1'>
17:49:08	      <B[1] 0D>
17:49:08	      <L[1]
17:49:08	        <L[2]
17:49:08	          <A[16] '0039640900000000'>
17:49:08	          <L[1]
17:49:08	            <U1[1] 1>
17:49:08	          >
17:49:08	        >
17:49:08	      >
17:49:08	      <L[3]
17:49:08	        <U1[1] 1>
17:49:08	        <A[16] 'HandlingOnlyTest'>
17:49:08	        <L[0]
17:49:08	        >
17:49:08	      >
17:49:08	      <Bool[4] true>
17:49:08	      <L[0]
17:49:08	      >
17:49:08	    >
17:49:08	  >
17:49:08	>
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
17:49:08	[RECEIVED MESSAGE]
17:49:08	S16F16 [00000022]
17:49:08	<L[2]
17:49:08	  <L[1]
17:49:08	    <A[3] 'PJ1'>
17:49:08	  >
17:49:08	  <L[2]
17:49:08	    <Bool[4] True>
17:49:08	    <L[0]
17:49:08	    >
17:49:08	  >
17:49:08	>
```

9.2.1.13 S14F9 Create Object Request – ControlJob H → E

Host creates Control Job with auto start to start processing

```
17:49:12	[SENDING MESSAGE]
17:49:12	S14F9 [00000023] W
17:49:12	<L[3]
17:49:12	  <A[3] 'CJ1'>
17:49:12	  <A[10] 'ControlJob'>
17:49:12	  <L[3]
17:49:12	    <L[2]
17:49:12	      <A[16] 'CarrierInputSpec'>
17:49:12	      <L[1]
17:49:12	        <A[16] '0039640900000000'>
17:49:12	      >
17:49:12	    >
17:49:12	    <L[2]
17:49:12	      <A[18] 'ProcessingCtrlSpec'>
17:49:12	      <L[1]
17:49:12	        <L[3]
17:49:12	          <A[3] 'PJ1'>
17:49:12	          <L[0]
17:49:12	          >
17:49:12	          <L[0]
17:49:12	          >
17:49:12	        >
17:49:12	      >
17:49:12	    >
17:49:12	    <L[2]
17:49:12	      <A[11] 'StartMethod'>
17:49:12	      <Bool[4] true>
17:49:12	    >
17:49:12	  >
17:49:12	>
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
17:49:12	[RECEIVED MESSAGE]
17:49:12	S14F10 [00000023]
17:49:12	<L[3]
17:49:12	  <A[3] 'CJ1'>
17:49:12	  <L[3]
17:49:12	    <L[2]
17:49:12	      <A[16] 'CarrierInputSpec'>
17:49:12	      <L[1]
17:49:12	        <A[16] '0039640900000000'>
17:49:12	      >
17:49:12	    >
17:49:12	    <L[2]
17:49:12	      <A[18] 'ProcessingCtrlSpec'>
17:49:12	      <L[1]
17:49:12	        <L[3]
17:49:12	          <A[3] 'PJ1'>
17:49:12	          <L[0]
17:49:12	          >
17:49:12	          <L[0]
```

```
17:49:12            >
17:49:12          >
17:49:12        >
17:49:12      >
17:49:12      <L[2]
17:49:12        <A[11] 'StartMethod'>
17:49:12        <Bool[4] True>
17:49:12      >
17:49:12    >
17:49:12    <L[2]
17:49:12      <U1[1] 0>
17:49:12      <L[0]
17:49:12      >
17:49:12    >
17:49:12  >
```

10 Abbreviations

List of abbreviations used in this document:

Abbreviation	Description
NA	Not Applicable